TS5000 Robot Controller

Communication Manual

Rev. 13

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IMPORTANT

- Make sure that this manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading this manual, keep it in an easily accessible place for future reference.

Shibaura Machine

This manual applies to the controller TS5000.

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Introduction

Thank you for purchasing SHIBAURA MACHINE's robot system.
This manual describes the communication with the TS5000 Series Robot Controller.
Communication with the controller can be used to obtain various controller information and to perform control and settings for the controller.
Before using this manual, refer to the Safety Manual, and be sure that you fully understand the implemented safety measures.

This manual consists of the following sections.

Part 1: Overview
This provides an overview of communication with the TS5000.

Part 2: Specifications
This describes the specifications for non-protocol communication and host command communication.

Part 3: Settings
This describes the parameters.

Manufacturer

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If you have purchased this controller outside of Japan, please contact the supplier at the time of purchase.

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Usage Range

The product has been developed for use in a wide range of automation tasks such as transport work including the movement and alignment of workpieces and assembly work such as insertion of parts and screw tightening.
The customer should also use this product for the purpose of automating these tasks and for incorporating into automation equipment.

This system was not designed for use in operations where contact is made with people (operations in medical treatment, nursing care, etc.), for incorporating into equipment used for sustaining human life (life support devices, infant incubators, etc.), for incorporating into equipment used for transporting people (automobile and other transportation equipment, airplanes and other aerospace equipment, roller coasters and other amusement park equipment), for equipment used for protecting human life or the human body (safety devices, etc.), or similar equipment.
This product is not intended to be incorporated in equipment for controlling large heat quantities (such as nuclear power controllers and incinerator controllers).
This product is prohibited from being used for tasks or incorporated in equipment where a failure or malfunction could likely pose a direct threat to human life or harm to the human body. The customer is responsible for providing any warranties if this product is used.

Usage Procedures

This section describes the usage requirements for the TS5000 robot. Usage of the product in a state outside the required conditions could result in unexpected risks. Be sure to always use the robot in an environment that meets the requirements

1. Main Specifications of the Product — Refer to the robot manual for your robot.
2. Usage Restrictions: — Tables 1 to 6 each provide the “Restricted Specifications Sheet” indicating the usage restrictions.

Table 1. Restricted Specifications Sheet

1. Purpose and application for using product (usage restrictions)	
1) Intended use	To make the robot perform operations for specific purposes (item transport, assembly, inspection, machining, etc.) (1) Transportation [1] Obtain a transportation route to the unpacking site. [2] Use a forklift or other equipment to lift the packing box. [3] Move the packing box to the unpacking site. (2) Unpacking [1] Obtain the space required for unpacking. [2] Cut the packing straps and unpack. [3] Cut the packing straps securing the robot. [4] Peel off the tape from the controller packing box, and unpack. [5] Check that the robot and controller are not damaged. [6] Check the accessories and their quantities. - When transferring the robot to the installation site [7] Dispose of, or store and manage the packing materials. - When storing the robot [7] Secure the robot in place in the same way as the state at delivery, and repack. [8] Move the packing box to the storage site. (3) Transfer - Robot [1] Check the orientation of the robot during transfer. [2] Check that the robot fixing jigs are mounted. [3] If using a crane, mount the supplied eyebolts. [4] Check the tightness of the robot fixing jig fixing bolts. [5] Use a crane or cart to transfer to the installation site. - Controller [1] Use a cart to transfer to the installation site.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(4) Installation [1] Use bolts to secure the robot to the frame and other parts. [2] Fabricate any needed cables. [3] Remove the robot fixing jigs and eyebolts. [4] Install the controller and connect it to the robot. [5] Push the robot by hand to move to a location where it is easy to perform work. [6] Mount the hand effector. [7] Connect the required external devices. (5) Teaching [1] Move the robot to the desired position or orientation by guiding it with a teach pendant or pushing it by hand. [2] Teach the position where the robot will operate. [3] Create the operation sequence (program, etc.). [4] Make sure that the signals and data are transferred to and from the external equipment. [5] Check operation at low speed (250 mm/sec or less). [6] While gradually increasing the operating speed, make sure that the operation matches the specific purpose. (6) Operation [1] Start robot operation. [2] Make sure that operation is being performed to match the specific purpose at the specified speed. [3] Stop robot operation. (7) Repacking [1] Remove the hand effector. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Obtain the space needed for packing. [6] Use a crane to transfer the robot, and mount the supplied eyebolts. [7] Transfer the robot and controller to the packing site. [8] Pack the robot and controller so that they are in the original delivery state. [9] Obtain a transportation route to the transportation equipment (truck, etc.). [10] Use a forklift to lift the packing box. [11] Load the packing box onto the transportation equipment.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(8) Maintenance Work performed when the primary power is off: - Robot [1] Check the bolts for any looseness and backlash and re-tighten them. [2] Make sure that the cable clamp is not loose. [3] Check for any damage. [4] Clean off any dust. [5] Inspect and apply corrosion inhibitor. [6] Check for any wear on the cable and air tube. [7] Check that there is no misalignment between the motion of each joint axis and the origin position. - Controller [1] Replace the memory backup batteries in the controller. Work performed when the primary power is on: [1] Check the grease state and apply grease (when the servo power is off). [2] Check for any abnormal vibrations and sounds during operation (when the servo power is on). [3] Check the teaching point for any misalignment (when the servo power is on). [4] Replace the backup batteries for the robot position detector (when the servo power is off). (9) Disposal [1] Remove the hand effector. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] To use a crane to transfer the robot, mount the supplied eyebolts. [6] Transfer the robot and controller to the disposal site. [7] Dispose of the robot using a method compliant with your country’s laws and regulations.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
2) Foreseeable misuse	- Use for work which applies excessive impact to the robot or to which excessive impact is applied by the robot - Modification of robot, controller, and optional parts, or supplying power while these covers are removed - Operation in other installation state than the specified (floor, suspended, and wall-mounted) - Operation on a frame or other structure that does not satisfy the specified strength - Operation under conditions not satisfying the specified installation requirements - Entry into the hazard zone of the robot by mistakenly assuming that operations would have stopped although the stop is just for waiting for an operation signal during operation - Usage under conditions where the specification values are exceeded or usage under improper settings - Stop of the robot caused by power off during operation or when the servo power is on - Unintended operation caused by inadvertently pressing of a button - Operating without checking operation in MANUAL mode - Operating without checking the tightness of the robot and hand set bolts - Unintended operation due to improper wiring - Unintended operation due to incorrect program selection - Riding on or leaning against the robot or hanging on the arm
3) Unexpected start	- Malfunction due to excessive noise - Robot start due to start signal unexpectedly sent from peripheral equipment to the robot - Unintended start due to abnormal communication data - Malfunction due to voltage fluctuation - Power on due to powering after power outage - Malfunction due to failure or other cause

Table 2. Restricted Specifications Sheet

2. Replacement or other modification of program components (time restrictions)	
1) Mechanical restrictions	- Before operating the robot, perform operation at a low speed (20% or less of maximum speed) for five (5) minutes - Replace consumable parts as required or periodically by conducting daily inspection, regular inspection (every 3 and 6 months and every year) and overhauls (every 5 years).
2) Electrical restrictions	

Table 3. Restricted Specifications Sheet

3. Movable range of product (restriction on space)		
1) Work envelope	1st axis 2nd axis 3rd axis 4th axis Additional axis — —	Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with individual specifications. — —
2) Interface	Transportation: The packing box shall be able to be transferred by a transportation worker. Unpacking: The packing box shall be able to be carried in. Space of at least 2 m^3 shall be secured. Transfer: The robot shall be able to be hoisted by a crane. The robot shall be able to be lifted by two or more people and loaded onto a cart or similar device. There shall be no obstacles on the route from the unpacking site to the installation site. Installation: The product shall be installed at a location suitable for the outline and work envelope of the robot. The product shall be installed at a location suitable for the outline and that provides adequate ventilation of the controller. The cable shall be installed at a location where it will not be stepped on by operators. Teaching: Teaching work shall be able to be performed from outside the hazard zone of the robot. Space shall be provided so that, if a person enters the hazard zone, the person will not get entrapped by the robot or peripheral equipment. Operation: A fence or enclosure shall be installed to prevent any operator from approaching the hazard zone of the robot. The robot shall be able to be operated from outside the hazard zone of the robot. An operator in the fence or enclosure shall be visible from the operation location. Repacking: Work shall be able to be performed to remove the hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the packing site. The repacking work shall be able to be performed. A forklift or other equipment shall be able to be used to lift the packing box. Maintenance: Space required for maintenance shall be obtained. After maintenance, operation shall be able to be checked from outside the hazard zone of the robot. Space shall be provided so that, in the hazard zone, a person will not get entrapped by the robot or peripheral equipment. Disposal: Work shall be able to be performed to remove the hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the disposal location.	

Table 3. Restricted Specifications Sheet (continued)

3. Movable range of product (restriction on space)	
3) Work environment	- A fence, enclosure, or other protection shall be provided that surrounds the hazard zone of the robot to prevent people from entering the zone during operation. - A display indicating the robot operation status shall be provided to indicate whether the robot is operating or stopped. - Lighting shall be provided for enabling visual checks by operators. - To prevent operators from slipping, the floor shall not be covered with oil or similar substances. - No noises that impair attention shall be emitted.

Table 4. Restricted Specifications Sheet

4. Life cycle of product
In the product life cycle of assembly, shipment testing, transportation, unpacking, transfer, installation, teaching, operation, repacking, maintenance, and disposal, risk assessment is carried out for the life cycle of “transfer and installation” (transportation, unpacking, transfer, installation, and repacking), “use” (teaching, operation, and maintenance), and “disposal.”

Table 5. Restricted Specifications Sheet

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
1) Transportation worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for forklifts and other transfer equipment work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
2) Unpacking worker	Knowledge	Contents of the instruction manual
	Requirement	—
3) Transfer worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
4) Installation worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
5) Teaching worker Work supervisor	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations
	Requirement	Person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
6) Operator	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations
	Requirement	—
7) Repacking worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer

Table 5. Restricted Specifications Sheet (continued)

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
8) Maintenance worker Work supervisor	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations Storage procedures for target robot
	Requirement	Person who has received training for robot testing work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
9) Disposal worker	Knowledge	Contents of instruction manual, operating procedures for target robot, and waste disposal methods
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
10) People passing nearby Other workers	- Measures shall be taken to prevent entry into the fence, enclosure, or other protection surrounding the hazard zone of the robot. - Measures shall be taken to make the robot stop operation when someone approaches the hazard zone. - If another worker enters the hazard zone of the robot to perform work, robot operation shall be stopped, the drive power turned off, and a teaching worker or maintenance worker shall supervise the situation to ensure safety.	

Table 6. Restricted Specifications Sheet

6. Other

Warranty

We conduct strict testing and inspection of this product before delivery to ensure that it satisfies our performance standards.
We warrant this product to be free from defects under the following terms and conditions.

1. Warranty Period

The warranty period of this product is one of the following periods, whichever is shortest.

(1) 18 months after shipping from our factory
(2) 12 months after installation at customer site

2. Warranty scope

(1) The warranty is provided for this product. The warranty covers only the specifications and functions included in the product specifications, catalog, and instruction manual. Any secondary or incidental damages caused by failure of this product are not covered by the warranty under any circumstances.
(2) We will repair a faulty product free of charge during the warranty period only if this product has been operated and used in accordance with the instruction manual supplied with the product. A fee will be charged for any repairs after the warranty period has expired.

3. Disclaimers

The following cases are outside the scope of the warranty.

(1) Any failure or damage arising from careless use or misuse, or usage that is not described in the instruction manual.
(2) Problems due to deterioration over time or usage wear (natural fading of the paint coating, deterioration of consumable parts [1)]
(3) Irregularities confined to sensory quality such as noise not recognized as affecting functions
(4) When any modification or disassembly without our consent was performed
(5) Any failure or damage arising from inadequate maintenance and inspection or improper repair
(6) Any failure or damage due to an act of God, fire, and other force majeure
(7) Internal data such as programs and points created or modified by customers
(8) When this product was purchased in Japan and taken outside the country

1) Consumable parts refer to the replacement parts for maintenance as listed in the target controller manual.

4. Safety Information

(1) SHIBAURA MACHINE shall provide no guarantees for the standard performance of this product when this product is used outside the specifications.
(2) SHIBAURA MACHINE will not be liable for any injuries, damage, faults, or other problems that arise by failing to follow the dangers, warnings, cautions and notes provided in the instruction manual.
(3) Be aware that the dangers, warnings, cautions, notes and other safety information in the instruction manual apply to the safety risks foreseeable by SHIBAURA MACHINE, and they do not

cover every possible scenario.

Standards, laws, and regulations pertaining to the safety of industrial robots

The international industrial standard ISO/DIS 12100 "Safety of machinery" describes the risk reduction measures for machines and equipment.
Step 1: Inherently safe design measures - Limitations on force, speed, energy, etc.
Step 2: Safeguarding - Installation of safety fences, etc.
Complementary protective measures - Installation of emergency stop devices, etc.
Step 3: Information for use - Warning signs, alarms, instruction manuals, etc.

Based on this, the guidelines known as "Guide 51" were defined by classifying the various standards for the ISO/IEC international standards in a hierarchical structure.
Safety standards for industrial robots are standardized as separate safety standards for machinery in Type C.

- ISO 10218, ISO 10218-1

Be sure to follow the standards, laws, and regulations set forth in the country where the industrial robot will be used.

Safety Information

The instruction manual for the controller contains important safety information for ensuring the safe and correct operation of this device and for preventing hazards to the operators and others and property damage.
Be sure that you fully understand the following information (signal words and symbols) before reading the main text, and always follow the provided safety instructions.

[Explanation of indications]

Signal word and symbol	Meaning
⚠ DANGER	Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
⚠ WARNING	Indicates a potentially hazardous situation which, if not avoided, will result in death or serious injury.
⚠ CAUTION	Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury[1].
NOTICE	Indicates a situation which could result in physical damage[2] only. Indicates advice on how to proceed.

1) Injury refers to wounds, burns, electric shocks, and similar cases that do not require hospitalization or prolonged outpatient treatment.
2) physical damage refers to extensive damage to property or materials.

⚠ DANGER
● Be sure to carefully read the safety manual described in the target controller manual in order to ensure the safety of the controller from transportation to disposal. Performing an operation different from what is described in the manual could result in harm to people.

Service Network

Please refer to our distributor's handouts

Overview

This provides an overview of communication with the TS5000.

1. Introduction

This manual describes the Ethernet communication between the robot controller and an external computer.
The transfer procedures include non-protocol communication and host command communication, and each procedure has the following features.

Non-protocol communication

(1) Input program variables from an external device
(2) Output program variables or messages

Host command communication

(1) Start and stop operation of the robot
(2) Program reset, step reset, cycle reset and output signal reset
(3) Servo ON/OFF
(4) Upload and download program files
(5) Erase program files
(6) Select programs
(7) Monitor the status of the robot
(8) Monitor robot alarms
(9) Monitor communication data in non-protocol communication and dedicated communication

Host command communication is a collective term for communication that uses control communication ports, monitor communication ports, and periodic monitor communication ports.

The communication interface has the configuration shown in “Fig. 1.1 Communication interface configuration”.

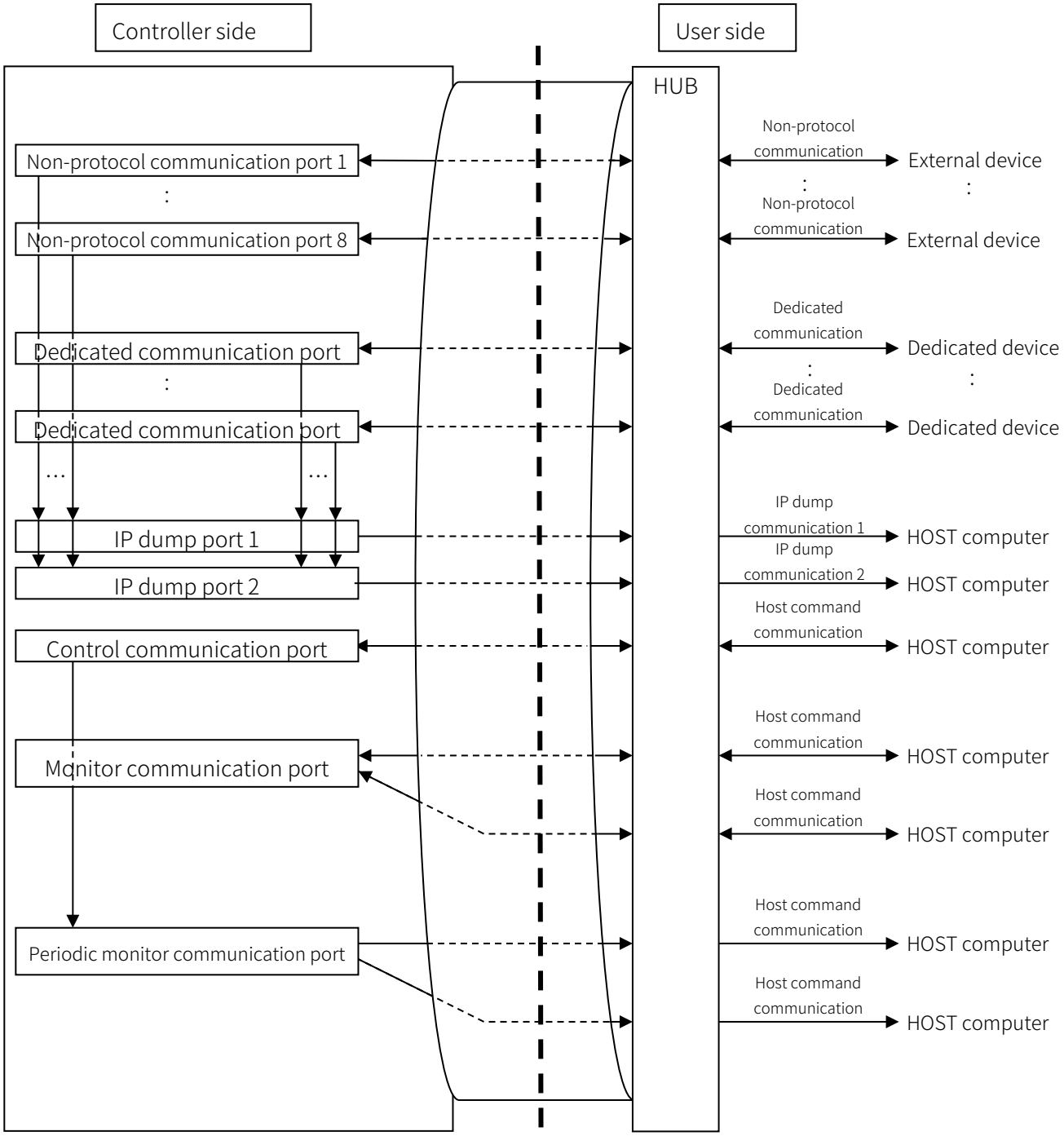


Fig. 1.1 Communication interface configuration

1.1 Status Transitions

The status transition diagram for the controller is shown in “Figure 1.2 Status transition diagram”. According to the status of the controller, the status of a command changes between valid and invalid. For details on the commands that can and cannot be executed for each status, see "4.3.1 Commands and Operation Modes".

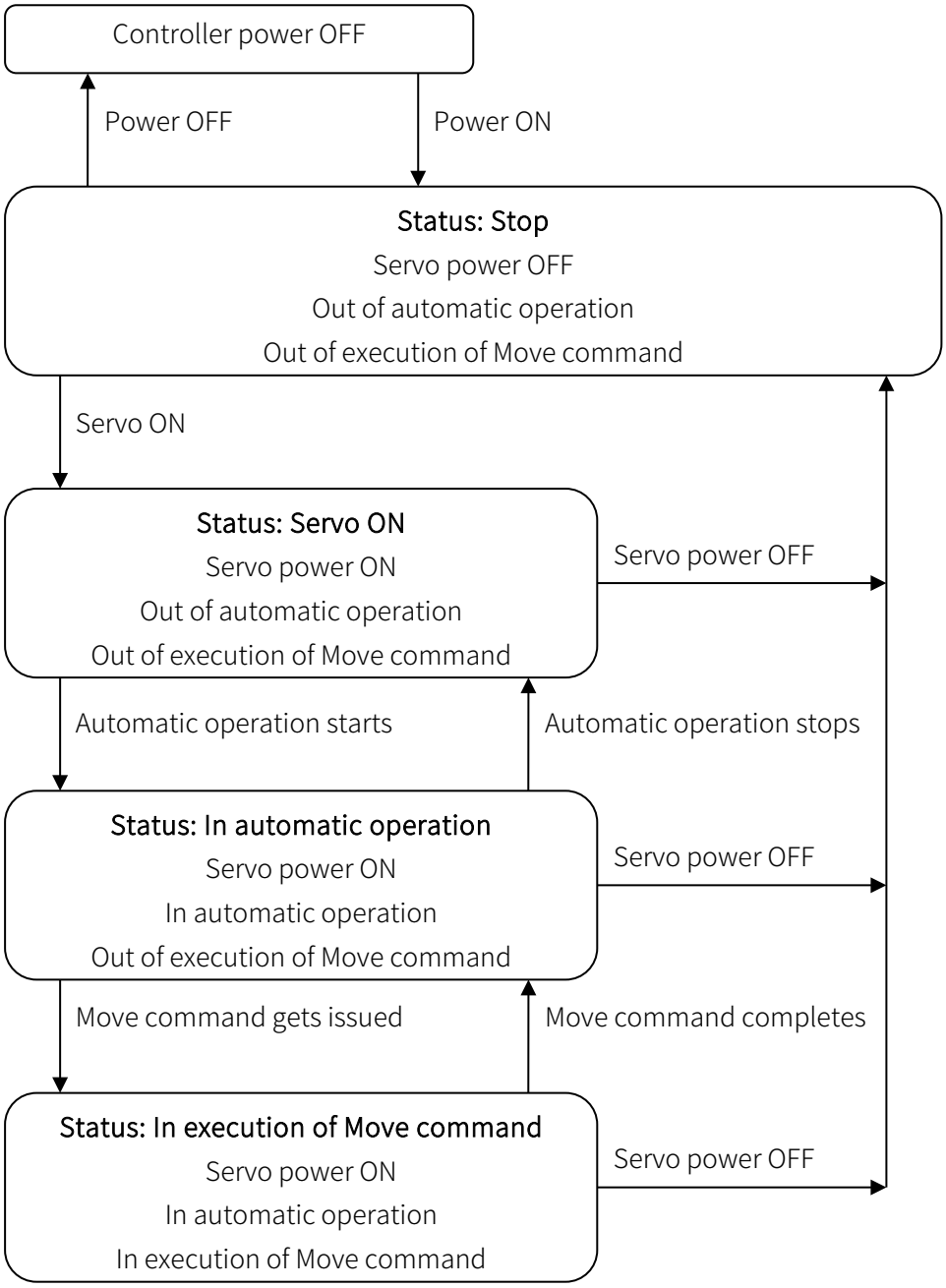


Figure 1.2 Status transition diagram

Specifications

This describes the specifications for communication with the TS5000.

2. Ethernet Specifications and Settings

In non-protocol communication, a non-protocol communication port is used to perform non-protocol communication with an external device. In host command communication, a control communication port, monitor communication port, periodic monitor communication port, or IP dump port is used to communicate with a host computer. In dedicated communication, a dedicated communication port is used to communicate with a dedicated communication device.
This function uses TCP/IP protocol to enable communication on the same local area network with computers and visual equipment on the Ethernet.

2.1 Connections

The robot controller has an automatic recognition function for enabling use of either a straight cable or crossover cable.

2.2 Communication Specifications

Table 2.1 Communication specifications

Item		Specification				
Interface		1000BASE-X				
Communication rate		1000 Mbps				
Function		Communication system	Setting role	Number of available ports	Number of simultaneous connectible clients	Port number
	Non-protocol communication port	TCP / UDP	Server / Client (Selectable)	8	1 *1	10000 to 65535 (selectable)
	Dedicated communication port	TCP / UDP	Server / Client (Selectable)	2	1 *1	10000 to 65535 (selectable)
	Control communication port	TCP	Server (Fixed)	1	1	2000 (Fixed)
	Dedicated monitor communication port	TCP	Server (Fixed)	1	2 *2	2001 (Fixed)
	Periodic monitor communication port	TCP	Server (Fixed)	1	2 *2	2002 (Fixed)
	IP dump port 1	TCP	Server (Fixed)	1	1	2003 (Fixed)
	IP dump port 2	TCP	Server (Fixed)	1	1	2004 (Fixed)

*1 Only one client can connect to ports that are set to the Server function. Connection from multiple clients is not allowed.

*2 Up to two clients can be connected simultaneously to the same server port of the controller.

2.3 Communication Parameters

The Ethernet communication settings are defined in the Ethernet parameter file (ETHERNET.PAR) stored in the robot controller.

[E000] ROBOT CONTROLLER IP ADDRESS

Controller IP address

This sets the IP address of the controller.

Parameter key	IP-ADDRESS
Parameter data	=(xxxx), (xxxx), (xxxx), (xxxx)
Data type	Integer
Data count	4
Data range	-
Default	=192, 168, 0, 124

Setting example

```
'[E000] ROBOT CONTROLLER IP ADDRESS
IP-ADDRESS = 192, 168, 0, 124
```

MEMO

- *Instead of a dot (.), a comma (,) is used to separate numbers.*
- *Be careful that the IP addresses within the network are not duplicated.*
- *The controller cannot connect to the network unless the IP address is set correctly.*
- *The IP address range of "223.255.0.0" to "223.255.255.255" is reserved for the system by the controller. When setting the IP address for the controller, set an IP address in the range outside those reserved for the system.*

[E001] SUBNET MASK

Controller subnet mask settings

This sets the controller subnet mask.

Parameter key	SUBNETMASK
Parameter data	=(xxxx), (xxxx), (xxxx), (xxxx)
Data type	Integer
Data count	4
Data range	-
Default	=255, 255, 255, 0

Setting example

```
‘[E001] SUBNET MASK
SUBNETMASK = 255, 255, 255, 0
```

MEMO

- *The controller cannot connect to the network unless the subnet mask is set correctly.*

[E002] DEFAULT GATEWAY

Controller default gateway setting

This sets the controller default gateway.

Parameter key	GATEWAY
Parameter data	=(xxxx), (xxxx), (xxxx), (xxxx)
Data type	Integer
Data count	4
Data range	-
Default	=192, 168, 0, 1

Setting example

```
‘[E002] DEFAULT GATEWAY
GATEWAY = 192, 168, 0, 1
```

MEMO

- *The controller cannot connect to the network unless the default gateway is set correctly.*

[E003] ROBOT COMMUNITY NAME

Controller name

This sets the controller name.

Parameter key	ROBO-COMMUNITY-NAME
Parameter data	= "Name"
Data type	Character string
Data count	1
Data range	-
Default	=”TS5K”

Setting example

```
‘[E003] ROBOT COMMUNITY NAME
ROBO-COMMUNITY-NAME = "TS5K"
```

MEMO

- *Set robot controller names to unique values on the same network.*

[E004] OPEN MODE IP1 IP2 IP3 IP4 IP5 IP6 IP7 IP8

IP open mode

Select from IP1 to IP8 mode.

Parameter key	OPEN-MODE	
Parameter data	=(IP1), (IP2), (IP3), (IP4), (IP5), (IP6), (IP7), (IP8)	
Data type	Integer	
Data count	1	
Data range	0 to 3	
Description	=0	Not used
	=1	This sets the controller to a TCP/IP server for performing communication.
	=2	This sets the controller to a TCP/IP client for performing communication.
	=3	The controller uses UDP to perform communication.
Default	=1, 0, 0, 0, 0, 0, 0, 0	

Setting example

```
‘[E004] OPEN MODE IP1 IP2 IP3 IP4 IP5 IP6 IP7 IP8
OPEN-MODE = 1, 2, 2, 0, 0, 0, 0, 0
```

MEMO

- *The controller cannot connect to the network unless the server, client, UDP, and other parameters are set correctly.*

[E005] OWN PORT NUMBER

Controller port number

Set the port number assigned to controller IP1 to IP8.

Parameter key	OWN-PORT-NO[num[1)]]
Parameter data	= (Port number)
Data type	Integer
Data count	1
Data range	10000 to 65535

1) An integer value from 0 to 7 is entered for "num".

Setting example

```
‘[E005] OWN PORT NUMBER
‘{IP1}
OWN-PORT-NO[0] = 10000
‘{IP2}
OWN-PORT-NO[1] = 10001
‘{IP3}
OWN-PORT-NO[2] = 10002
‘{IP4}
OWN-PORT-NO[3] = 10003
‘{IP5}
OWN-PORT-NO[4] = 10004
‘{IP6}
OWN-PORT-NO[5] = 10005
‘{IP7}
OWN-PORT-NO[6] = 10006
‘{IP8}
OWN-PORT-NO[7] = 10007
```

MEMO

- *This is used when the OPEN MODE that was set in [E004] is Server (=1).*
- *Set so that the port numbers are not duplicated.*
- *The controller cannot connect to the network unless the port number is set correctly.*

[E006] IP ADDRESS OF DESTINATION

IP address of destination

This sets the connection target IP address for IP1 to IP8.

Parameter key	DEST-IP-ADR[num[1)]]
Parameter data	=(xxxx), (xxxx), (xxxx), (xxxx)
Data type	Integer
Data count	4
Data range	-

1) An integer value from 0 to 7 is entered for "num".

Setting example

```
‘[E006] IP ADDRESS OF DESTINATION
‘{IP1}
DEST-IP-ADR[0] = 192, 168, 0, 150
‘{IP2}
DEST-IP-ADR[1] = 192, 168, 0, 151
‘{IP3}
DEST-IP-ADR[2] = 192, 168, 0, 152
‘{IP4}
DEST-IP-ADR[3] = 192, 168, 0, 153
‘{IP5}
DEST-IP-ADR[4] = 192, 168, 0, 154
‘{IP6}
DEST-IP-ADR[5] = 192, 168, 0, 155
‘{IP7}
DEST-IP-ADR[6] = 192, 168, 0, 156
‘{IP8}
DEST-IP-ADR[7] = 192, 168, 0, 157
```

MEMO

- *The controller cannot connect to the network unless the IP address is set correctly.*

[E007] PORT NUMBER OF DESTINATION

Connection target port number

This sets the connection target port number for IP1 to IP8.

Parameter key	DEST-PORT-NO [num[1)]]
Parameter data	= (Connection target port number)
Data type	Integer
Data count	4
Data range	-

1) An integer value from 0 to 7 is entered for "num".

Setting example

```
‘[E007] PORT NUMBER OF DESTINATION
‘{IP1}
DEST-PORT-NO[0] = 10000
‘{IP2}
DEST-PORT-NO[1] = 10001
 ‘{IP3}
DEST-PORT-NO[2] = 10002
 ‘{IP4}
DEST-PORT-NO[3] = 10003
 ‘{IP5}
DEST-PORT-NO[4] = 10004
 ‘{IP6}
DEST-PORT-NO[5] = 10005
 ‘{IP7}
DEST-PORT-NO[6] = 10006
 ‘{IP8}
DEST-PORT-NO[7] = 10007
```

MEMO

- *This is used when the OPEN MODE that was set in [E004] is Client (=2).*
- *The controller cannot connect to the network unless the port number is set correctly.*

[E008] TCP STATUS ALARM IP1 IP2 IP3 IP4 IP5 IP6 IP7 IP8

TCP status alarm

This sets the controller IP1 to IP8 alarm level that occurs when the Ethernet connection is disconnected while running a program.

Parameter key	TCP-STATUS-ALM
Parameter data	=(IP1), (IP2), (IP3), (IP4), (IP5), (IP6), (IP7), (IP8)
Data type	Integer
Data count	8
Data range	One of 0, 1, 2, 4, or 8
Default	=0, 0, 0, 0, 0, 0, 0, 0

Setting example

‘[E009] TCP STATUS ALARM IP1 IP2 IP3 IP4 IP5 IP6 IP7 IP8
TCP-STATUS-ALM = 8, 4, 2, 0, 0, 0, 0, 0

MEMO

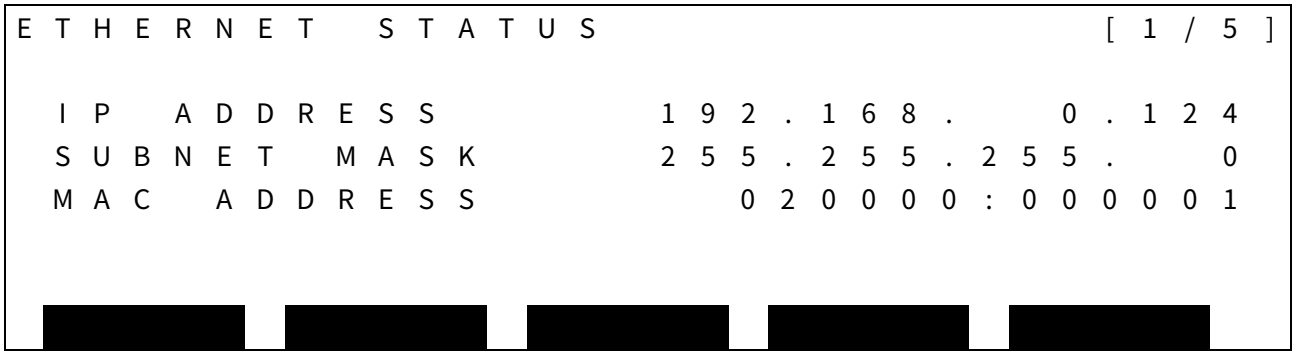
No alarm occurs when the value is set to 0.

2.4 Ethernet Status Display [ETHER]

This shows the controller Ethernet information and the status of each Ethernet port.

The controller IP address, subnet mask, default gateway, and non-protocol communication port status are displayed.

(Page 1)



- IP ADDRESS : IP address of the robot controller itself that is written in the "ETHERNET.PAR" [E01] file
- SUBNET MASK : Subnet mask of the network that is written in the "ETHERNET.PAR" [E03] file
- MAC ADDRESS : MAC address of the robot controller itself

(Page 2)

ETHERNET STATUS			[2/5]
	OwnPort	Connect IP	Port
IP1	10000	192.168.0.150	10000
IP2	10001	192.168.0.151	10001
IP3	10002	192.168.0.152	10002
IP4	10003	192.168.0.153	10003

(Page 3)

ETHERNET STATUS			[3/5]
	OwnPort	Connect IP	Port
IP5	10004	192.168.0.154	10004
IP6	10005	192.168.0.155	10005
IP7	10006	192.168.0.156	10006
IP8	10007	192.168.0.157	10007

- IP1 to IP8 : Ethernet channel number
- OwnPort : Robot port number
- Port : Port number of the destination
- Connect IP: IP address of the destination

(Page 4)

```
ETHERNET STATUS                                    [ 4 / 5 ]
        App              Mode                Status
IP1     SCOL             TCP/S          Established
IP2     SCOL             TCP/S          Established
IP3     SCOL             TCP/S          Established
IP4     SCOL             TCP/S          Established
```

(Page 5)

```
ETHERNET STATUS                                    [ 5 / 5 ]
        App              Mode                Status
IP5     SCOL             TCP/S          Established
IP6     SCOL             TCP/S          Established
IP7     SCOL             TCP/S          Established
IP8     SCOL             TCP/S          Established
```

- IP1 to IP8 : Ethernet channel number
- App. : Name of application using the channel
  This shows the connection when controlling from robot language using non-protocol communication.
- Mode : Protocol type, open status

TCP/S indicates a server connection using TCP protocol, and TCP/C indicates a client connection using TCP protocol.

- Status : Ethernet port status

The statuses below are displayed. For the specific meaning of each status, refer to any commercially-available reference that provides explanations of TCP/IP.

Closed
Listen
Syn-Sent
Syn-Received
Established
Fin-Wait1
Fin-Wait2
Close wait
Closing
Last ACK
Time Wait

The valid keys in the ETHER screen and their actions are shown in the following table.

Key	Description
NEXT key	Shows the next page.
ALT key + NEXT key	Shows the previous page.
ESC key	Returns to the utility menu screen.

MEMO

- *Check that the Ethernet cable between the controller and external device is connected correctly.*
- *After setting the parameters, turn the AC power off and then on again.*

2.5 Confirmation with a Windows Computer

This describes how to confirm the connection using the DOS command "ping".
From the Start menu in Windows, go to the Command Prompt, and specify the IP address of the robot controller as shown below.

C: |> ping [Robot controller IP address]

If there is a transmission problem, the result "unreachable" or "Request Time out" is displayed as shown in Fig. 2.1.
If transmission is performed normally, the result "Reply from …" is displayed as shown in Fig. 2.2.

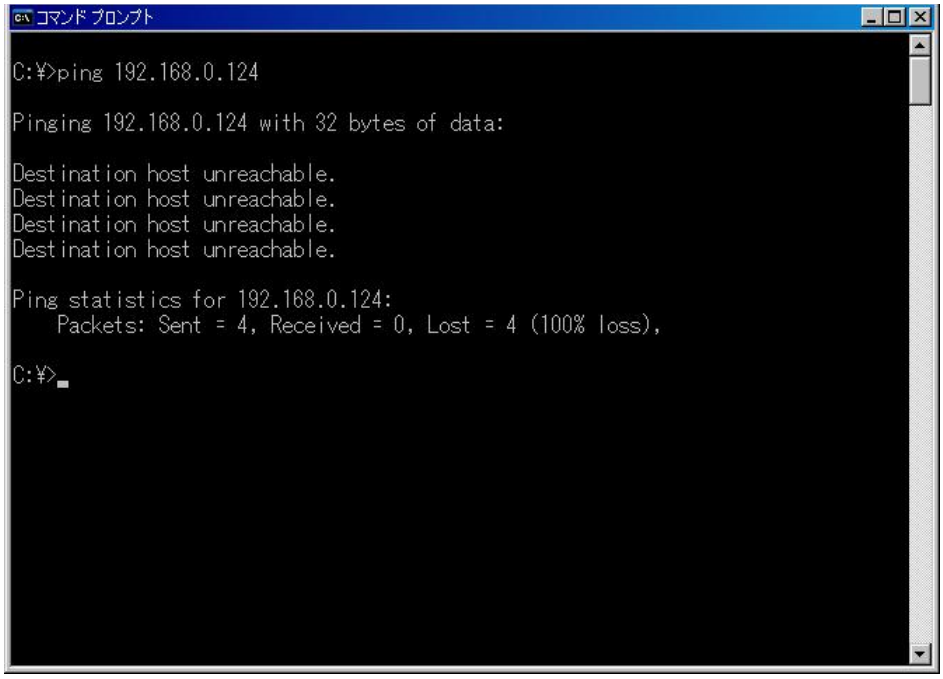


Fig. 2.1 Ping execution screen (Problem result)

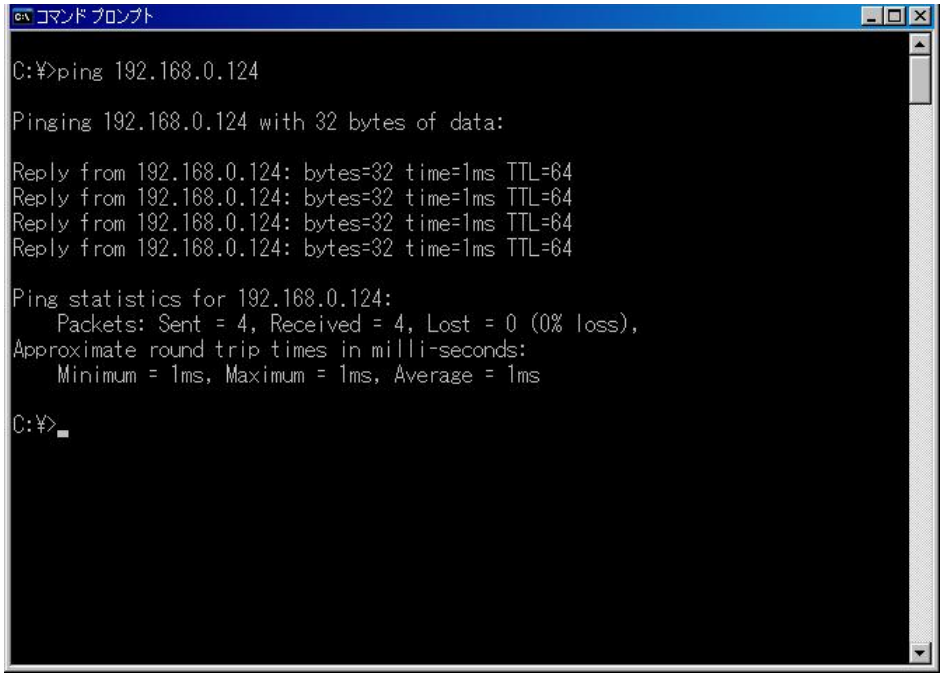


Fig. 2.2 Ping execution screen (Normal result)

3. Non-Protocol Communication

Non-protocol communication uses a non-protocol communication port to exchange data with a robot language program.
The following is the communication data format for sending data to the robot controller using non-protocol communication.

Character string CR
Character string,
character string CR

These character strings are based on ASCII format. At the end of the character strings, add CR, which indicates the end of the character strings. Separating character strings with commas enables you to send more than one kind of character strings.

The functions described in host command communication cannot be used in non-protocol data communication. Also, no processes for responses or timeout checks are performed for the data received by the controller.

3.1 Communication with a Robot Language Program

Data communication with a robot language program can be performed using the PRINT and INPUT commands. For details on robot language, see "STA1163-0_Robot Language Manual (SCOL2)".

3.1.1 PRINT Command

The PRINT command allows you to send out a specified character string from the controller. The controller transfers character strings or values of variable specified by the PRINT command.
The format of the PRINT command is shown below:

PRINT ␣ [<channel>,] {<character string>|<expression>|}[, {<character string>|<expression>}]
.....[, CR]

<channel>: Specifies the communication port over which the data is to be transmitted. One of the following should be specified as the channel.

IP1 : Ethernet IP1 port
IP2 : Ethernet IP2 port
IP3 : Ethernet IP3 port
IP4 : Ethernet IP4 port
IP5 : Ethernet IP5 port
IP6 : Ethernet IP6 port
IP7 : Ethernet IP7 port
IP8 : Ethernet IP8 port
TP : Screen output to teach pendant

Unless <channel> is specified, data is transmitted to the teach pendant.

<character string>: The character string to be transmitted is specified by enclosing that string in double quotation marks (").
<expression>: Expressions made up of constants, variables, arithmetic operands and functions may be specified.
CR: CR is used when the record end code CR (0x0D) is added to the last of sending data.
Commas are used to separate any multiple character strings or expressions specified in the PRINT command. Character strings are enclosed inside double quotation marks, and everything inside of those double quotation marks is transmitted in ASCII code. Expressions are first solved, and the result is sent as a 12-character block of fixed length with the result pushed over to the right of that block. Should the result of the expression be an integer, that result is sent as a Base 10 number having a maximum of ten places (digits). Should the result of the expression be a real number, the result is sent as a number having an integral part with a maximum of four (4) digits and a decimal part with a maximum of three (3) digits (for a maximum of eight (8) places counting the decimal point). One space in front of the number is allocated for the sign (+ or -) of that number, although the sign is omitted if it is plus (+). The number is sent in a 12-character block, with the number pushed over to the right. The remaining spaces are filled with space codes (0x20). The number itself is sent in ASCII code. The commas used to separate character strings or expressions in a PRINT command are themselves not transmitted. When multiple character strings and/or expressions are specified with the PRINT command, the controller will send out these character strings and/or expressions as a single text.

Ex.: Transmission using the IP1 port
PRINT IP1, “INPUT DATA = ” , -1000.0/3, CR

Ex.: When the above example program is executed, the following data are transmitted.
「INPUT DATA = -333.333CR」

*In the PRINT command, if user parameter [U013] PRINT ALIGNMENT is set to 1, data is output left-justified without spaces. If [U013] PRINT ALIGNMENT is set to 0, data other than numbers will be output as spaces in a block with a fixed length of 12 characters.
Refer below for an example.

```
'[U013] PRINT ALIGNMENT PRINT-ALIGNMENT-SW = 1
```

* To use the PRINT command to output multiple character strings and/or expressions delimited by a comma ‘,’, add a comma enclosed by double quotation marks ‘”’ separately from delimiter ‘,’. They can be output delimited by a comma.

Ex.: Transmission of comma (,) character from IP1 port
PRINT IP1, A,“,”, B, CR

*To output control characters with the PRINT command, set the user parameter [U029] PRINT/INPUT CONTROL CODE CONVERSION to 1, or execute "ENABLE CTRLCODE" in the program and enter the ASCII code of the control character you wish to output as a two-digit hexadecimal number enclosed in "<>" as shown below.

Ex.: Control character output
PRINT IP1, "<02>ABCDE<0A><03>"

Ex.: If the above program example is executed, the data to be sent is as follows
「STXABCDELFETX」

3.1.2 INPUT Command

As opposed to the PRINT command, which is used for transmitting data, the INPUT command is used for receiving data. The only data which may be received by the controller are integers and real numbers. Data received by the controller is put into a variable (in a robot language program) specified by the INPUT command. This data can be referred to later in the program to operate the robot.

INPUT ␣ [<channel>,] <variable>, [<variable>] ...

<channel>:

Specifies the communication channel over which the data is to be received. One of the following should be specified as the channel.

IP1 : Ethernet IP1 port
IP2 : Ethernet IP2 port
IP3 : Ethernet IP3 port
IP4 : Ethernet IP4 port
IP5 : Ethernet IP5 port
IP6 : Ethernet IP6 port
IP7 : Ethernet IP7 port
IP8 : Ethernet IP8 port
TP : Key input from teach pendant

Unless <channel> is specified, the controller receives data from the teach pendant.

<variable>:

Specifies the variable in the robot program into which the data is to be entered.

Ex.: Waiting for input to N1 and N2 from IP1 port
INPUT IP1, N1, N2

Ex.: Waiting for input to N1 and N2 from IP1
INPUT IP1, N1, N2

Commas are used to separate two (2) or more variables specified in the INPUT command. The controller waits until data comes in over the communication channel specified by the INPUT command. Add a record end code CR (0x0D) to the end of data which is sent to the controller. When multiple units of data are to be sent to the controller, the individual data units should be separated with commas before being transmitted. When more units of data are received by the controller than was specified with the INPUT command, the surplus data are ignored and used for the next INPUT command. If fewer units of data are received by the controller than was specified with the INPUT command, the controller waits until the short data reach.

*To receive control characters with the INPUT command and utilize them in the program, set the user parameter [U029] PRINT/INPUT CONTROL CODE CONVERSION to 1, or execute "ENABLE CTRLCODE" in the program. When data is received in a string type variable in this state, with character codes in the range of 0x00 to 0x1F and 0x7F and above, it is converted into a two-digit hexadecimal string enclosed in angle brackets (<>) and held in the string type variable. (Pay attention to the size of the string type variable, as it holds a one-character control character converted into four characters.) However, CR (0x0D) and LF (0x0A) are handled as delimiters and cannot be included in variables.

Ex.: When receiving "ABCDENUL LF"
INPUT IP1, BUF IF BUF == "ABCDE<00>" THEN ...

MEMO

- *The controller starts reading any data sent to it only after an INPUT command is executed. Data are set in the ring buffer, and the input data are picked up according to the request of the INPUT statement. Any data received before the INPUT command is executed may be ignored. Take careful precautions when determining the data transmission timing.*
- *Should program execution be suspended while the controller is waiting for data to come in (as directed by an INPUT command), the execution of the INPUT command will be cancelled. Should the program be resumed, program execution will start from the step following the INPUT command. Any variables for which data was not yet received when the program was suspended will be treated as 0.*
  You should keep this in mind when writing your robot language program, i.e., you should arrange your program in such a way that it will still function properly even should the controller (mistakenly) treat input values as 0. One way to do this is to have the controller ask the host computer for confirmation (retransmission) whenever the controller receives a 0. Another way to do this is to add on check-sum data (to the data to be transmitted) and check the validity of the received data. (For programming examples, see "Section. 3.2 Programming examples for communication with a robot language program.")
- *When the INPUT command was executed in step operation mode using MANUAL mode, program execution is in a standby status until data is received.*
  The same holds true when an INPUT command is executed directly.
- *When an INPUT command is to be used to receive multiple variables transmitted as one text from the host computer, individual numerals should be separated in the text with commas.*

NOTICE

· Before sending the INPUT data, check that the numbers and character strings are correct.
The robot that has performed unexpected motion may collide with peripheral equipment.

3.1.3 Clearing Communication Buffer

If the number of data the controller has received is larger than the number of data specified by the INPUT command, such data are stored in the communication buffer and used at request of the next INPUT command. If data is left in the communication buffer, unexpected data will reach by the INPUT command. To clear the data left in the communication buffer, output character string "BUFFRESET" to relevant communication port.

Example) When clearing the communication buffer of the TP port:
PRINT␣TP, “BUFFRESET”

Example) When clearing the communication buffer of the IP1 port:
PRINT␣IP1, “BUFFRESET”

3.2 Programming Examples for Communication with a Robot Language Program

It is possible to utilize communication functions in a robot language program in order to do such things as specifying program branches, correcting the position of the robot to be performed, etc.
Listed below are several programming examples showing how to do this.

3.2.1 Program Branching

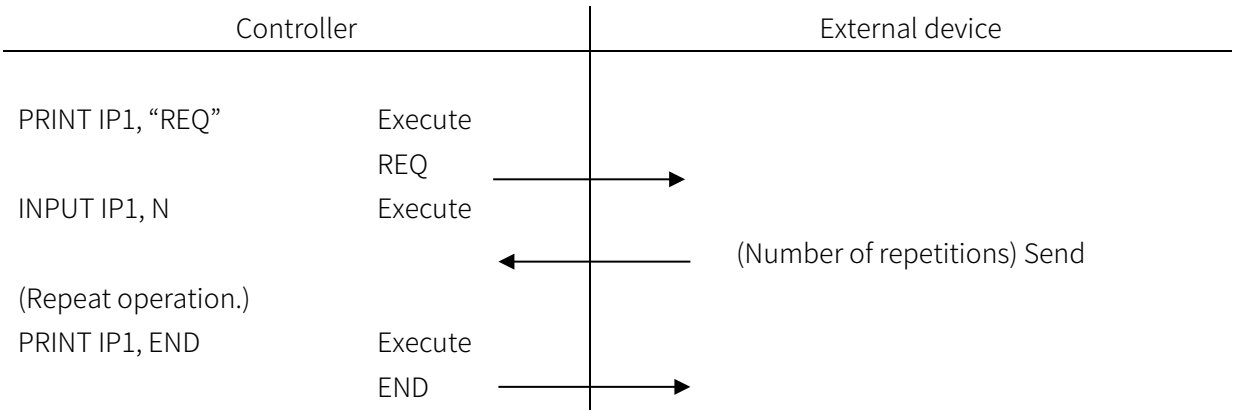
The INPUT command can be utilized to specify the number of times a certain action is to be repeated, to select a task for execution, etc.

Specifying the number of times an action is to be repeated (IP1)

```
PROGRAM REPEAT
    N=0
START:
    PRINT IP1, "REQ"
    INPUT IP1, N
    IF N==0 THEN GOTO START
    FOR K=1 TO N
        MOVE A1
        MOVE A2
        . . .
    NEXT K
    PRINT IP1, "END"
END
```

In this program, the robot will repeat a certain series of motions as many times as the external device tells it to.
First, the controller will send the character string REQ to the external device. The external device will reply by sending back the number of times the operation is to be repeated.
The controller will read in this number as the variable N, and will use this variable in the FOR statement.
Should the program execution have been suspended while the controller was waiting receiving data in the INPUT command, the variable N would be (mistakenly) taken as 0 when the program is resumed. In such a case, branch is executed to the top address of the program by asking the external device for a retransmission.
When the task is completed, the controller will send the character string END to the external device.

Data exchange



Selecting a task to be performed (IP1)

```
PROGRAM SELECT
    K=0
START:
    PRINT IP1, "STR"
    INPUT IP1, K
    GOTO(K) L1, L2, L3
    PRINT IP1, "NG"
    GOTO FIN
L1:
    (Task 1)
    GOTO FIN
L2:
    (Task 2)
    GOTO FIN
L3:
    (Task 3)
    GOTO FIN
FIN:
END
```

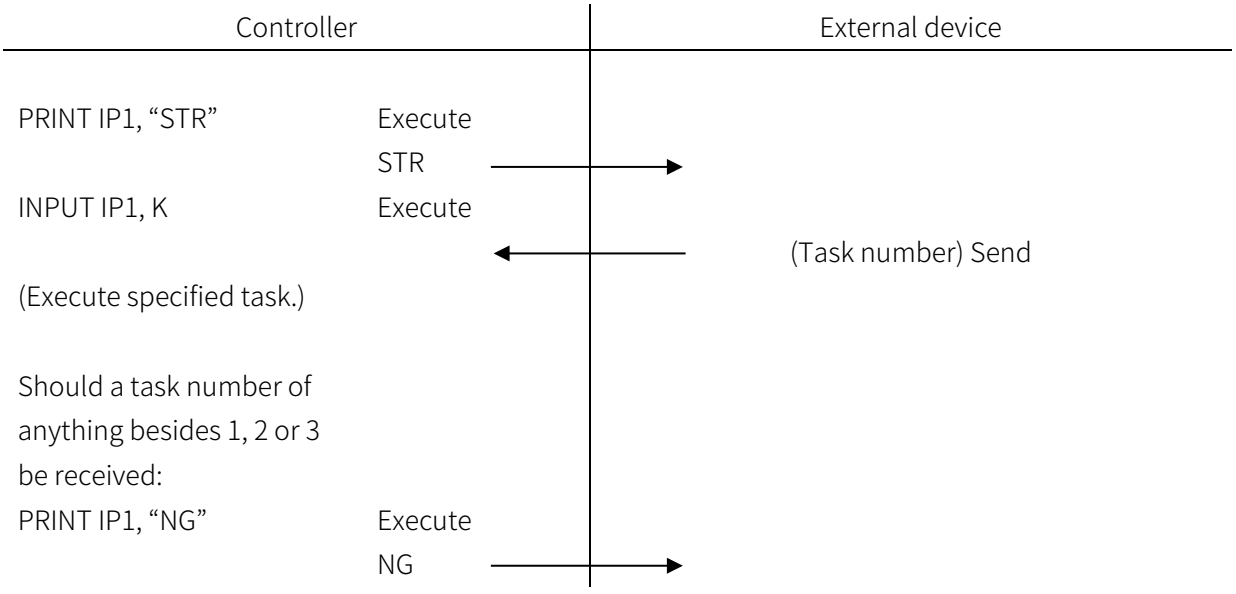
In this program, the external device tells the controller what task is to be performed.

First, the controller will send the character string STR to the external device. The external device will send back a number which specifying task the controller (robot) is to perform. The controller reads in this number as variable K, which is then used in the GOTO ( ) command to branch the program to the appropriate task.

Task 1 will be performed if the value of K is 1, Task 2 if the value is 2, and Task 3 if the value is 3. If the value of K is anything else, the controller will send the character string NG (non-acknowledge) back to the external device.

Should program execution have been suspended while the controller was waiting for the number to come in, variable K will be taken as 0 when the program is resumed. In this case also, the controller will send the character string NG back to the host computer.

Data exchange



3.2.2 Correcting the Position of the Robot

Data received from the host computer can be used to correct the position of the robot.

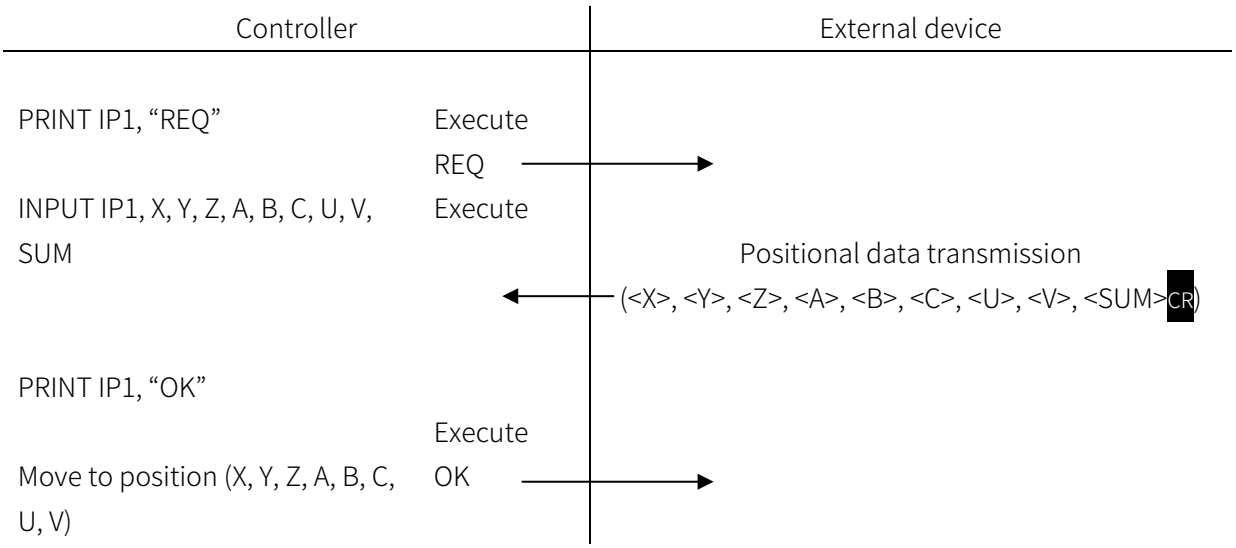
Directly specifying position (IP1)

```
PROGRAM DIRECT
   X=0.0
   Y=0.0
   Z=0.0
   A=0.0
   B=0.0
   C=0.0
   U=0.0
   V=0.0
   PRINT IP1, "REQ"
   INPUT X, Y, Z, A, B, C, U, V, SUM
   IF SUM==X+Y+Z+A+B+C+U+V+1 THEN GOTO ACTION
   PRINT IP1, "NG"
   GOTO FIN
ACTION:
   PRINT IP1, "OK"
   P1=POINT(X, Y, Z, A, B, C, U, V,
00000)
   MOVE P1
FIN:
END
```

In this program, the controller moves the robot to a position specified by the external device. The controller sends the character string REQ to the external device. The external device will respond by sending back the values for positional data X, Y, Z, A, B, C, U, and V, and also the sum of these values plus 1 (as check-sum data).

The controller will then see if the received data is correct by adding all the positional data, adding 1 to the result, and determining if check-sum value agrees with the value. If it does agree, the controller will send the character string OK. If the received data is wrong, the controller will send the character string NG back to the external device and quit the program. The controller will create positional data P1 using that data and then tell the robot to move to that point.

Data exchange

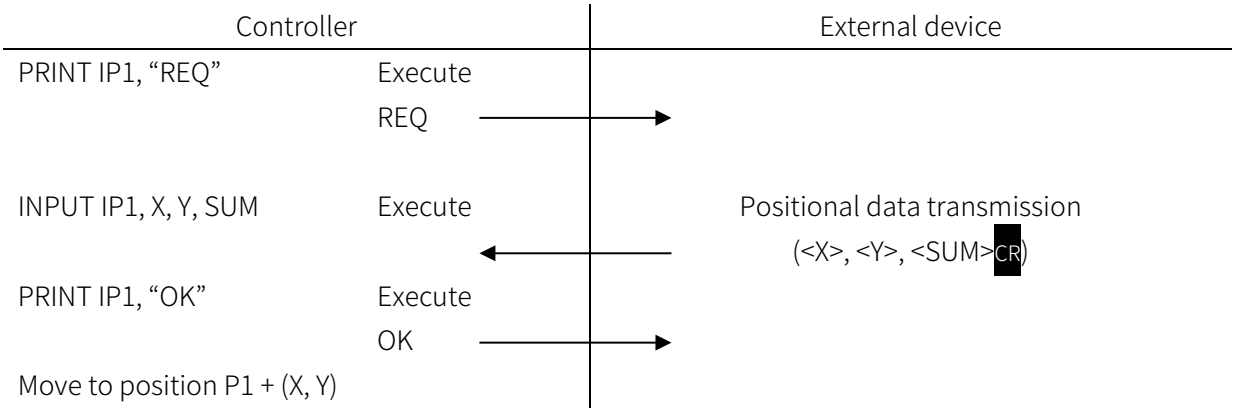


Specifying a relative position (IP1)

```
PROGRAM RELATIV
   X=0.0
   Y=0.0
   SUM=0.0
   MOVE P1
   PRINT IP1, “REQ”
   INPUT IP1, X, Y, SUM
   IF SUM==X+Y+1   THEN GOTO ACTION
   PRINT IP1, “NG”
   GOTO FIN
ACTION:
   PRINT IP1, “OK”
   MOVE P1+POINT(X, Y)
FIN:
END
```

In this program, the controller moves the robot to a position specified by the external device.
The controller sends the character string REQ to the external device.
The external device will respond by sending back the values for coordinate positions X and Y, and also the sum of these values plus 1 (as check-sum data). The controller will then see if the received data is correct by adding X and Y, adding 1 to the result, value agrees with the check-sum value. If it does agree, the controller will send back the character string OK to the external device.
If the received data is wrong, the controller will send the character string NG back to the external device and quit the program.
The controller will add the X and Y values of the transmission to the X and Y values of positional data P1 and make the robot move to the new position thus created.

Data exchange



4. Host Command Communication

In host command communication, commands are issued to the robot controller for starting and stopping the robot, transferring program files, and monitoring various statuses.
This communication also enables creating and editing of programs, debug startup support, and maintenance using the TSAssist PC programmer.

The characters displayed as  in this chapter are ASCII code control characters. See "6.1 ASCII Code Table".

4.1 Communication Protocol

Host command communication includes three types of communication protocols: Control communication protocol, periodic monitor protocol, and IP dump protocol.

4.1.1 Control Communication Protocol

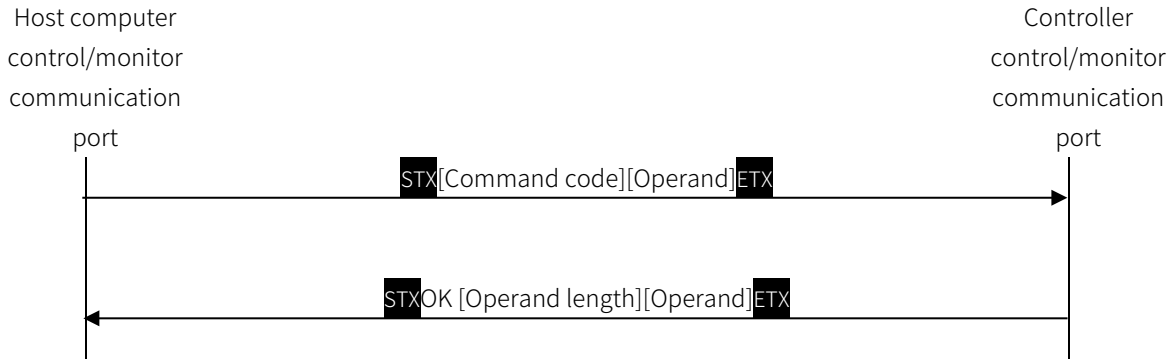
This is a control communication protocol that is used when executing processes in the controller.

When the host computer sends a request command and the controller receives the request command, the controller executes the process for the request command, and after execution is completed, the execution results are sent as a response command.

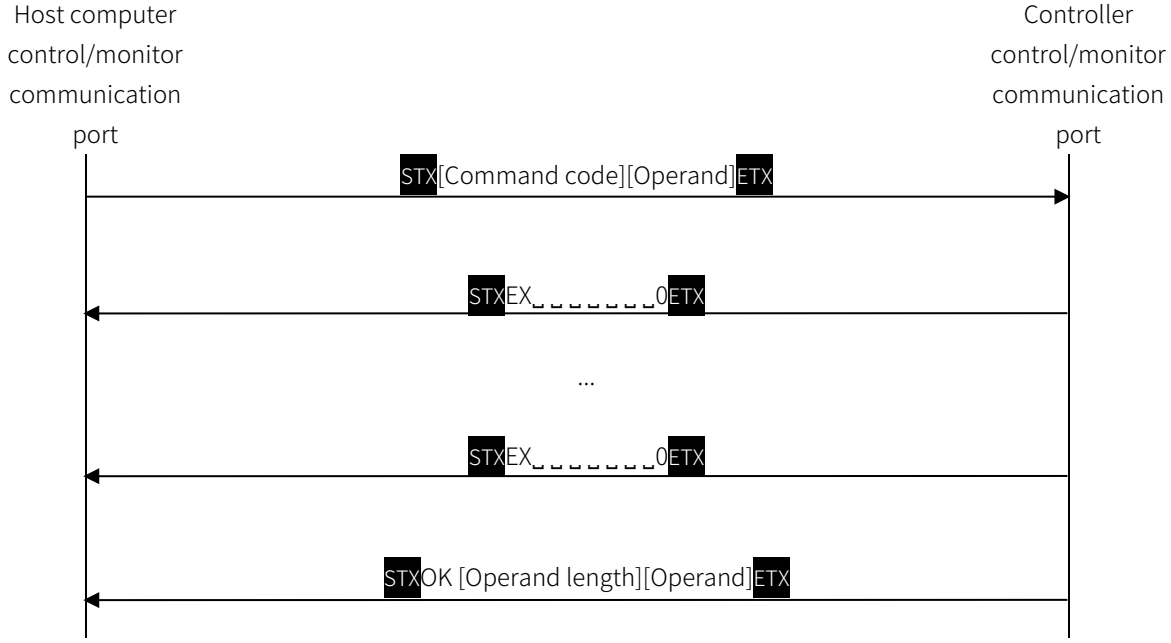
If the host computer sent a request command while the host computer was waiting for a response from the controller for a previously-sent request command, the request command that was received during processing is discarded.
Before sending a new request command from the host computer, be sure that a response was received from the controller for the previous command.

The control communication protocol is used by the control communication port and monitor communication port. The control communication port receives all request commands. The monitor communication port receives only the request commands that can be monitored instead of the control command.
Monitor communication ports are used when performing asynchronous communication with control communication ports.
Even if the control communication port is waiting for a response from the controller of the host computer (when the host computer is executing the process for a request command), requests are received from the monitor ports, and responses are returned.

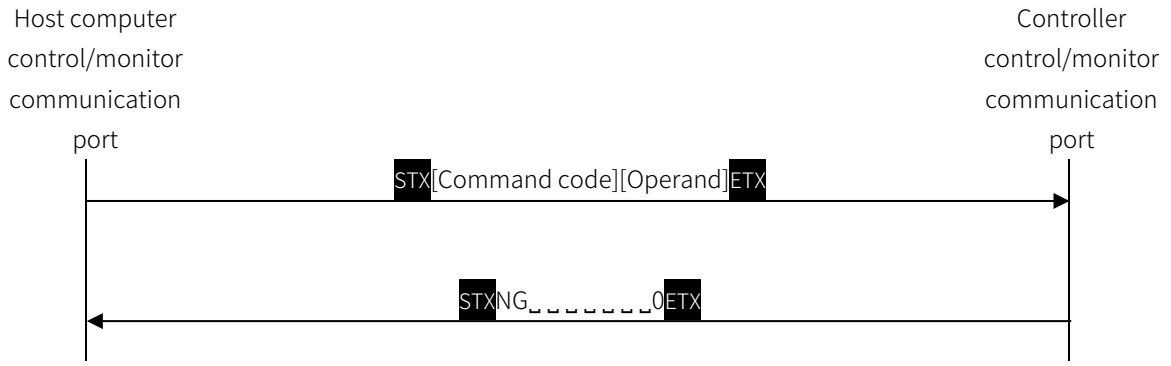
During normal process execution (when normal response is returned within 2 seconds)



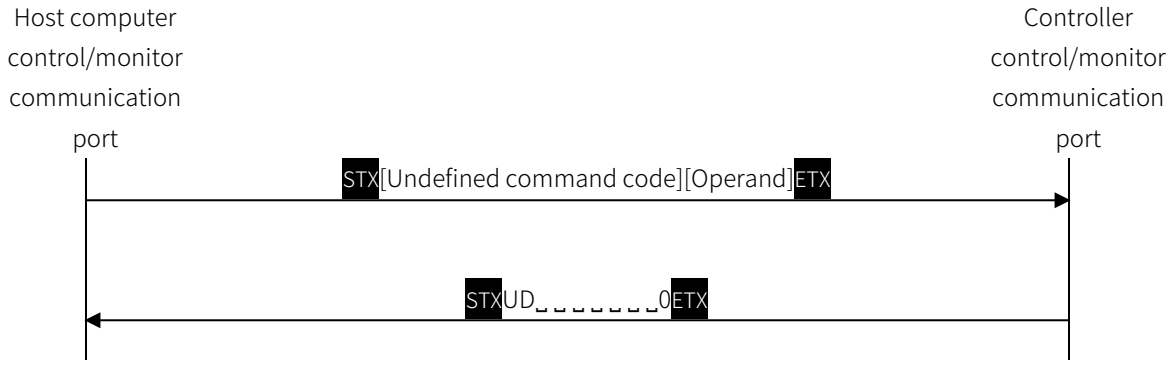
During normal process execution (when response cannot be sent within 2 seconds)



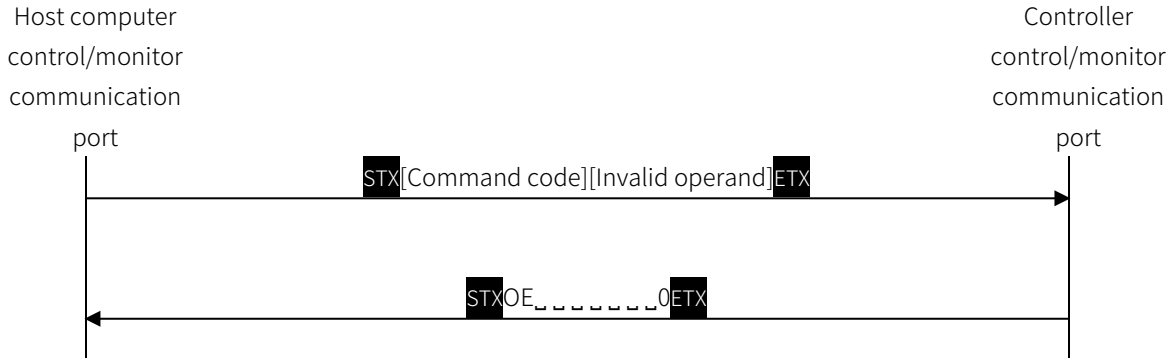
When error occurs in process execution



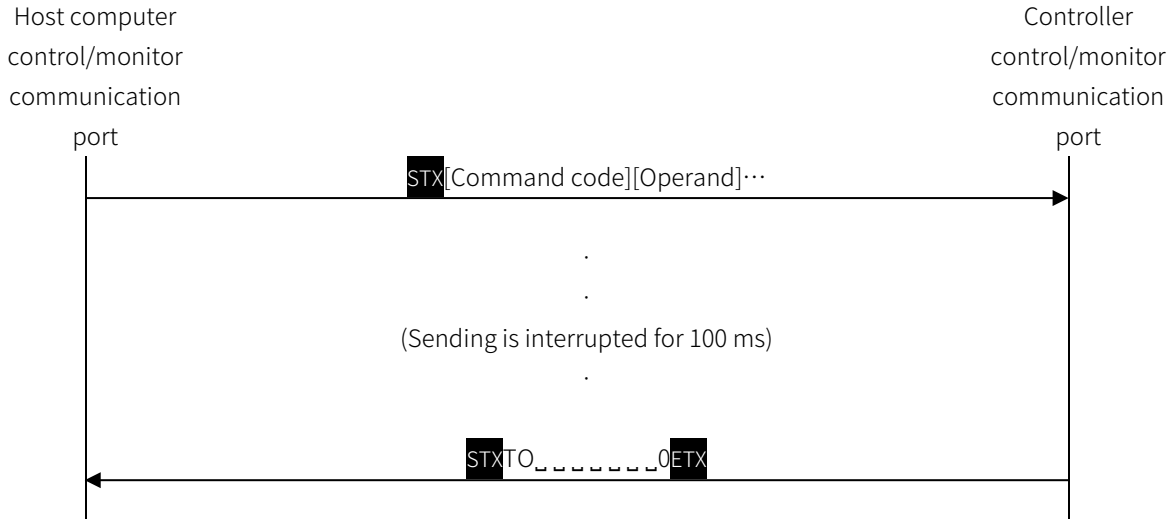
When an undefined command is sent



When operand is invalid



When timeout occurs



When the control request command is split into multiple packets, the controller waits for 100 ms only to receive the next packet. If the next packet is not received within 100 ms, a timeout error is returned.

4.1.2 Periodic Monitor Protocol

Periodic monitor protocol is used to periodically obtain the response results only from the controller.

When request commands that have response commands to be obtained periodically are registered, the controller continually sends request command responses only that were registered periodically as periodic monitor results commands to the periodic monitor port.
Only request commands that can be registered are monitorable command.

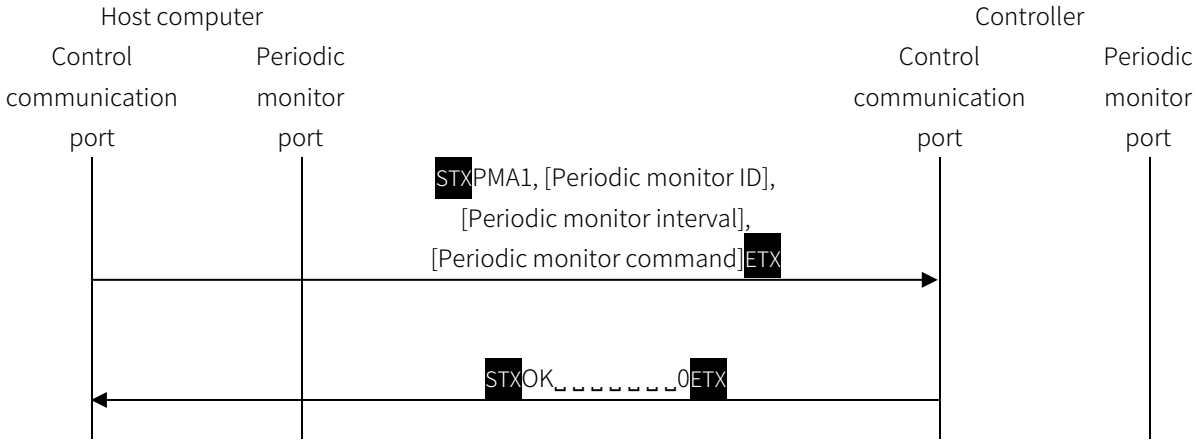
When the host computer sends the commands, monitor intervals, and monitor IDs to be monitored periodically as operands of the Register Periodic Monitor command, the commands to be monitored are registered, and the registration pass/fail results are sent as responses.

The host computer can delete a specified command from the periodic monitor by sending the monitor ID to be deleted as a parameter of the periodic monitor delete command.

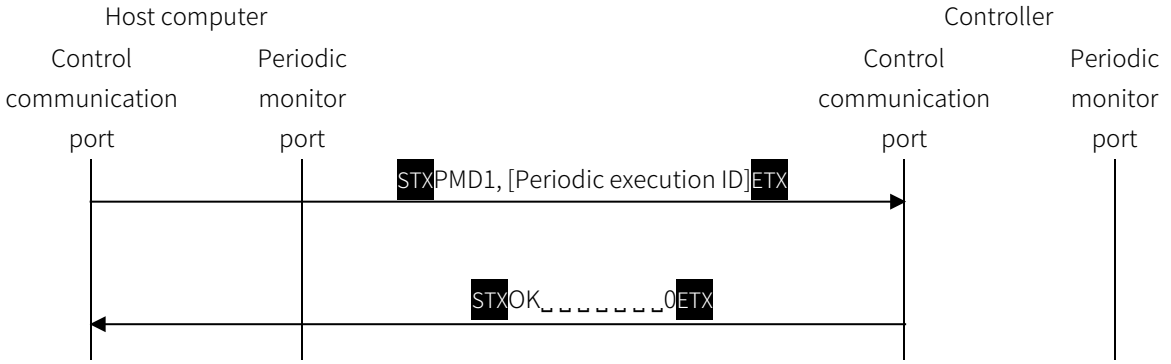
When a Start Periodic Monitor Operation command is sent, the monitor information for the command that was registered by the Register Periodic Monitor command is sent to the host computer from the periodic monitor communication port. If the controller registered multiple request commands at the same timing, the multiple response results are stored and sent starting from the newest monitor IDs to the periodic monitor results.

When the End Periodic Monitor Operation command is sent, the periodic monitor function is stopped.

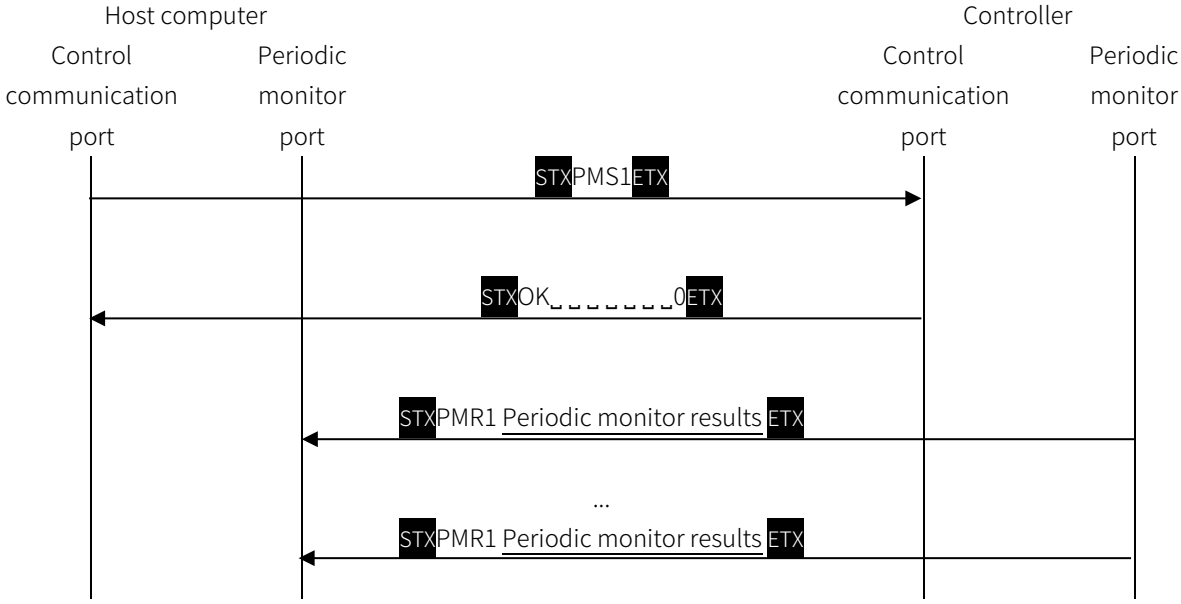
During registration process



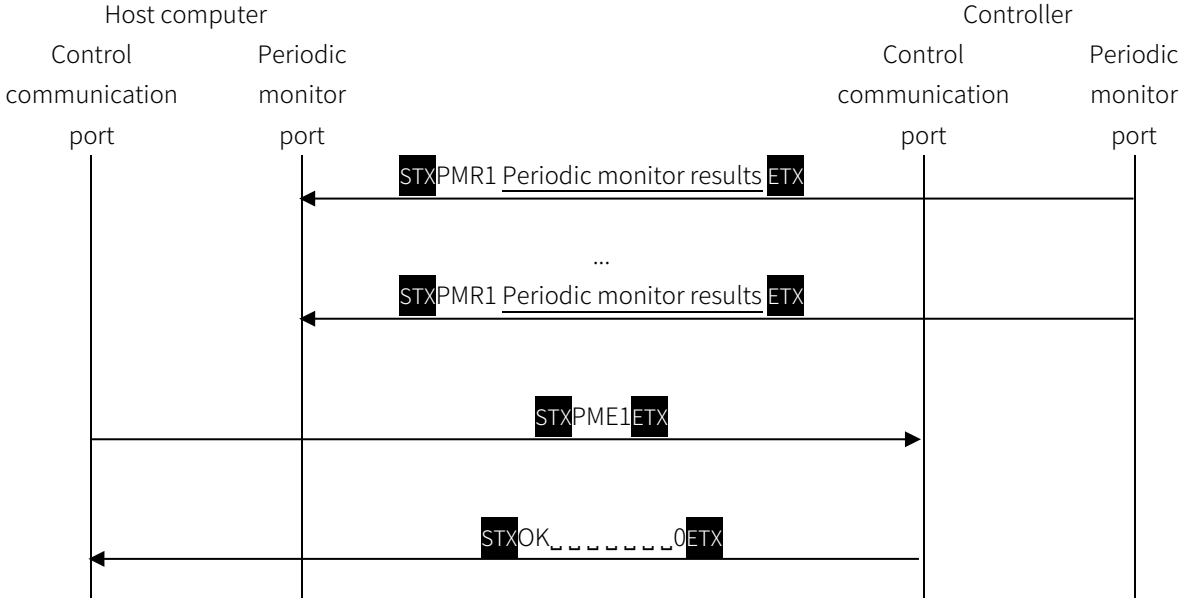
During deletion process



During start process



During stop process

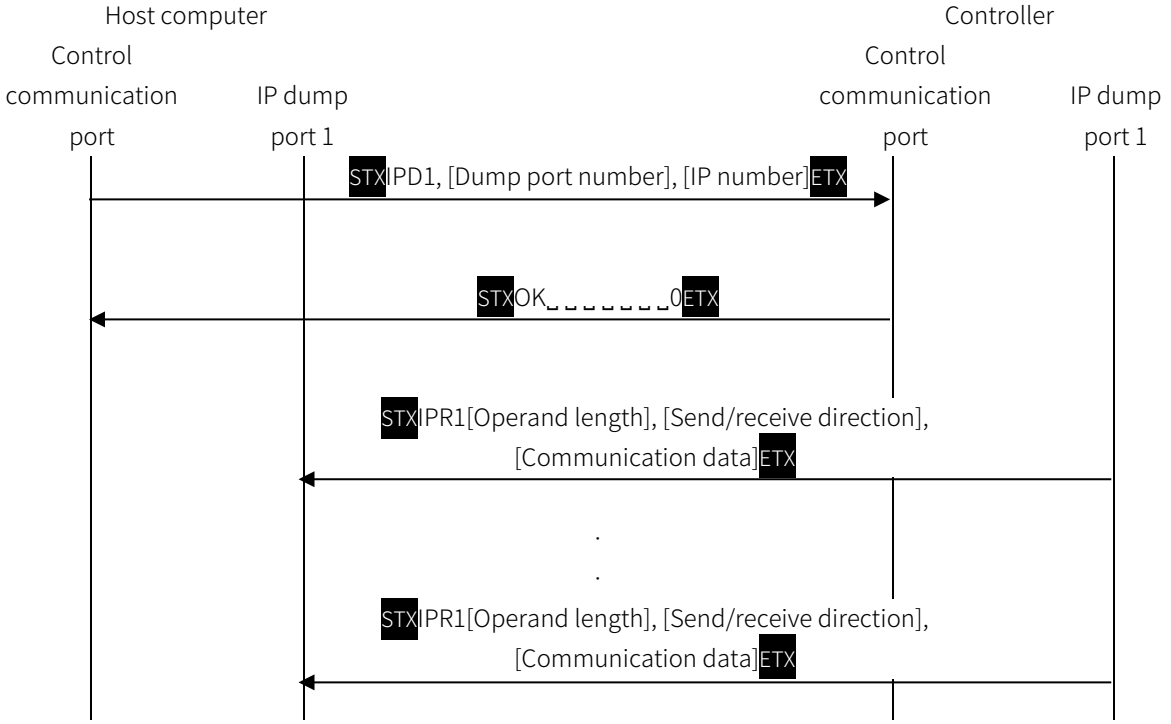


4.1.3 IP Dump Protocol

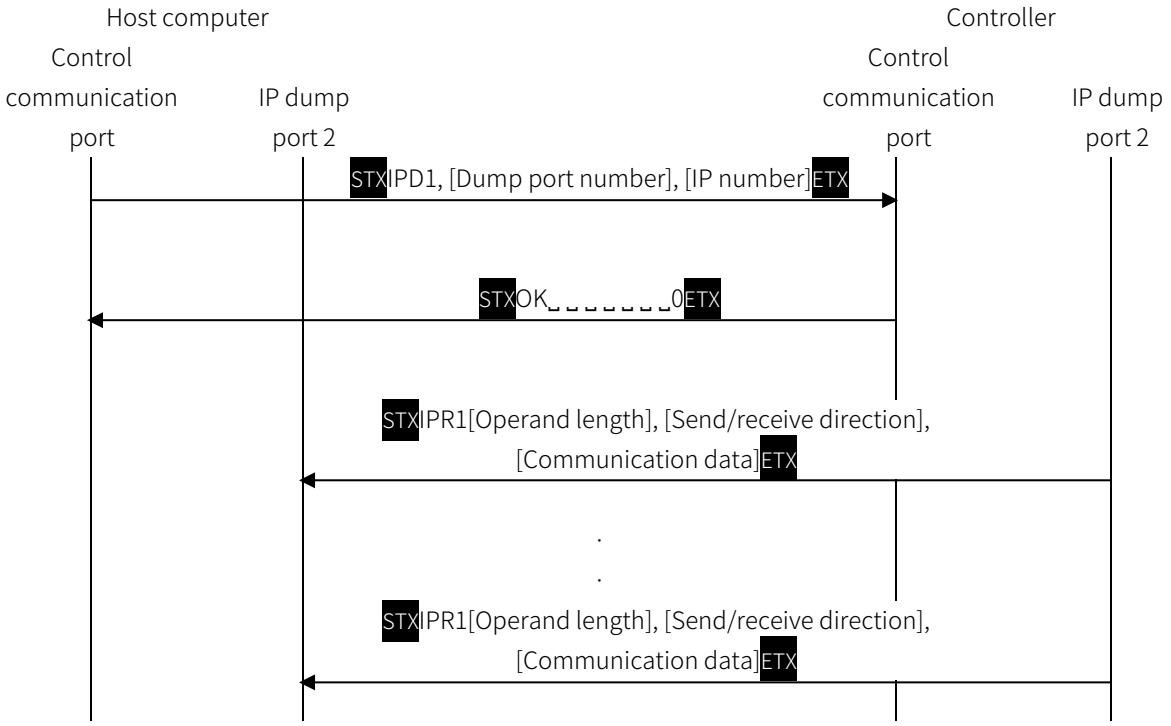
The controller uses IP dump protocol to obtain communication data that uses non-protocol communication and special communication.

When the communication channels to be dumped are set by the IP dump setting command, the communication data for non-protocol communication or special communication that was set by the IP dump setting command is sent from the IP dump port to the host computer.

IP dump port 1 start process



IP dump port 2 start process

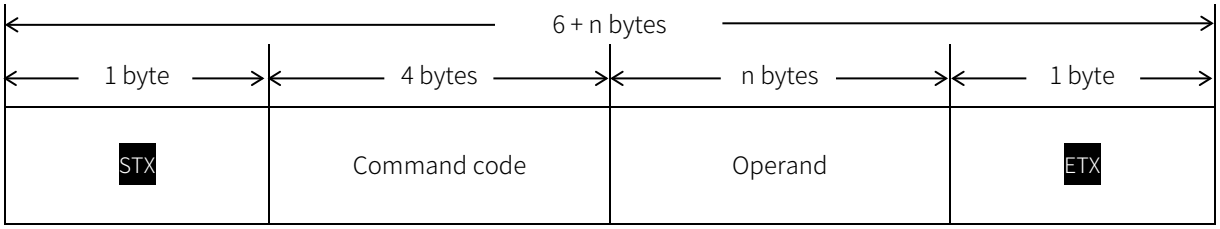


4.2 Transmission Format

This communication is performed in command units.

4.2.1 Request Command Format

The format for commands that request processes is as follows.

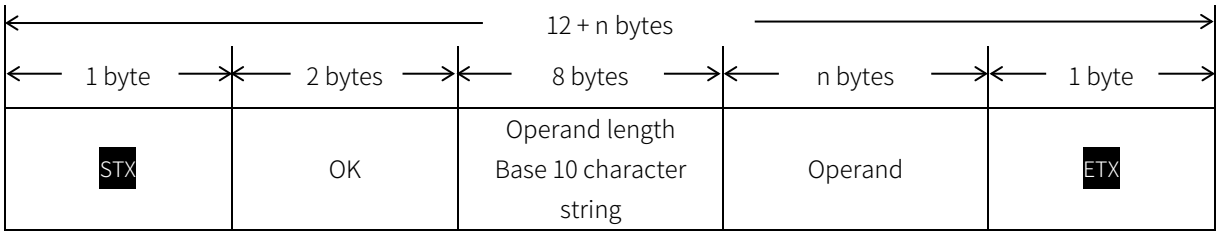


If the command includes an operand, the operand starts with a comma (,). If the command does not include an operand, there is no character string forming the operand, and the operand length is 0.

4.2.2 Response Command Format

Normal response command format

The internal format for response commands that send notification when the process cannot be executed normally is as follows.



When the operand length is set to a base 10 character string, spaces are filled in from the start so that the value is 8 bytes.

MEMO

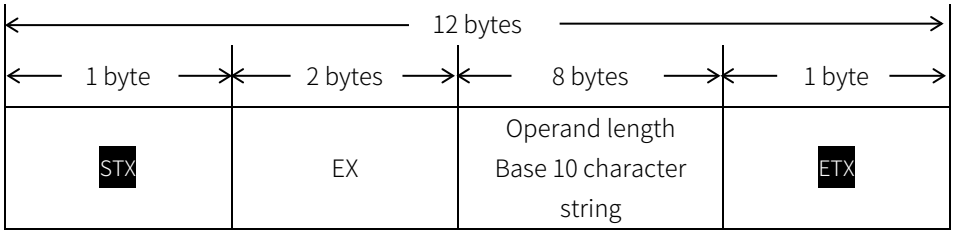
- *If the response command exceeds the maximum packet size that can be received at one time by the host computer, the controller automatically divides the response command into multiple packets and sends them.*
- *The host computer must be aware that the response command is received from multiple packets that were divided. Also, the maximum packet size that can be received at one time by the host computer varies depending on the specific host computer.*

Response Command Format During Process Execution

For the Copy File command, Write File command, and Select Program command, it may take some time from when the controller receives the request command until a normal response or error response is received.

When a Copy File command (CPY2), Write File command (FLW2), or Select Program command (SEL2) is issued, notification is sent that the process is being executed at 2-second intervals as a function for preventing a host computer timeout.

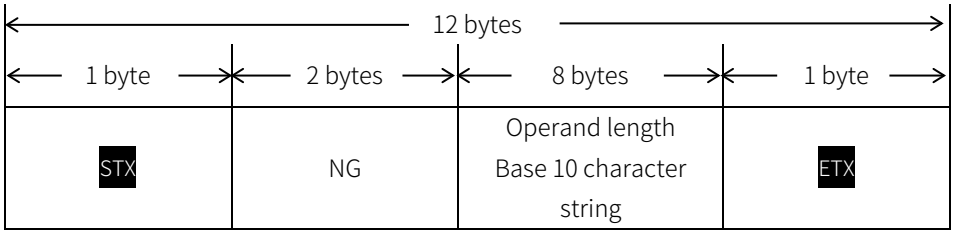
The internal format for commands that send notification that a process is being executed is as follows.



The operand length is always "␣␣␣␣␣␣␣0".

Error response command format

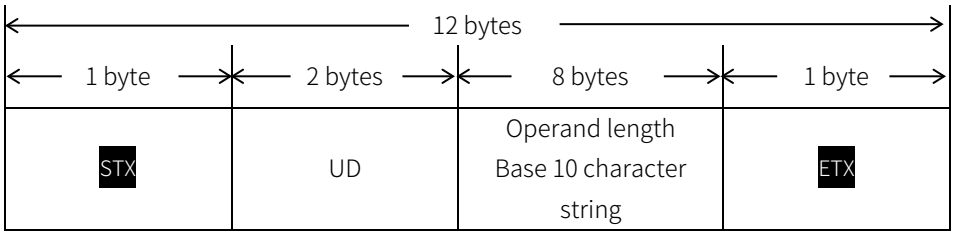
The internal format for response commands that send notification when a process has failed is as follows.



The operand length is always "␣␣␣␣␣␣␣0".

Undefined response command format

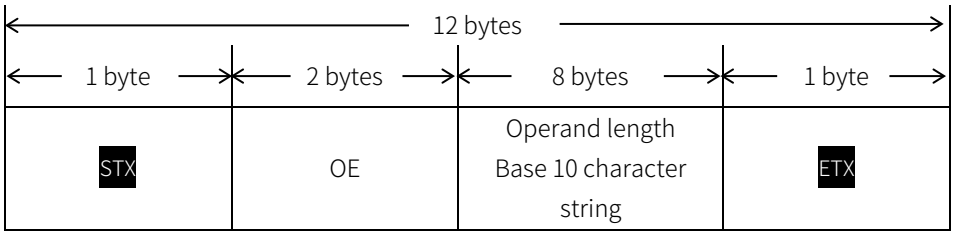
The response format for notifying when the requested command is an undefined command is as follows.



The operand length is always "␣␣␣␣␣␣␣0".

Operand error response command format

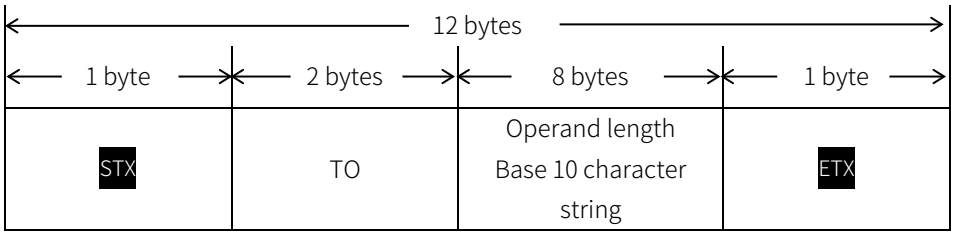
The response command for notifying when the requested command operand contains an error is as follows.



The operand length is always "␣␣␣␣␣␣␣0".

Timeout response command format

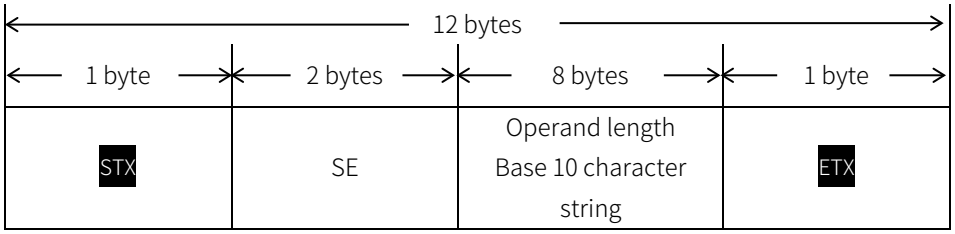
The response command for notifying when the controller did not receive the next packet even after 100 ms had elapsed for response commands from the host computer that were divided into multiple packets is as follows.



The operand length is always "␣␣␣␣␣␣␣0".

System error response command format

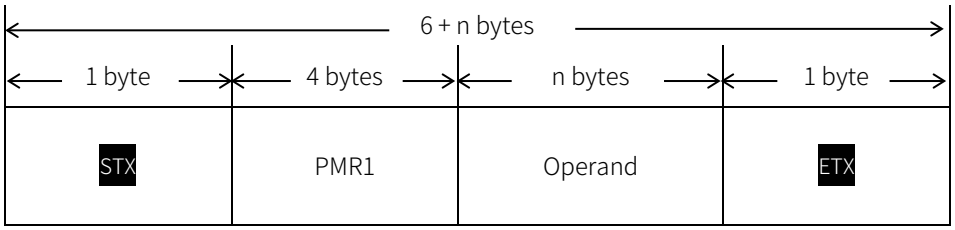
The response command for notifying when the controller did not execute due to some sort of a system error is as follows.



The operand length is always "␣␣␣␣␣␣␣0".

Periodic monitor results command format

The format for commands that send the periodic execution registration results to the host computer is as follows.



For details on operands, see "PMR1 Periodic Monitor Results".

4.3 Commands

4.3.1 Commands and Operation Modes

The operation modes where each command is enabled are shown in "Table 4.1 Commands and Operation Modes".

Communication through the control communication ports, monitor communication ports, and periodic monitor communication ports used in host command communication are enabled regardless of the master mode, but the functions are limited depending on the master mode.

Table 4.1 Commands and Operation Modes

Master Mode		MANUAL			AUTO			
Status		Stopped	Servo On	During Automatic Operation	Stopped	Servo On	During Automatic Operation	Register Periodic Monitor
Command	Command Description							
ARH1	Reset Alarm History	✓	✓	✓	✓	✓	✓	
ARN1	Reset Current Alarm	✓	✓	✓	✓	✓	✓	
ALH1	With Alarm History Message	✓	✓	✓	✓	✓	✓	✓
ALH2	Without Alarm History Message	✓	✓	✓	✓	✓	✓	✓
ALH3	With Alarm History Designated Language Message	✓	✓	✓	✓	✓	✓	✓
ALN1	With Current Alarm Message	✓	✓	✓	✓	✓	✓	✓
ALN2	Without Current Alarm Message	✓	✓	✓	✓	✓	✓	✓
ALN3	With Current Alarm Designated Language Message	✓	✓	✓	✓	✓	✓	✓
BOF1	Buzzer Off	✓	✓	✓	✓	✓	✓	
BRK1	Brake (Immediate Stop)			✓			✓	
CDS1	Set Coordinates				✓	✓	✓	
CPY1	Copy File				✓	✓	✓	
CPY2	Copy File				✓	✓	✓	
DEL1	Delete File				✓	✓	✓	
DIR1	Obtain Directory Information	✓	✓	✓	✓	✓	✓	
DOC1	Run DO Statement					✓		
DOC2	Run DO Statement					✓		
EIM1	Obtain Emergency Stop Information	✓	✓	✓	✓	✓	✓	✓
FLR1	Read File	✓	✓	✓	✓	✓	✓	
FLW1	Write File	✓	✓	✓	✓	✓	✓	
FLW2	Write File	✓	✓	✓*3	✓	✓	✓*3	
IOR1	Read Signal Information (All Areas)	✓	✓	✓	✓	✓	✓	✓

Master Mode		MANUAL			AUTO			
Status		Stopped	Servo On	During Automatic Operation	Stopped	Servo On	During Automatic Operation	Register Periodic Monitor
Command	Command Description							
IOR2	Read input signal information (Specified Signal Number)	✓	✓	✓	✓	✓	✓	✓
IOR3	Read output signal information (Specified Signal Number)	✓	✓	✓	✓	✓	✓	✓
IOW1	Write Signal Information (General-purpose Output)				✓	✓	✓	
IOW2	Write Signal Information (General-purpose Output)				✓	✓	✓	
IPC1	Close IP Port	✓	✓	✓	✓	✓	✓	
IPD1	Set IP Dump	✓	✓	✓	✓	✓	✓	
IPO1	Open IP Port	✓	✓	✓	✓	✓	✓	
IPR1	IP Dump Results							
JOC1	Select Guidance Coordinates	✓	✓	✓	✓	✓		
JOG1	Execute Jogging					✓		
JOI1	Execute Inching					✓		
JOI2	Execute Inching					✓		
JOM1	Set Guidance Amount				✓	✓		
JOS1	Set Guidance Speed				✓	✓		
JPR1	Obtain Jog Payload	✓	✓	✓	✓	✓	✓	✓
JPS1	Set Jog Payload				✓	✓	✓	
LCR0	Read Load Cell Information	✓	✓	✓	✓	✓	✓	✓
LCZ0	Set Load Cell Origin	✓	✓		✓	✓		
MDE1	Set Execution Mode				✓	✓	✓	
MER1	Read Memory Variable	✓	✓	✓	✓	✓	✓	✓
MEW1	Write Memory Variable				✓	✓	✓	
MNT1	Obtain Maintenance Information	✓	✓	✓	✓	✓	✓	✓
MNT2	Obtain Maintenance Information	✓	✓	✓	✓	✓	✓	✓
MNT3	Obtain Maintenance Information	✓	✓	✓	✓	✓	✓	✓
MVP1	Start Teaching Point Movement (MOVE) (PTP Movement)					✓		
MVP2	Teaching Point Movement (Joint Movement)					✓		
MVS1	Start Teaching Point Movement (MOVES) (CP Movement)					✓		
MVXC	Continue Teaching Point Movement and Obtain Status					✓*1		
OVR1	Set Override				✓	✓	✓	
PLC1	TCmini (Internal PLC) Monitor	✓	✓	✓	✓	✓	✓	✓

Master Mode		MANUAL			AUTO			
Status		Stopped	Servo On	During Automatic Operation	Stopped	Servo On	During Automatic Operation	Register Periodic Monitor
Command	Command Description							
PLC2	Obtain TCmini (Internal PLC) Status	✓	✓	✓	✓	✓	✓	✓
PMA1	Register Periodic Monitor	✓	✓	✓	✓	✓	✓	
PMD1	Delete Periodic Monitor	✓	✓	✓	✓	✓	✓	
PME1	End Periodic Monitor Operation	✓	✓	✓	✓	✓	✓	
PMF1	Instant Execute Periodic Monitor	✓	✓	✓	✓	✓	✓	
PMR1	Periodic Monitor Results							
PMS1	Start Periodic Monitor Operation	✓	✓	✓	✓	✓	✓	
PSJ1	Obtain Joint Position Information	✓	✓	✓	✓	✓	✓	✓
PSW1	Obtain World Position Information	✓	✓	✓	✓	✓	✓	✓
RCH1	Determine If Specified Position Reached	✓	✓	✓	✓	✓	✓	
RCH2	Determine If Specified Joint Angle Reached	✓	✓	✓	✓	✓	✓	
REN1	Rename File				✓	✓	✓*2	
RST1	Reset	✓	✓		✓	✓		
RUN1	Start Automatic Operation					✓		
SAL1	Save Alarm History to SD Card	✓	✓	✓	✓	✓	✓	
SEL1	Select Program	✓	✓		✓	✓		
SEL2	Select Program	✓	✓		✓	✓		
SOF1	Servo Off	✓	✓	✓	✓	✓	✓	
SON1	Servo On				✓			
STB1	Obtain Board Information	✓	✓	✓	✓	✓	✓	✓
STB2	Obtain Board Information	✓	✓	✓	✓	✓	✓	✓
STC1	Obtain Controller Status Information	✓	✓	✓	✓	✓	✓	✓
STE1	Obtain Ethernet Status Information	✓	✓	✓	✓	✓	✓	✓
STP1	Stop Automatic Operation			✓			✓	
TRQ1	Obtain Motor Torque Information	✓	✓	✓	✓	✓	✓	✓
VER1	Read Version	✓	✓	✓	✓	✓	✓	✓
VLJ1	Obtain Joint Speed Information	✓	✓	✓	✓	✓	✓	✓
VLW1	Obtain Hand Speed	✓	✓	✓	✓	✓	✓	✓
VLW2	Obtain Hand Combined Speed	✓	✓	✓	✓	✓	✓	✓
WHL1	Obtain Global Variable List	✓	✓	✓	✓	✓	✓	
WHR1	Read Global Variable	✓	✓	✓	✓	✓	✓	✓
WHW1	Write Global Variable	✓	✓	✓	✓	✓	✓	

✓: Enabled, Blank: Disabled

For commands sent from the host computer to the robot, ✓ indicates that the command can be received.
For commands sent from the controller to the computer, ✓ indicates that the command can be sent.

*1 The command can be received during teaching point movement only.
*2 The currently-selected file cannot be renamed.

MEMO

- *External operation signal inputs are also limited based on the master mode. For details, see the "Interface Manual".*

4.3.2 Command Descriptions

ARH1 Reset Alarm History

Function

This clears the alarm history of the controller.

If the request cannot be received, an error response is returned.

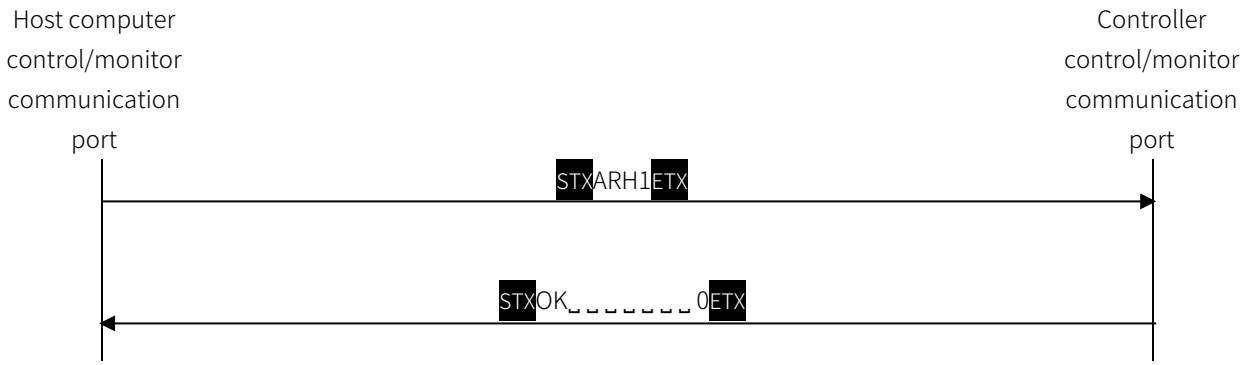
Request Format

STXARH1ETX

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command for clearing the alarm history is as follows.

STXARH1ETX
STXOK␣␣␣␣␣␣␣␣0ETX

ARN1 Reset Current Alarm History

Function

This clears the current alarm in the controller.

If the request cannot be received, an error response is returned.

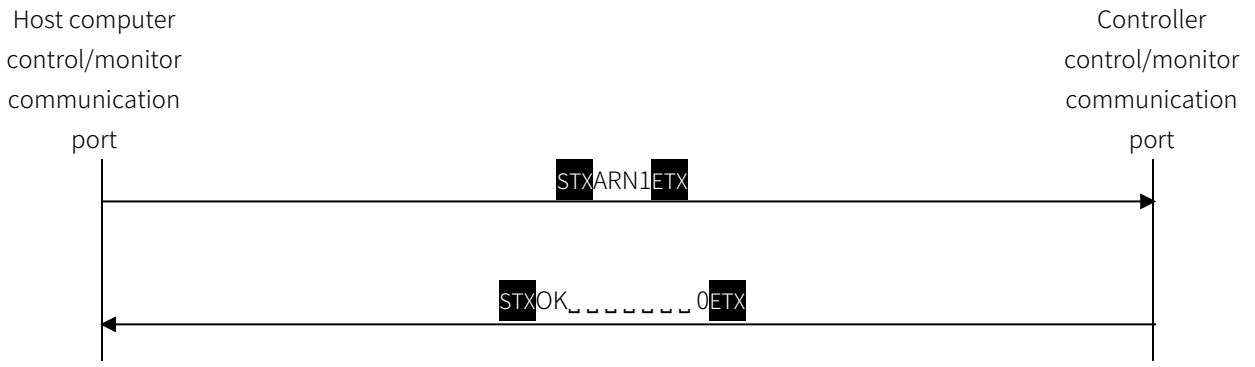
Request Format

STXARN1ETX

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

This command for clearing the current alarm in the controller is as follows.

STXARN1ETX
STXOK␣␣␣␣␣␣␣␣0ETX

ALH1 With Alarm History Message

Function

Monitorable command

This transfers the alarm history of the controller to the host computer with messages.

If no alarm has occurred, 0 is returned for the "Alarm count".

Request Format

STXALH1ETX

Response Format

STXOK [Operand length], [Alarm count] Alarm information ETX

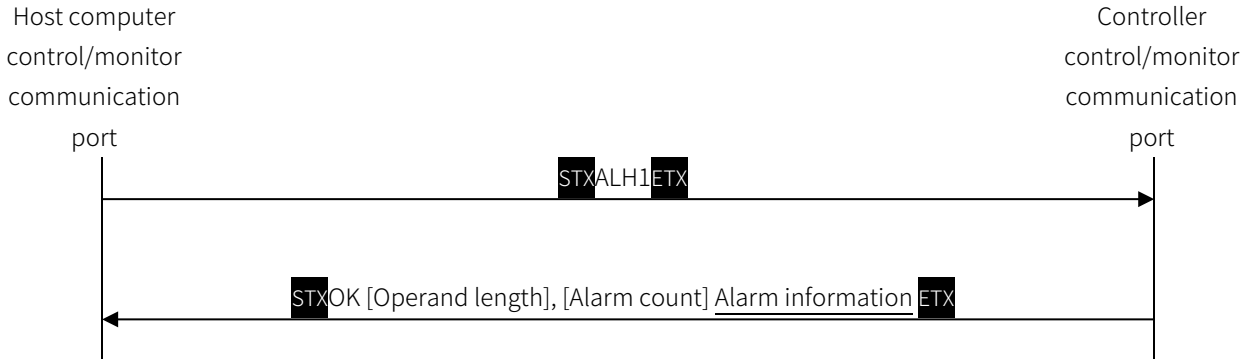
In the alarm information, the following information is stored consecutively for the quantity in [Alarm count].

CR[Alarm number], [Message], [Date]␣ [Time]

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	Alarm count	Variable length	Number of alarm history events (base 10 format) Added to start of file only.
3	Alarm number	7 Fixed length	The alarm history number has the following format. “XXX-YYY” XXX: Main code YYY: Subcode For descriptions of the alarms, see the Alarm Manual.
4	Message	Variable length	Shows the message for the alarm that occurred.
5	Date	11 Fixed length	Shows the date that the alarm occurred. “Mon-DD-YYYY” Mon: Month January: Jan, February: Feb March: Mar, April: Apr May: May, June: Jun July: Jul, August: Aug September: Sep, October: Oct November: Nov, December: Dec DD: Date YYYY: Year (Western calendar year)

No.	Name	Size (bytes)	Description
6	Time	8 Fixed length	Shows the time that the alarm occurred. “HH:MM:SS” HH: Hours (24-hour format) MM: Minutes SS: Seconds

Communication Example



Send/Receive Example

If the two alarm events “Emergency Stop SW ON” and “Safety SW ON” are contained in the alarm history, the alarm history appears as follows.

STXALH1ETX
STXOK␣␣␣␣␣␣89, 2CR008-014, Emergency␣Stop␣SW␣ON, 17-06-15␣10:32:18CR008-017, Safety␣SW␣ON, 17-06-15␣10:29:26ETX

If no alarms are found in the alarm history, the alarm history appears as follows.
STXALH1ETX
STXOK␣␣␣␣␣␣␣2, 0ETX

ALH2 Without Alarm History Message

Function

Monitorable command

This transfers the alarm history information of the controller to the host computer.

If no alarm has occurred, 0 is returned for the "Alarm count".

Request Format

STXALH1ETX

Response Format

STXOK [Operand length], [Alarm count] Alarm information ETX

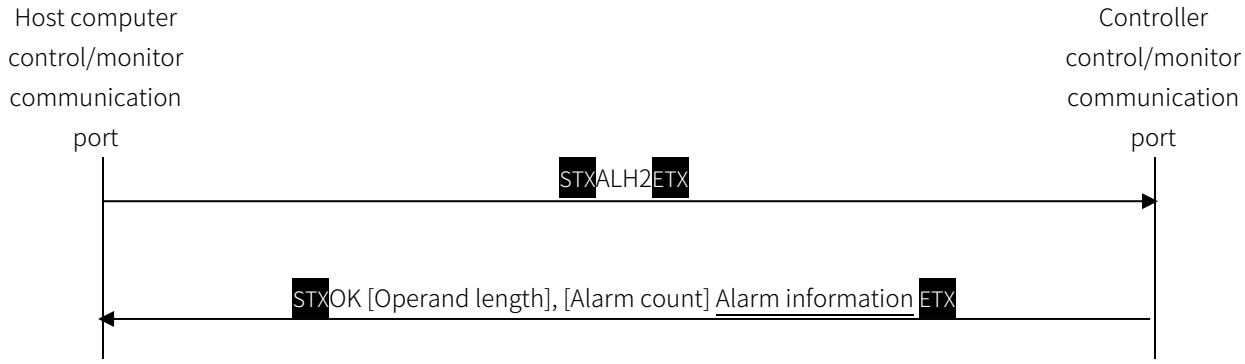
In the alarm information, the following information is stored consecutively for the quantity in [Alarm count].

CR[Alarm number], [Date]␣[Time]

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	Alarm count	Variable length	Number of alarm history events (base 10 format) Added to start of file only.
3	Alarm number	7 Fixed length	The alarm history number has the following format. “XXX-YYY” XXX: Main code YYY: Subcode For descriptions of the alarms, see the “Alarm Manual”.

No.	Name	Size (bytes)	Description
4	Date	11 Fixed length	Shows the date that the alarm occurred. “Mon-DD-YYYY” Mon: Month January: Jan, February: Feb, March: Mar, April: Apr, May: May, June: Jun, July: Jul, August: Aug, September: Sep, October: Oct, November: Nov, December: Dec DD: Date YYYY: Year (Western calendar year)
5	Time	8 Fixed length	Shows the time that the alarm occurred. “HH:MM:SS” HH: Hours (24-hour format) MM: Minutes SS: Seconds

Communication Example



Send/Receive Example

If the two alarm events “Emergency Stop SW ON” and “Safety SW ON” are contained in the alarm history, the alarm history appears as follows.

STXALH2ETX
STXOK␣␣␣␣␣␣55, 2CR008-014, 17-06-15␣10:32:18CR008-017, 17-06-15␣10:29:26ETX

If no alarms are found in the alarm history, the alarm history appears as follows.

STXALH2ETX
STXOK␣␣␣␣␣␣␣2, 0ETX

ALH3 With Alarm History Designated Message

Function

Monitorable command

This transfers the alarm history of the controller with a message with the designated language to the host computer.

If no alarm has occurred, 0 is returned for the “Alarm count”.

*Supported version: X8KCS-02H or later

Request Format

STXALH3，[Language type]ETX

No.	Name	Size (bytes)	Description
1	Language Type	1 Fixed length	Specifies message language to be obtained. 0: English 1: Japanese 2: Chinese (Simplified)

Response Format

STXOK [Operand length], [Alarm count] Alarm information ETX

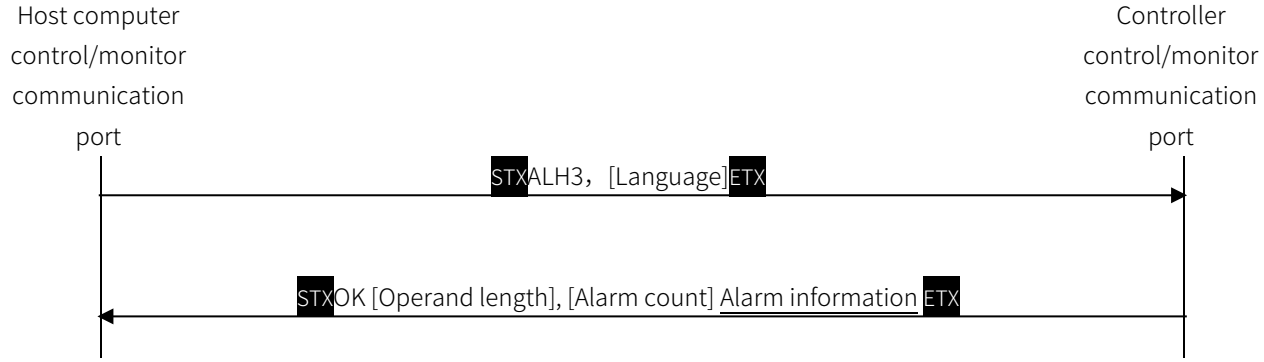
In the alarm information, the following information is stored consecutively for the quantity in [Alarm count].

CR[Alarm number], [Message], [Date]␣[Time]

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	Alarm count	Variable length	Number of alarm history events (base 10 format) Added to start of file only.
3	Alarm number	7 Fixed length	The alarm history number has the following format. “XXX-YYY” XXX: Main code YYY: Subcode For descriptions of the alarms, see the “Alarm Manual”.
4	Message	Variable length	Shows the message for the alarm that occurred.

NO.	Name	Size (bytes)	Description
5	Date	11 Fixed length	Shows the date that the alarm occurred. “Mon-DD-YYYY” Mon: Month — January: Jan, February: Feb, March: Mar, April: Apr, May: May, June: Jun, July: Jul, August: Aug, September: Sep, October: Oct, November: Nov, December: Dec DD: Date YYYY: Year (Western calendar year)
6	Time	8 Fixed length	Shows the time that the alarm occurred. “HH:MM:SS” HH: Hours (24-hour format) MM: Minutes SS: Seconds

Communication Example



Send/Receive Example

If the alarm history contains two alarms “Emergency Stop SW ON” and “Safety SW ON,” the alarm history appears as follows.

STXALH3,0ETX
STXOK␣␣␣␣␣␣␣89,2CR008-014,Emergency␣Stop␣SW␣ON,17-06-15␣10:32:18CR008-017,Safety␣SW␣ON,17-06-15␣10:29:26ETX

If no alarms are found in the alarm history, the alarm history appears as follows.

STXALH3,0ETX
STXOK␣␣␣␣␣␣␣␣2,0ETX

ALN1 With Current Alarm Message

Function

Monitorable command

This transfers a maximum of 32 currently-occurring alarms in the controller to the host computer with messages.

If no alarm has occurred, 0 is returned for the "Alarm count".

Request Format

STXALN1ETX

Response Format

STXOK [Operand length], [Alarm count] Alarm information ETX

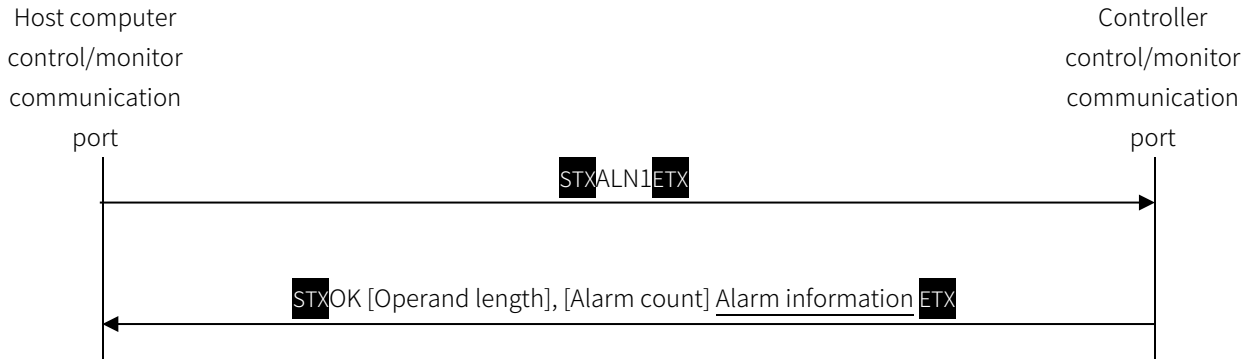
In the alarm information, the following information is stored consecutively for the quantity in [Alarm count].

CR[Alarm number], [Message], [Date]␣[Time]

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	Alarm count	Variable length	Number of currently-occurring alarms (base 10 format) Added to start of file only.
3	Alarm number	7 Fixed length	The alarm history number has the following format. “XXX-YYY” XXX: Main code YYY: Subcode For descriptions of the alarms, see the Alarm Manual.
4	Message	Variable length	Shows the message for the alarm that occurred.

No.	Name	Size (bytes)	Description
5	Date	11 Fixed length	Shows the date that the alarm occurred. “Mon-DD-YYYY” Mon: Month January: Jan / February: Feb March: Mar / April: Apr May: May / June: Jun July: Jul / August: Aug September: Sep / October: Oct November: Nov / December: Dec DD: Date YYYY: Year (Western calendar year)
6	Time	8 Fixed length	Shows the time that the alarm occurred. “HH:MM:SS” HH: Hours (24-hour format) MM: Minutes SS: Seconds

Communication Example



Send/Receive Example

If the two alarms “Emergency Stop SW ON” and “Safety SW ON” occurred, they appear as shown below in the alarm history.

STXALN1ETX

STXOK␣␣␣␣␣␣87, 2CR008-014, Emergency␣Stop␣SW␣ON, 17-06-15␣10:32:18CR008-017, Safety␣SW␣ON, 17-06-15␣10:29:26ETX

ALN2 Without Current Alarm Message

Function

Monitorable command

This transfers a maximum of 32 currently-occurring alarms in the controller to the host computer.

If no alarm has occurred, 0 is returned for the “Alarm count”.

*Supported version: X8KCS-02H or later

Request Format

STXALN2ETX

Response Format

STXOK [Operand length], [Alarm count] Alarm information ETX

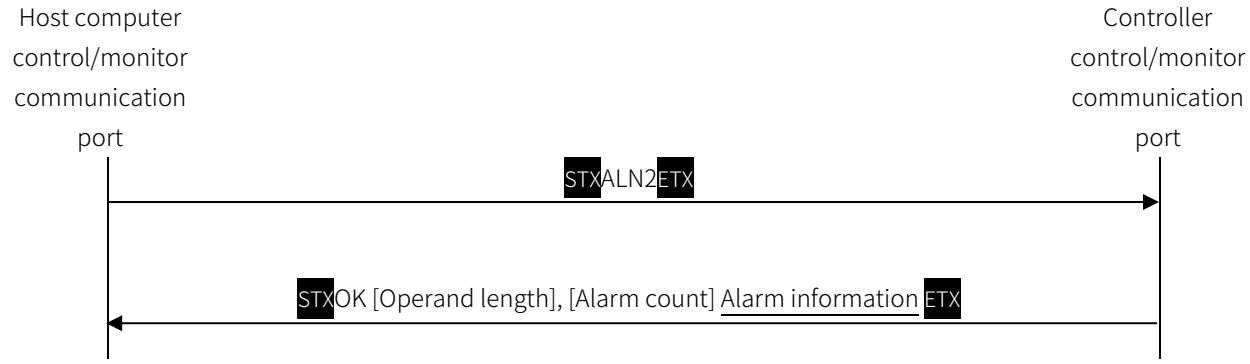
In the alarm information, the following information is stored consecutively for the quantity in [Alarm count].

CR[Alarm number], [Date] ␣ [Time]

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	Alarm count	Variable length	Number of currently-occurring alarms (base 10 format) Added to start of file only.
3	Alarm number	7 Fixed length	The alarm history number has the following format. “XXX-YYY” XXX: Main code YYY: Subcode For descriptions of the alarms, see the Alarm Manual.

No.	Name	Size (bytes)	Description
4	日付	11 Fixed length	Shows the date that the alarm occurred. “Mon-DD-YYYY” Mon: Month January: Jan / February: Feb March: Mar / April: Apr May: May / June: Jun July: Jul / August: Aug September: Sep / October: Oct November: Nov / December: Dec DD: Date YYYY: Year (Western calendar year)
5	Time	8 Fixed length	Shows the time that the alarm occurred. “HH:MM:SS” HH: Hours (24-hour format) MM: Minutes SS: Seconds

Communication Example



Send/Receive Example

If two alarms “Emergency Stop SW ON” and “Safety SW ON” occurred, they appear as shown below in the alarm history.

STXALN2ETX

STXOK␣␣␣␣␣␣␣55,2CR008-014,17-06-15␣10:32:18CR008-017,17-06-15␣10:29:26ETX

ALN3 With Current Alarm Designated Message

Function

Monitorable command

This transfers a maximum of 32 currently-occurring alarms in the controller with a message with the designated language to the host computer.

If no alarm has occurred, 0 is returned for the “Alarm count”.

*Supported version: X8KCS-02H or later

Request Format

STXALN3, [Language type]ETX

No.	Name	Size (bytes)	Description
1	Language	1 Fixed length	Specifies message language to be obtained. 0: English 1: Japanese 2: Chinese (Simplified)

Response Format

STXOK [Operand length], [Alarm count] Alarm information ETX

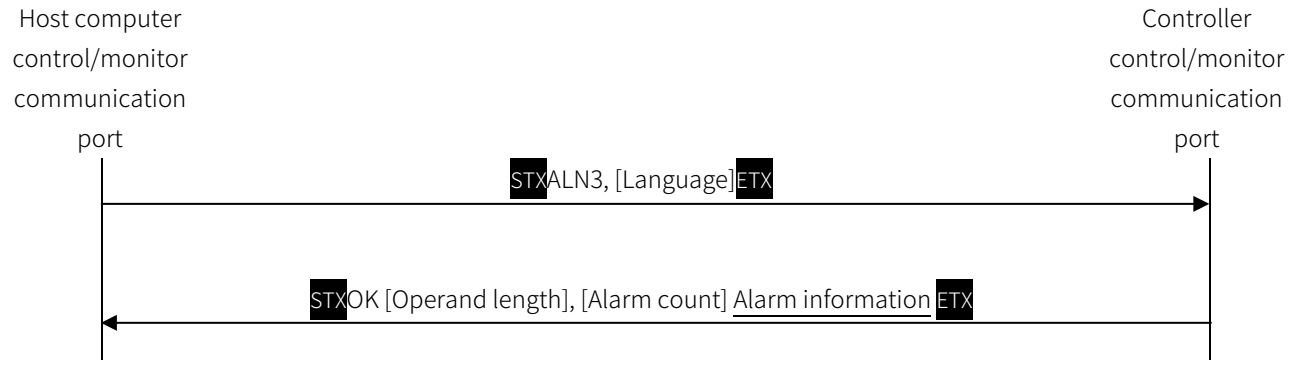
In the alarm information, the following information is stored consecutively for the quantity in [Alarm count].

CR[Alarm number], [Message], [Date]␣[Time]

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	Alarm count	Variable length	Number of currently-occurring alarms (base 10 format) Added to start of file only.
3	Alarm number	7 Fixed length	The alarm history number has the following format. “XXX-YYY” XXX: Main code YYY: Subcode For descriptions of the alarms, see the Alarm Manual.
4	Message	Variable length	Shows the message for the alarm that occurred.

No.	Name	Size (bytes)	Description
5	Date	11 Fixed length	Shows the date that the alarm occurred. "Mon-DD-YYYY" Mon: Month January: Jan / February: Feb March: Mar / April: Apr May: May / June: Jun July: Jul / August: Aug September: Sep / October: Oct November: Nov / December: Dec DD: Date YYYY: Year (Western calendar year)
6	Time	8 Fixed length	Shows the time that the alarm occurred. "HH:MM:SS" HH: Hours (24-hour format) MM: Minutes SS: Seconds

Communication Example



Send/Receive Example

If two alarms "Emergency Stop SW ON" and "Safety SW ON" occurred, they appear as shown below in the alarm history.

STXALN3,0ETX
STXOK␣␣␣␣␣␣␣89,2CR008-014,Emergency␣Stop␣SW␣ON,17-06-15␣10:32:18CR008-017,Safety␣SW␣ON,17-06-15␣10:29:26ETX

If no alarms are found in the alarm history, the alarm history appears as follows.

STXALN3,0ETX
STXOK␣␣␣␣␣␣␣␣2,0ETX

BOF1 Buzzer Off

Function

This stops the buzzer, which sounds when an alarm is detected.

If a new alarm occurs after the buzzer is stopped, the buzzer sounds again.

If the request cannot be received, an error response is returned.

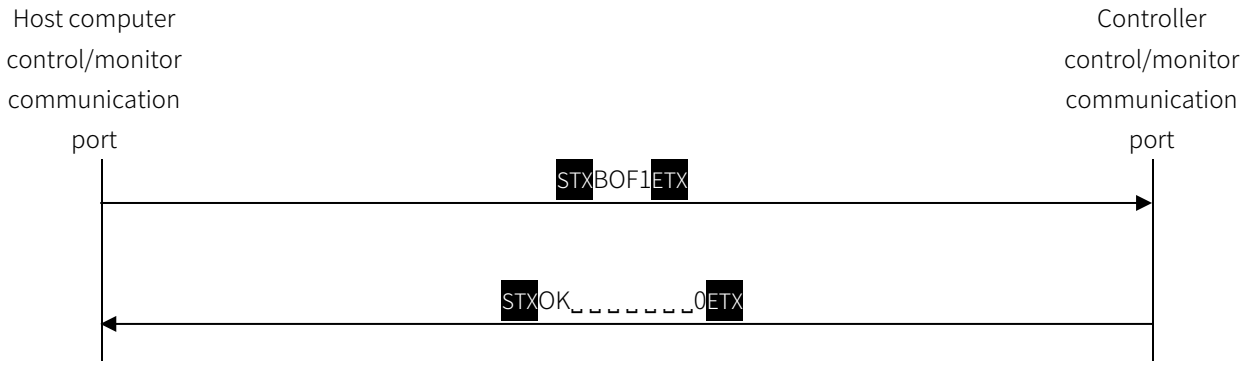
Request Format

STXBOF1ETX

Response Format

STXOK_ _ _ _ _ _ _ _0ETX

Communication Example



Send/Receive Example

The command to stop a buzzer that is sounding is as follows.

STXBOF1ETX
STXOK_ _ _ _ _ _ _ _0ETX

BRK1 Brake (Immediate Stop)

Function

This immediately stops automatic operation of the controller.

If the request cannot be received, an error response is returned.

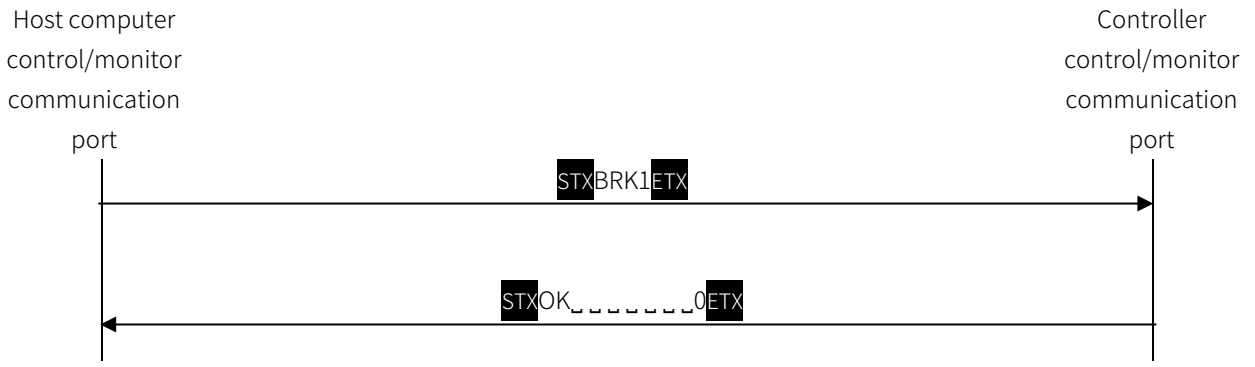
Request Format

STXBRK1ETX

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command to immediately stop the automatic operation state is as follows.

STXBRK1ETX
STXOK␣␣␣␣␣␣␣␣0ETX

MEMO

** This command can be received only during automatic operation (RUN status).*

CDS1 Set Coordinates

Function

This sets the tool coordinates, work coordinates, and base coordinates of the controller.
If the request cannot be received, an error response is returned.

Request Format

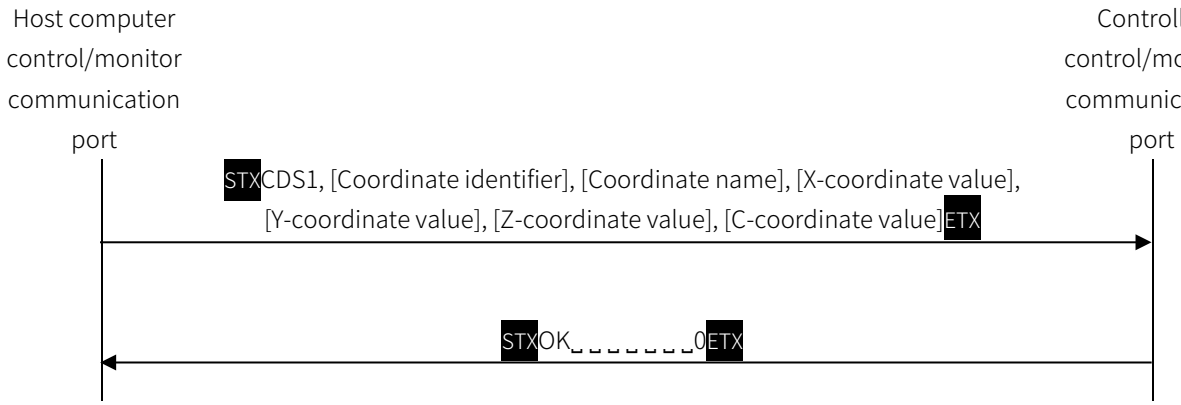
STXCDS1, [Coordinate identifier], [Coordinate name], [X-coordinate value], [Y-coordinate value], [Z-coordinate value], [C-coordinate value]ETX

No.	Name	Size (bytes)	Description
1	Coordinate identifier	1 Fixed length	Specify one of the following numbers for the coordinate identifier. 1: Tool coordinates 2: Work coordinates 4: Base coordinates
2	Coordinate name	Variable length	The coordinate name is a maximum of 34 characters. The coordinate name created by the user is in all uppercase.
3 to 6	Coordinate value	Variable length	For the coordinate value, specify a real number with up to 3 decimal places and in the range from 9999.999 to -9999.999.

Response Format

STXOK␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command to set each guidance coordinates as the work coordinates "X" is as follows.

STXCDS1, 2, X, 10.0, 20.0, 0.0, 0.0ETX
STXOK␣␣␣␣␣␣␣0ETX

The command to set the tool coordinates to the initial status is as follows.
STXCDS1, 1, .system, 0, 0, 0, 0ETX
STXOK␣␣␣␣␣␣␣0ETX

The command to set the work coordinates to the initial status is as follows.
STXCDS1, 2, .world, 0, 0, 0, 0ETX
STXOK␣␣␣␣␣␣␣0ETX

The command to set the base coordinates to the initial status is as follows.
STXCDS1, 4, .world, 0, 0, 0, 0ETX
STXOK␣␣␣␣␣␣␣0ETX

NOTICE

· Before sending this command, be sure to check the effect of the target coordinates on the program. The robot that has performed unexpected motion may collide with peripheral equipment.

CPY1 Copy File

Function

This copies the files in the file system of the controller.
If a file with the same name already exists at the copy target location, an error response is returned.
If the request cannot be received, an error response is returned.

Request Format

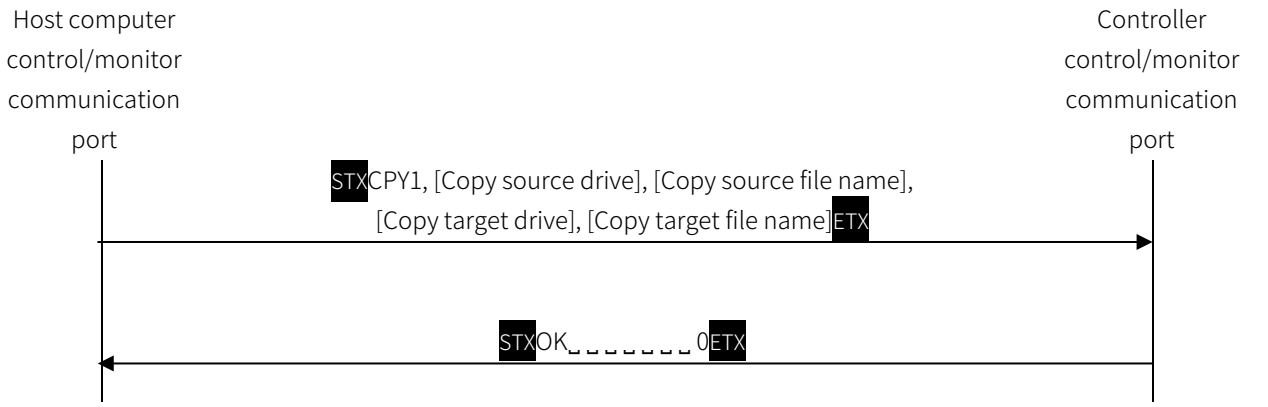
STXCPY1, [Copy source drive], [Copy source file name], [Copy target drive], [Copy target file name]ETX

No.	Name	Size (bytes)	Description
1	Copy source drive	1 Fixed length	Specifies the drive containing the copy source file. 1: ROM drive 2: RAM drive 3: SD card drive
2	Copy source file name	Variable length	File name of the copy source The file name must be in one of the following formats. · name.extension · name The name must be from 1 to 8 characters. The extension must be from 0 to 3 characters. When omitting the extension, the period (.) must also be omitted.
3	Copy target drive	1 Fixed length	Specifies the drive containing the copy target file. 1: ROM drive 2: RAM drive
4	Copy target file name	Variable length	File name of the copy target The file name must be in one of the following formats. · name.extension · name The name must be from 1 to 8 characters. The extension must be from 0 to 3 characters. When omitting the extension, the period (.) must also be omitted.

Response Format

STXOK␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command when copying the file "PRG1" in the ROM drive of the controller to the file "PRG1" in the RAM drive is as follows.

STXCPY1, 1, PRG1, 2, PRG1ETX
STXOK␣␣␣␣␣␣␣0ETX

CPY2 Copy File

Function

This copies the files in the file system of the controller.
If a file with the same name already exists at the copy target location, an error response is returned.
If the request cannot be received, an error response is returned.
If the Copy File process takes longer than 2 seconds, the controller sends a "Process execution in progress" response command at 2-second intervals to indicate that the process is being executed.

Request Format

STXCPY2, [Copy source drive], [Copy source file name], [Copy target drive], [Copy target file name]ETX

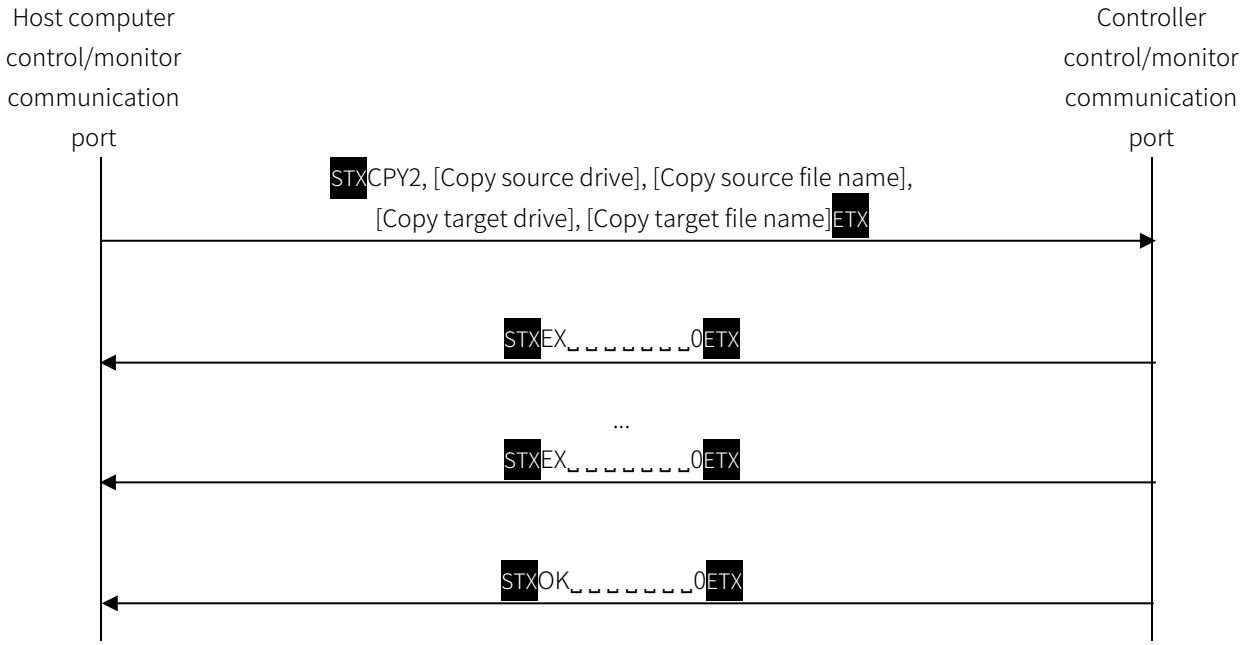
No.	Name	Size (bytes)	Description
1	Copy source drive	1 Fixed length	Specifies the drive containing the copy source file. 1: ROM drive 2: RAM drive 3: SD card drive
2	Copy source file name	Variable length	File name of the copy source The file name must be in one of the following formats. ・name.extension ・name The name must be from 1 to 8 characters. The extension must be from 0 to 3 characters. When omitting the extension, the period (.) must also be omitted.
3	Copy target drive	1 Fixed length	Specifies the drive containing the copy target file. 1: ROM drive 2: RAM drive

No.	Name	Size (bytes)	Description
4	Copy target file name	Variable length	File name of the copy target The file name must be in one of the following formats. ・name.extension ・name The name must be from 1 to 8 characters. The extension must be from 0 to 3 characters. When omitting the extension, the period (.) must also be omitted.

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX
STXEX␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command when copying the file "PRG1" in the ROM drive of the controller to the file "PRG1" in the RAM drive is as follows.

STXCPY2, 1, PRG1, 2, PRG1ETX
STXEX␣␣␣␣␣␣␣␣0ETX
STXEX␣␣␣␣␣␣␣␣0ETX
STXOK␣␣␣␣␣␣␣␣0ETX

DEL1 Delete File

Function

This deletes files from the controller.

If the request cannot be received, an error response is returned.

Request Format

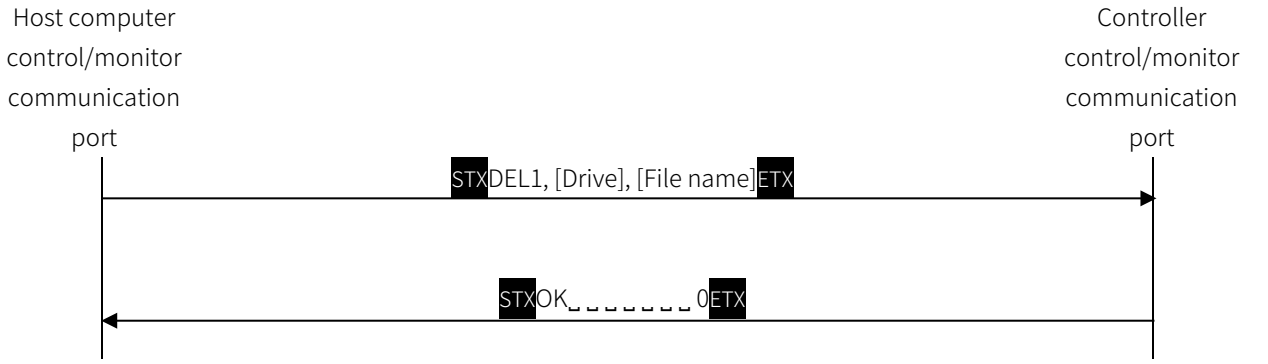
STXDEL1, [Drive], [File name]ETX

No.	Name	Size (bytes)	Description
1	Drive	1 Fixed length	Specifies the drive containing the file to be deleted. 1: ROM drive 2: RAM drive 3: SD card drive
2	File name	Variable length	Name of file to be deleted The file name must be in one of the following formats. ・name.extension ・name The name must be from 1 to 8 characters. The extension must be from 0 to 3 characters. When omitting the extension, the period (.) must also be omitted.

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command when deleting the file "PRG1" from the ROM drive of the controller is as follows.

STXDEL1, 1, PRG1ETX
STXOK␣␣␣␣␣␣␣0ETX

NOTICE

· Do not delete parameter files.
The robot that has performed unexpected motion may collide with peripheral equipment.
When a parameter file is deleted, the parameters are reset.

DIR1 Obtain Directory Information

Function

This transfers the directory information of the controller to the host computer.

Request Format

STXDIR1, [Drive]ETX

No.	Name	Size (bytes)	Description
1	Drive	1 Fixed length	Specifies the drive where the directory information is obtained from. 1: ROM drive 2: RAM drive 3: SD card drive

Response Format

STXOK [Operand length], [File count] Directory information ETX

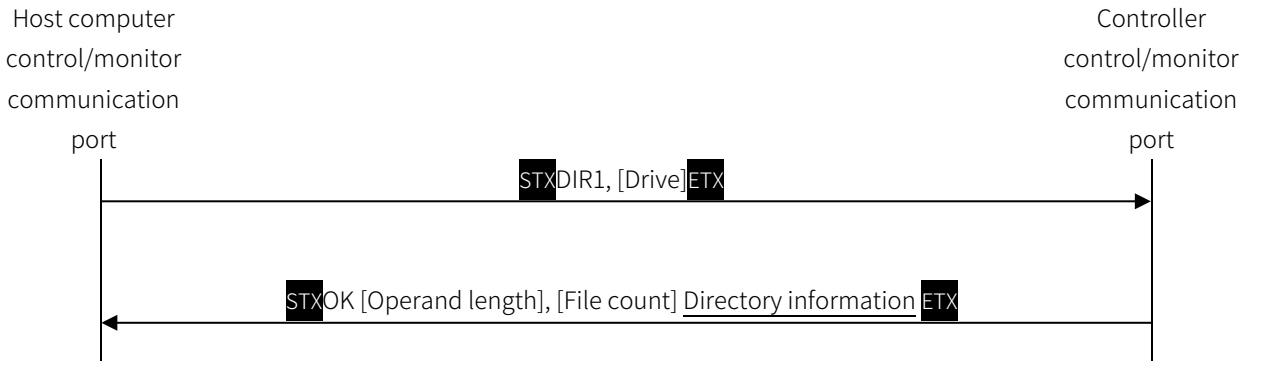
In the directory information, the following information is stored consecutively for the quantity in [File count].

CR[File name], [Size], [Date]␣[Time]

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	File count	Variable length	File count sent as response
3	File name	Variable length	File name in alphanumeric characters and starting with a letter
4	Size	Variable length	Byte size of file
5	Date	11 Fixed length	Shows the date that the file was updated “Mon-DD-YYYY” Mon: Month January: Jan, February: Feb March: Mar, April: Apr May: May, June: Jun July: Jul, August: Aug September: Sep, October: Oct November: Nov, December: Dec DD: Date YYYY: Year (Western calendar year)

No.	Name	Size (bytes)	Description
6	Time	8 Fixed length	Shows the time that the file was updated “HH:MM:SS” HH: Hours (24-hour format) MM: Minutes SS: Seconds

Communication Example



Send/Receive Example

The command for a 20-byte file with the file name "PRG1" and 30-byte file with the file name "PRG2" in the controller ROM drive is as follows.

STXDIR1, 1ETX

STXOK␣␣␣␣␣␣19, 2CRPRG1, 20CRPRG2, 30ETX

If there are no files in the drive, the command is as follows.

STXDIR1, 1ETX

STXOK␣␣␣␣␣␣␣2, 0ETX

EIM1 Obtain Emergency Stop Information

Function

This obtains position and speed information when the emergency stop signal is input.

*Supported version: X8KCS-03A or later

Request Format

STXEIM1，[Axis number]ETX

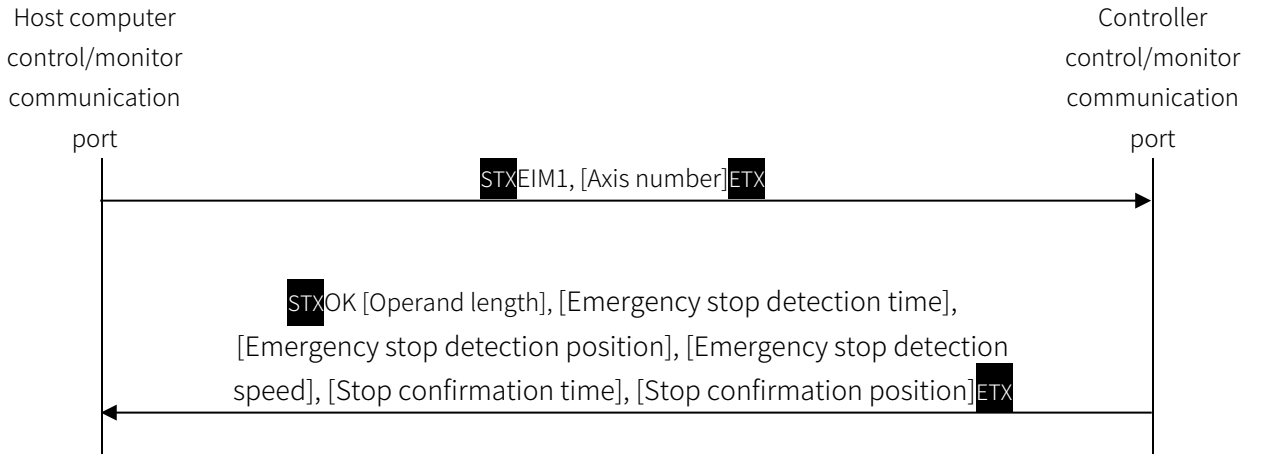
No.	Name	Size (bytes)	Description
1	Axis number	1 Fixed length	Specifies the joint axis number of the robot whose information is obtained.

Response Format

STXOK [Operand length], [Emergency stop detection time], [Emergency stop detection position], [Emergency stop detection speed], [Stop confirmation time], [Stop confirmation position]ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	Emergency stop detection time	Variable length	System time when the last emergency stop was detected [ms] （Elapsed time from start）
3	Emergency stop detection position	Variable length	Position of the specified joint axis when the last emergency stop was detected [deg] or [mm]
4	Emergency stop detection speed	Variable length	Speed of the specified joint axis when the last emergency stop was detected [deg/s] or [mm/s]
5	Stop confirmation time	Variable length	System time when stop state was entered [ms]
6	Stop confirmation position	Variable length	Position of j the specified joint axis when stop state was entered [deg] or [mm]

Communication Example



Send/Receive Example

The command for obtaining information when the first axis comes to an emergency stop is as follows.

STXEIM1,1ETX

STXOK␣␣␣␣␣␣33,10000,30.000,60.000,10005,31.000ETX

FLR1 Read File

Function

This reads the content of a file in the controller and transfers it to the host computer.
If the request cannot be received, or if file reading is aborted, the controller sends an abort response command.
If the request cannot be received, an error response is returned.
If reading of the file failed during the read file process, an OK command is returned, but the operand length of the response command will not match the data length that was received.

Request Format

STXFLR1, [Drive], [File name]ETX

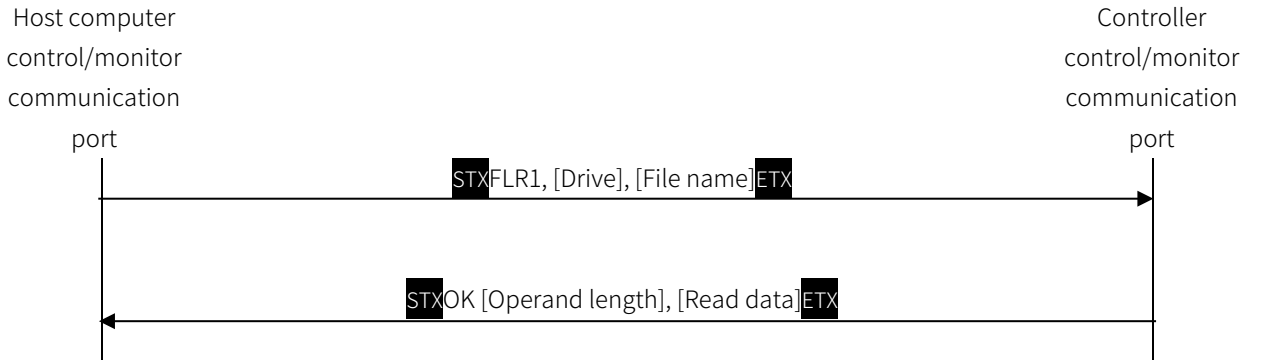
No.	Name	Size (bytes)	Description
1	Drive	1 Fixed length	Specify the controller reading target drive. 1: ROM drive 2: RAM drive 3: SD card drive
2	File name	Variable length	File name to be read from the controller The file name must be in one of the following formats. · name.extension · name The name must be from 1 to 8 characters. The extension must be from 0 to 3 characters. When omitting the extension, the period (.) must also be omitted.

Response Format

STXOK [Operand length], [Read data]ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX) This is the read data size + 1.
2	Read data	Variable length	Data read from the file

Communication Example



Send/Receive Example

The command when issuing a read request for the "PRG1" file from the controller ROM drive is as follows.

File name: PRG1

```
PROGRAM␣MAINCR
␣SUB1CR
CR
ENDCR
```

STXFLR1, 1, PRG1ETX
STXOK␣␣␣␣␣␣25, PROGRAM␣MAINCR␣SUB1CR CRENDCR ETX

FLW1 Write File

Function

This transfers the file content and writes it to the controller.
If the file already exists at the target location, the file is overwritten.
If the request cannot be received, an error response is returned.

Request Format

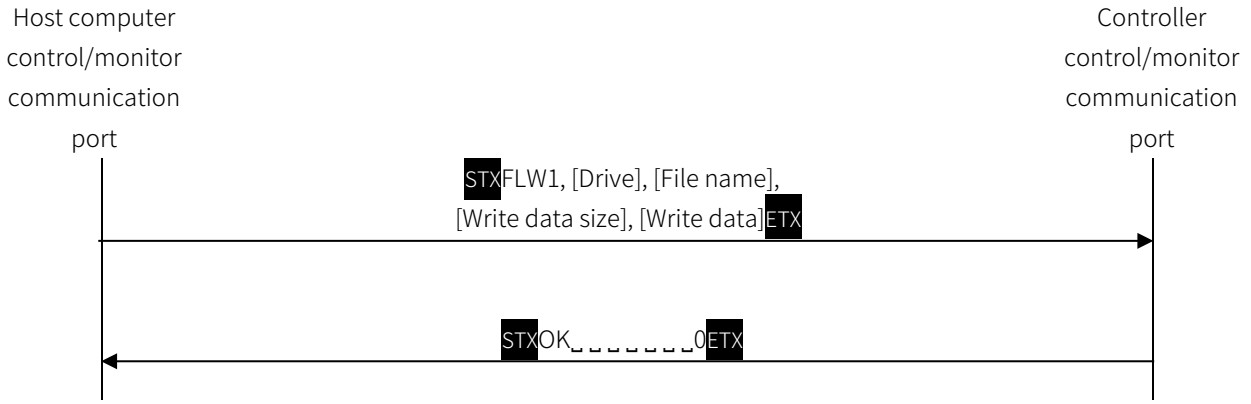
STXFLW1, [Drive], [File name], [Write data size], [Write data]ETX

No.	Name	Size (bytes)	Description
1	Drive	1 Fixed length	Specify the controller write target drive. 1: ROM drive 2: RAM drive 3: SD card drive
2	File name	Variable length	File name to be written to the controller The file name must be in one of the following formats. · name.extension · name The name must be from 1 to 8 characters. The extension must be from 0 to 3 characters. When omitting the extension, the period (.) must also be omitted.
3	Write data size	Variable length	Data length of [Write data] This is the file size. A file up to maximum size of 8 MB can be created.
4	Write data	Variable length	Data to be written to file

Response Format

STXOK␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command when creating the "PRG1" file in the ROM drive of the controller and issuing a write request for 24 bytes is as follows.

File name: PRG1

```
PROGRAM␣MAINCR
␣SUB1CR
CR
ENDCR
```

STXFLW1, 1, PRG1, 24, PROGRAM␣MAINCR␣SUB1CRCRENDCRETX
STXOK␣␣␣␣␣␣␣␣0ETX

MEMO

* *The controller determines the end of the file from [Write data size] of the request command.*
* *If communication is interrupted for a period of 100 ms, the controller determines that file writing was aborted, and the incomplete writing data is discarded. If a file already exists at the target location, the state of the existing file before file writing is retained.*
* *All files saved in the RAM drive are deleted when the AC power is turned off. To retain the files, save them in the ROM drive.*

NOTICE

- When writing parameter files, do not set to values other than those shown in the Parameter Manual. Otherwise, the robot that has performed unexpected motion may collide with peripheral equipment.
- Files can be written to the ROM drive for about 10, 000 times. When the writing limit is exceeded, the file writing process could fail or other problems can occur. For this reason, in systems where autorun programs are overwritten frequently, the programs should be written to the RAM drive.

FLW2 Write File

Function

This transfers the file content and writes it to the controller.
If the file already exists at the target location, the file is overwritten.
If the request cannot be received, an error response is returned.
If the Write File process takes longer than 2 seconds, the controller sends a "Process execution in progress" response command at 2-second intervals to indicate that the process is being executed.

Request Format

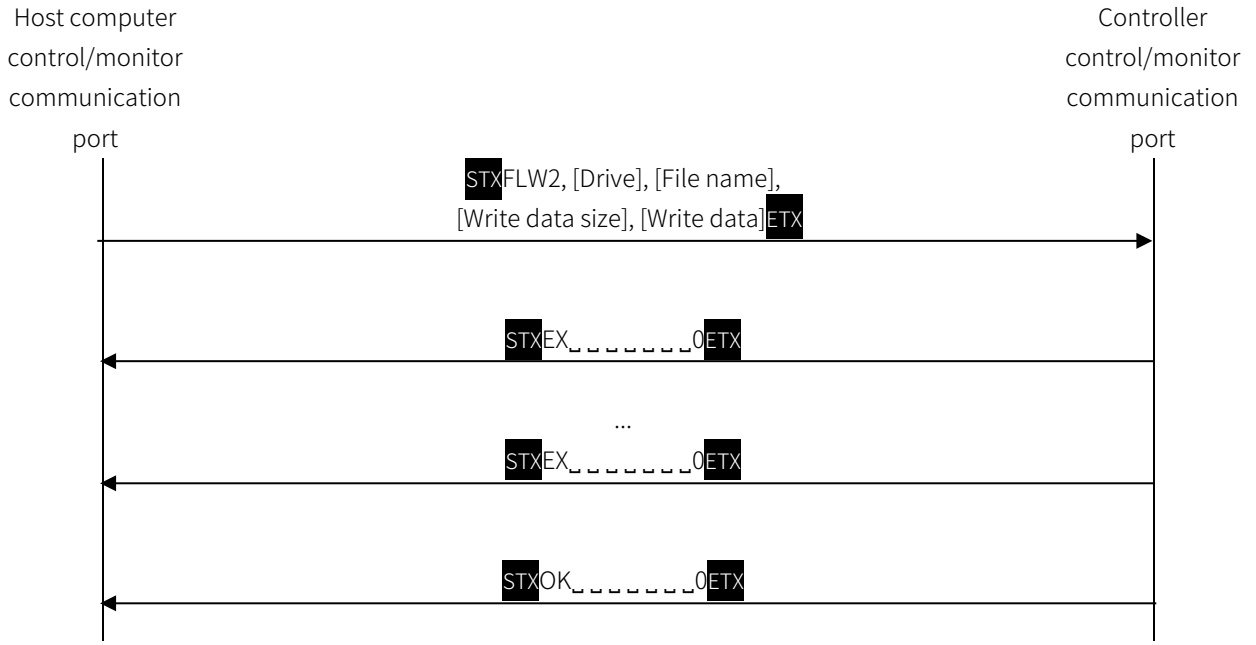
STXFLW2, [Drive], [File name], [Write data size], [Write data]ETX

No.	Name	Size (bytes)	Description
1	Drive	1 Fixed length	Specify the controller write target drive. 1: ROM drive 2: RAM drive 3: SD card drive
2	File name	Variable length	File name to be written to the controller The file name must be in one of the following formats. · name.extension · name The name must be from 1 to 8 characters. The extension must be from 0 to 3 characters. When omitting the extension, the period (.) must also be omitted.
3	Write data size	Variable length	Data length of [Write data] This is the file size. A file up to maximum size of 8 MB can be created.
4	Write data	Variable length	Data to be written to file

Response Format

STXOK␣␣␣␣␣␣␣0ETX
STXEX␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command when creating the "PRG1" file in the ROM drive of the controller and issuing a write request for 24 bytes is as follows.

File name: PRG1

```
PROGRAM␣MAINCR
␣SUB1CR
CR
ENDCR
```

```
STXFLW1, 1, PRG1, 24, PROGRAM␣MAINCR␣SUB1CR CRENDCR ETX
STXEX␣␣␣␣␣␣␣0ETX
STXEX␣␣␣␣␣␣␣0ETX
STXOK␣␣␣␣␣␣␣0ETX
```

MEMO

* *The controller determines the end of the file from [Write data size] of the request command.*
* *If communication is interrupted for a period of 100 ms, the controller determines that file writing was aborted, and the incomplete writing data is discarded. If a file already exists at the target location, the state of the existing file before file writing is retained.*
* *All files saved in the RAM drive are deleted when the AC power is turned off. To retain the files, save them in the ROM drive.*

NOTICE

- When writing parameter files, do not set to values other than those shown in the Parameter Manual. Otherwise, the robot that has performed unexpected motion may collide with peripheral equipment.
- Files can be written to the ROM drive for about 10, 000 times. When the writing limit is exceeded, the file writing process could fail or other problems can occur. For this reason, in systems where autorun programs are overwritten frequently, the programs should be written to the RAM drive.

IOR1 Read Signal Information (All Areas)

Function

Monitorable command

This transfers the I/O information of the controller to the host computer.

Request Format

STXIOR1ETX

Response Format

STXOK [Operand length], Signal information ETX

In the signal information, the following information is stored.
[General-purpose input 1], [General-purpose input 2], [General-purpose input 3], [General-purpose input 4], [Extension input 1], [Extension input 2], [Extension input 3], [Extension input 4], [System input 1], [System input 2], [System input 3], [System input 4], [System input 5], [Fieldbus input 1], [Fieldbus input 2], [Fieldbus input 3], [Fieldbus input 4], [Fieldbus input 5], [Fieldbus input 6], [Fieldbus input 7], [Fieldbus input 8], [Fieldbus input 9], [Fieldbus input 10], [Fieldbus input 11], [Fieldbus input 12], [Fieldbus input 13], [Fieldbus input 14], [Fieldbus input 15], [Fieldbus input 16], [Fieldbus input 17], [Fieldbus input 18], [Fieldbus input 19], [Fieldbus input 20], [Fieldbus input 21], [Fieldbus input 22], [Fieldbus input 23], [Fieldbus input 24], [Fieldbus input 25], [Fieldbus input 26], [Fieldbus input 27], [Fieldbus input 28], [Fieldbus input 29], [Fieldbus input 30], [Fieldbus input 31], [Fieldbus input 32], [User input 1], [User input 2], [User input 3], [User input 4], [User input 5], [User input 6], [User input 7], [User input 8], [User input 9], [User input 10], [User input 11], [User input 12], [User input 13], [User input 14], [User input 15], [User input 16], [User input 17], [User input 18], [User input 19], [User input 20], [User input 21], [User input 22], [User input 23], [User input 24], [User input 25], [User input 26], [User input 27], [User input 28], [User input 29], [User input 30], [User input 31], [User input 32], [User input 33], [User input 34], [User input 35], [User input 36], [User input 37], [User input 38], [User input 39], [User input 40], [User input 41], [User input 42], [User input 43], [User input 44], [User input 45], [User input 46], [User input 47], [User input 48], [Reserved area input 1], [Reserved area input 2], [Reserved area input 3], [Reserved area input 4], [Reserved area input 5], [Reserved area input 6], [Reserved area input 7], [Reserved area input 8], [Reserved area input 9], [Reserved area input 10], [Reserved area input 11], [Reserved area input 12], [Reserved area input 13], [Reserved area input 14], [Reserved area input 15], [Reserved area input 16], [Reserved area input 17], [Reserved area input 18], [Reserved area input 19], [Reserved area input 20], [Reserved area input 21], [Reserved area input 22], [Reserved area input 23], [Reserved area input 24], [Reserved area input 25], [Reserved area input 26], [Reserved area input 27], [Reserved area input 28], [Reserved area input 29], [Reserved area input 30], [Reserved area input 31], [Reserved area input 32], [General-purpose output 1], [General-purpose output 2], [General-purpose output 3], [General-purpose output 4], [Extension output 1], [Extension output

2], [Extension output 3], [Extension output 4], [System output 1], [System output 2], [System output 3], [System output 4], [System output 5], [Fieldbus output 1], [Fieldbus output 2], [Fieldbus output 3], [Fieldbus output 4], [Fieldbus output 5], [Fieldbus output 6], [Fieldbus output 7], [Fieldbus output 8], [Fieldbus output 9], [Fieldbus output 10], [Fieldbus output 11], [Fieldbus output 12], [Fieldbus output 13], [Fieldbus output 14], [Fieldbus output 15], [Fieldbus output 16], [Fieldbus output 17], [Fieldbus output 18], [Fieldbus output 19], [Fieldbus output 20], [Fieldbus output 21], [Fieldbus output 22], [Fieldbus output 23], [Fieldbus output 24], [Fieldbus output 25], [Fieldbus output 26], [Fieldbus output 27], [Fieldbus output 28], [Fieldbus output 29], [Fieldbus output 30], [Fieldbus output 31], [Fieldbus output 32], [User output 1], [User output 2], [User output 3], [User output 4], [User output 5], [User output 6], [User output 7], [User output 8], [User output 9], [User output 10], [User output 11], [User output 12], [User output 13], [User output 14], [User output 15], [User output 16], [User output 17], [User output 18], [User output 19], [User output 20], [User output 21], [User output 22], [User output 23], [User output 24], [User output 25], [User output 26], [User output 27], [User output 28], [User output 29], [User output 30], [User output 31], [User output 32], [User output 33], [User output 34], [User output 35], [User output 36], [User output 37], [User output 38], [User output 39], [User output 40], [User output 41], [User output 42], [User output 43], [User output 44], [User output 45], [User output 46], [User output 47], [User output 48], [Reserved area output 1], [Reserved area output 2], [Reserved area output 3], [Reserved area output 4], [Reserved area output 5], [Reserved area output 6], [Reserved area output 7], [Reserved area output 8], [Reserved area output 9], [Reserved area output 10], [Reserved area output 11], [Reserved area output 12], [Reserved area output 13], [Reserved area output 14], [Reserved area output 15], [Reserved area output 16], [Reserved area output 17], [Reserved area output 18], [Reserved area output 19], [Reserved area output 20], [Reserved area output 21], [Reserved area output 22], [Reserved area output 23], [Reserved area output 24], [Reserved area output 25], [Reserved area output 26], [Reserved area output 27], [Reserved area output 28], [Reserved area output 29], [Reserved area output 30], [Reserved area output 31], [Reserved area output 32]

No.	Name	Size (bytes)	Description		
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)		
2	General-purpose input 1	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1 to Din16	General-purpose input
3	General-purpose input 2	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din17 to Din32	General-purpose input
4	General-purpose input 3	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din33 to Din48	General-purpose input
5	General-purpose input 4	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din49 to Din64	General-purpose input
6	Extension input 1	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din101 to Din116	Extension input
7	Extension input 2	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din117 to Din132	Extension input
8	Extension input 3	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din133 to Din148	Extension input
9	Extension input 4	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din149 to Din164	Extension input

No.	Name	Size (bytes)	Description		
10	System input 1	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din201 to Din216	Hand input
11	System input 2	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 3	Din217 to Din220	Alarm 8 level
			4 to 7	Din221 to Din224	Alarm 4 level
			8 to 11	Din225 to Din228	Alarm 2 level
			12 to 15	Din229 to Din232	Alarm 1 level
12	System input 3	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 1	Din233 to Din234	J_REMOTE
			2 to 3	Din235 to Din236	J_SPEED
			4 to 5	Din237 to Din238	J_COORD
			6 to 9	Din239 to Din242	J_AXIS
			10	Din243	J_DIRECT
			11	Din244	J_MOVE
			12	Din245	FEED_HOLD
			13 to 15	Din236 to Din248	
13	System input 4	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0	Din249	STROBE
			1	Din250	PRG_RST
			2	Din251	STEP_RST
			3	Din252	CYC_RST
			4	Din253	DO_RST
			5	Din254	ALM_RST
			6	Din255	RUN
			7	Din256	EX_SVON
			8	Din257	STOP
			9	Din258	CYCLE
			10	Din259	LOW_SPD
			11	Din260	BREAK
			12	Din261	SVOFF
			13	DIN262	BX_RST
			13 to 15	Din262 to Din264	
14	System input 5	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din265 to Din280	

No.	Name	Size (bytes)	Description		
15	Fieldbus input 1	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din301 to Din316	Fieldbus input
16	Fieldbus input 2	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din317 to Din332	Fieldbus input
17	Fieldbus input 3	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din333 to Din348	Fieldbus input
18	Fieldbus input 4	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din349 to Din364	Fieldbus input
19	Fieldbus input 5	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din401 to Din416	Fieldbus input
20	Fieldbus input 6	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din417 to Din432	Fieldbus input
21	Fieldbus input 7	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din433 to Din448	Fieldbus input
22	Fieldbus input 8	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din449 to Din464	Fieldbus input
23	Fieldbus input 9	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din501 to Din516	Fieldbus input
24	Fieldbus input 10	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din517 to Din532	Fieldbus input
25	Fieldbus input 11	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din533 to Din548	Fieldbus input
26	Fieldbus input 12	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din549 to Din564	Fieldbus input

No.	Name	Size (bytes)	Description		
27	Fieldbus input 13	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din601 to Din616	Fieldbus input
28	Fieldbus input 14	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din617 to Din632	Fieldbus input
29	Fieldbus input 15	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din633 to Din648	Fieldbus input
30	Fieldbus input 16	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din649 to Din664	Fieldbus input
31	Fieldbus input 17	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din701 to Din716	Fieldbus input
32	Fieldbus input 18	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din717 to Din732	Fieldbus input
33	Fieldbus input 19	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din733 to Din748	Fieldbus input
34	Fieldbus input 20	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din749 to Din764	Fieldbus input
35	Fieldbus input 21	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din801 to Din816	Fieldbus input
36	Fieldbus input 22	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din817 to Din832	Fieldbus input
37	Fieldbus input 23	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din833 to Din848	Fieldbus input
38	Fieldbus input 24	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din849 to Din864	Fieldbus input

No.	Name	Size (bytes)	Description		
39	Fieldbus input 25	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din901 to Din916	Fieldbus input
40	Fieldbus input 26	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din917 to Din932	Fieldbus input
41	Fieldbus input 27	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din933 to Din948	Fieldbus input
42	Fieldbus input 28	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din949 to Din964	Fieldbus input
43	Fieldbus input 29	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1001 to Din1016	Fieldbus input
44	Fieldbus input 30	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1017 to Din1032	Fieldbus input
45	Fieldbus input 31	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1033 to Din1048	Fieldbus input
46	Fieldbus input 32	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1049 to Din1064	Fieldbus input

No.	Name	Size (bytes)	Description		
47	User input 1	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1101 to Din1116	User input
48	User input 2	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1117 to Din1132	User input
49	User input 3	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1133 to Din1148	User input
50	User input 4	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1149 to Din1164	User input
51	User input 5	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1201 to Din1216	User input
52	User input 6	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1217 to Din1232	User input
53	User input 7	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1233 to Din1248	User input
54	User input 8	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1249 to Din1264	User input
55	User input 9	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1301 to Din1316	User input
56	User input 10	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1317 to Din1332	User input
57	User input 11	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1333 to Din1348	User input
58	User input 12	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1349 to Din1364	User input

No.	Name	Size (bytes)	Description		
59	User input 13	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1401 to Din1416	User input
60	User input 14	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1417 to Din1432	User input
61	User input 15	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1433 to Din1448	User input
62	User input 16	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1449 to Din1464	User input
63	User input 17	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1501 to Din1516	User input
64	User input 18	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1517 to Din1532	User input
65	User input 19	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1533 to Din1548	User input
66	User input 20	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1549 to Din1564	User input
67	User input 21	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1601 to Din1616	User input
68	User input 22	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1617 to Din1632	User input
69	User input 23	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1633 to Din1648	User input
70	User input 24	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1649 to Din1664	User input

No.	Name	Size (bytes)	Description		
71	User input 25	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1701 to Din1716	User input
72	User input 26	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1717 to Din1732	User input
73	User input 27	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1733 to Din1748	User input
74	User input 28	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1749 to Din1764	User input
75	User input 29	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1801 to Din1816	User input
76	User input 30	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1817 to Din1832	User input
77	User input 31	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1833 to Din1848	User input
78	User input 32	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1849 to Din1864	User input
79	User input 33	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1901 to Din1916	User input
80	User input 34	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1917 to Din1932	User input
81	User input 35	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1933 to Din1948	User input
82	User input 36	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din1949 to Din1964	User input

No.	Name	Size (bytes)	Description		
83	User input 37	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2001 to Din2016	User input
84	User input 38	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2017 to Din2032	User input
85	User input 39	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2033 to Din2048	User input
86	User input 40	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2049 to Din2064	User input
87	User input 41	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2101 to Din2116	User input
88	User input 42	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2117 to Din2132	User input
89	User input 43	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2133 to Din2148	User input
90	User input 44	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2149 to Din2164	User input
91	User input 45	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2201 to Din2216	User input
92	User input 46	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2217 to Din2232	User input
93	User input 47	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2233 to Din2248	User input
94	User input 48	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2249 to Din2264	User input

No.	Name	Size (bytes)	Description		
95	Reserved area input 1	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2301 to Din2316	Reserved area input
96	Reserved area input 2	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2317 to Din2332	Reserved area input
97	Reserved area input 3	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2333 to Din2348	Reserved area input
98	Reserved area input 4	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2349 to Din2364	Reserved area input
99	Reserved area input 5	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2401 to Din2416	Reserved area input
100	Reserved area input 6	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2417 to Din2432	Reserved area input
101	Reserved area input 7	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2433 to Din2448	Reserved area input
102	Reserved area input 8	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2449 to Din2464	Reserved area input
103	Reserved area input 9	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2501 to Din2516	Reserved area input
104	Reserved area input 10	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2517 to Din2532	Reserved area input
105	Reserved area input 11	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2533 to Din2548	Reserved area input
106	Reserved area input 12	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2549 to Din2564	Reserved area input

No.	Name	Size (bytes)	Description		
107	Reserved area input 13	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2601 to Din2616	Reserved area input
108	Reserved area input 14	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2617 to Din2632	Reserved area input
109	Reserved area input 15	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2633 to Din2648	Reserved area input
110	Reserved area input 16	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2649 to Din2664	Reserved area input
111	Reserved area input 17	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2701 to Din2716	Reserved area input
112	Reserved area input 18	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2717 to Din2732	Reserved area input
113	Reserved area input 19	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2733 to Din2748	Reserved area input
114	Reserved area input 20	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2749 to Din2764	Reserved area input
115	Reserved area input 21	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2801 to Din2816	Reserved area input
116	Reserved area input 22	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2817 to Din2832	Reserved area input
117	Reserved area input 23	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2833 to Din2848	Reserved area input
118	Reserved area input 24	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2849 to Din2864	Reserved area input

No.	Name	Size (bytes)	Description		
119	Reserved area input 25	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2901 to Din2916	Reserved area input
120	Reserved area input 26	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2917 to Din2932	Reserved area input
121	Reserved area input 27	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2933 to Din2948	Reserved area input
122	Reserved area input 28	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din2949 to Din2964	Reserved area input
123	Reserved area input 29	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din3001 to Din3016	Reserved area input
124	Reserved area input 30	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din3017 to Din3032	Reserved area input
125	Reserved area input 31	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din3033 to Din3048	Reserved area input
126	Reserved area input 32	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Din3049 to Din3064	Reserved area output

No.	Name	Size (bytes)	Description		
127	General-purpose output 1	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1 to Dout16	General-purpose output
128	General-purpose output 2	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout17 to Dout32	General-purpose output
129	General-purpose output 3	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout33 to Dout48	General-purpose output
130	General-purpose output 4	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout49 to Dout64	General-purpose output
131	Extension output 1	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout101 to Dout116	Extension output
132	Extension output 2	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout117 to Dout132	Extension output
133	Extension output 3	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout133 to Dout148	Extension output
134	Extension output 4	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout149 to Dout164	Extension output
135	System output 1	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout201 to Dout216	Hand output
136	System output 2	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout217 to Dout232	
137	System output 3	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout233 to Dout248	
138	System output 4	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0	Dout249 to Dout264	
139	System output 5	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout265 to Dout280	

No.	Name	Size (bytes)	Description		
140	Fieldbus output 1	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout301 to Dout316	
141	Fieldbus output 2	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout317 to Dout332	
142	Fieldbus output 3	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout333 to Dout348	
143	Fieldbus output 4	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout349 to Dout364	
144	Fieldbus output 5	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout401 to Dout416	
145	Fieldbus output 6	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout417 to Dout432	
146	Fieldbus output 7	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout433 to Dout448	
147	Fieldbus output 8	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout449 to Dout464	
148	Fieldbus output 9	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout501 to Dout516	
149	Fieldbus output 10	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout517 to Dout532	
150	Fieldbus output 11	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout533 to Dout548	
151	Fieldbus output 12	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout549 to Dout564	

No.	Name	Size (bytes)	Description		
152	Fieldbus output 13	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout601 to Dout616	
153	Fieldbus output 14	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout617 to Dout632	
154	Fieldbus output 15	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout633 to Dout648	
155	Fieldbus output 16	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout649 to Dout664	
156	Fieldbus output 17	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout701 to Dout716	
157	Fieldbus output 18	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout717 to Dout732	
158	Fieldbus output 19	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout733 to Dout748	
159	Fieldbus output 20	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout749 to Dout764	
160	Fieldbus output 21	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout801 to Dout816	
161	Fieldbus output 22	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout817 to Dout832	
162	Fieldbus output 23	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout833 to Dout848	
163	Fieldbus output 24	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout849 to Dout864	

No.	Name	Size (bytes)	Description		
164	Fieldbus output 25	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout901 to Dout916	
165	Fieldbus output 26	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout917 to Dout932	
166	Fieldbus output 27	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout933 to Dout948	
167	Fieldbus output 28	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout949 to Dout964	
168	Fieldbus output 29	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1001 to Dout1016	
169	Fieldbus output 30	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1017 to Dout1032	
170	Fieldbus output 31	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1033 to Dout1048	
171	Fieldbus output 32	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1049 to Dout1064	

No.	Name	Size (bytes)	Description		
172	User output 1	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1101 to Dout1116	User output
173	User output 2	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1117 to Dout1132	User output
174	User output 3	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1133 to Dout1148	User output
175	User output 4	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1149 to Dout1164	User output 1
176	User output 5	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1201 to Dout1216	User output 1
177	User output 6	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1217 to Dout1232	User output
178	User output 7	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1233 to Dout1248	User output
179	User output 8	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1249 to Dout1264	User output
180	User output 9	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1301 to Dout1316	User output
181	User output 10	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1317 to Dout1332	User output
182	User output 11	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1333 to Dout1348	User output
183	User output 12	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1349 to Dout1364	User output

No.	Name	Size (bytes)	Description		
184	User output 13	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1401 to Dout1416	User output
185	User output 14	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1417 to Dout1432	User output
186	User output 15	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1433 to Dout1448	User output
187	User output 16	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1449 to Dout1464	User output
188	User output 17	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1501 to Dout1516	User output
189	User output 18	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1517 to Dout1532	User output
190	User output 19	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1533 to Dout1548	User output
191	User output 20	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1549 to Dout1564	User output
192	User output 21	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1601 to Dout1616	User output
193	User output 22	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1617 to Dout1632	User output
194	User output 23	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1633 to Dout1648	User output
195	User output 24	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1649 to Dout1664	User output

No.	Name	Size (bytes)	Description		
196	User output 25	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1701 to Dout1716	User output
197	User output 26	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1717 to Dout1732	User output
198	User output 27	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1733 to Dout1748	User output
199	User output 28	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1749 to Dout1764	User output
200	User output 29	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1801 to Dout1816	User output
201	User output 30	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1817 to Dout1832	User output
202	User output 31	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1833 to Dout1848	User output
203	User output 32	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1849 to Dout1864	User output
204	User output 33	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1901 to Dout1916	User output
205	User output 34	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1917 to Dout1932	User output
206	User output 35	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1933 to Dout1948	User output
207	User output 36	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout1949 to Dout1964	User output

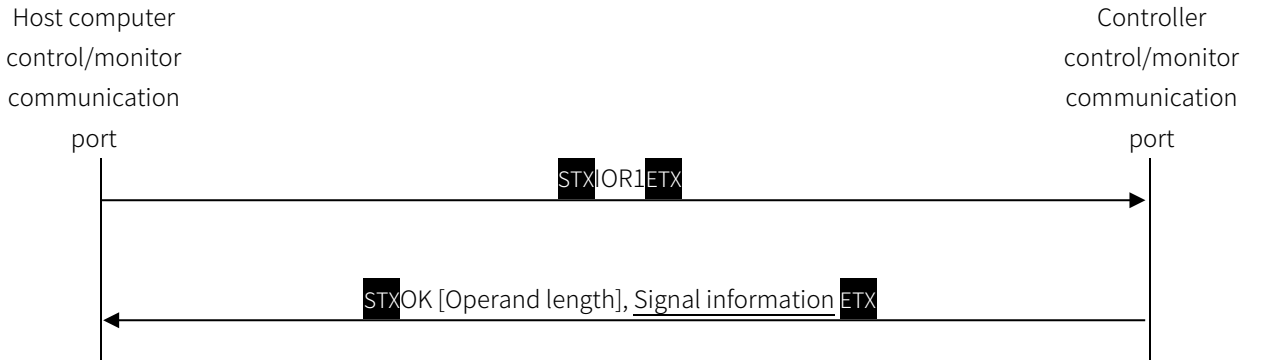
No.	Name	Size (bytes)	Description		
208	User output 37	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2001 to Dout2016	User output
209	User output 38	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2017 to Dout2032	User output
210	User output 39	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2033 to Dout2048	User output
211	User output 40	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2049 to Dout2064	User output
212	User output 41	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2101 to Dout2116	User output
213	User output 42	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2117 to Dout2132	User output
214	User output 43	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2133 to Dout2148	User output
215	User output 44	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2149 to Dout2164	User output
216	User output 45	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2201 to Dout2216	User output
217	User output 46	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2217 to Dout2232	User output
218	User output 47	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2233 to Dout2248	User output
219	User output 48	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2249 to Dout2264	User output

No.	Name	Size (bytes)	Description		
220	Reserved area output 1	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2301 to Dout2316	Reserved area output
221	Reserved area output 2	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2317 to Dout2332	Reserved area output
222	Reserved area output 3	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2333 to Dout2348	Reserved area output
223	Reserved area output 4	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2349 to Dout2364	Reserved area output
224	Reserved area output 5	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2401 to Dout2416	Reserved area output
225	Reserved area output 6	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2417 to Dout2432	Reserved area output
226	Reserved area output 7	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2433 to Dout2448	Reserved area output
227	Reserved area output 8	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2449 to Dout2464	Reserved area output
228	Reserved area output 9	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2501 to Dout2516	Reserved area output
229	Reserved area output 10	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2517 to Dout2532	Reserved area output
230	Reserved area output 11	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2533 to Dout2548	Reserved area output
231	Reserved area output 12	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2549 to Dout2564	Reserved area output

No.	Name	Size (bytes)	Description		
232	Reserved area output 13	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2601 to Dout2616	Reserved area output
233	Reserved area output 14	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2617 to Dout2632	Reserved area output
234	Reserved area output 15	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2633 to Dout2648	Reserved area output
235	Reserved area output 16	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2649 to Dout2664	Reserved area output
236	Reserved area output 17	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2701 to Dout2716	Reserved area output
237	Reserved area output 18	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2717 to Dout2732	Reserved area output
238	Reserved area output 19	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2733 to Dout2748	Reserved area output
239	Reserved area output 20	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2749 to Dout2764	Reserved area output
240	Reserved area output 21	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2801 to Dout2816	Reserved area output
241	Reserved area output 22	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2817 to Dout2832	Reserved area output
242	Reserved area output 23	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2833 to Dout2848	Reserved area output
243	Reserved area output 24	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2849 to Dout2864	Reserved area output

No.	Name	Size (bytes)	Description		
244	Reserved area output 25	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2901 to Dout2916	Reserved area output
245	Reserved area output 26	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2917 to Dout2932	Reserved area output
246	Reserved area output 27	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2933 to Dout2948	Reserved area output
247	Reserved area output 28	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout2949 to Dout2964	Reserved area output
248	Reserved area output 29	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout3001 to Dout3016	Reserved area output
249	Reserved area output 30	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout3017 to Dout3032	Reserved area output
250	Reserved area output 31	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout3033 to Dout3048	Reserved area output
251	Reserved area output 32	4	Hexadecimal character string 0000 to FFFF		
			Bit	Signal number	Signal name
			0 to 15	Dout3049 to Dout3064	Reserved area output

Communication Example



Send/Receive Example

If the signal status was obtained when the Din1 to Din64 status is as shown below, and the other signals are all OFF, the content is as shown below.

Signal number	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
Signal status	OFF	OFF	**ON**	**ON**	OFF	OFF	**ON**	OFF	OFF	OFF	OFF	**ON**	OFF	OFF	OFF	OFF
	3				2				1				0			

Signal number	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17
Signal status	OFF	**ON**	**ON**	**ON**	OFF	**ON**	**ON**	OFF	OFF	**ON**	OFF	**ON**	OFF	**ON**	OFF	OFF
	7				6				5				4			

Signal number	48	47	46	45	44	43	42	41	40	39	38	37	36	35	34	33
Signal status	**ON**	OFF	**ON**	**ON**	**ON**	OFF	**ON**	OFF	**ON**	OFF	OFF	**ON**	**ON**	OFF	OFF	OFF
	B				A				9				8			

Signal number	64	63	62	61	60	59	58	57	56	55	54	53	52	51	50	49
Signal status	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	OFF	**ON**	**ON**	OFF	**ON**	**ON**	**ON**	OFF	OFF
	F				E				D				C			

STXIOR1ETX

STXOK␣␣␣␣␣200, 3210, 7654, BA98, FEDC, 0000, 0000, 0000

IOR2 Read input signal information (Specified Signal Number)

Function

Monitorable command

Transfers I/O input signal information from the controller to the HOST computer.

Request Format

STXIOR2, [Signal number]ETX

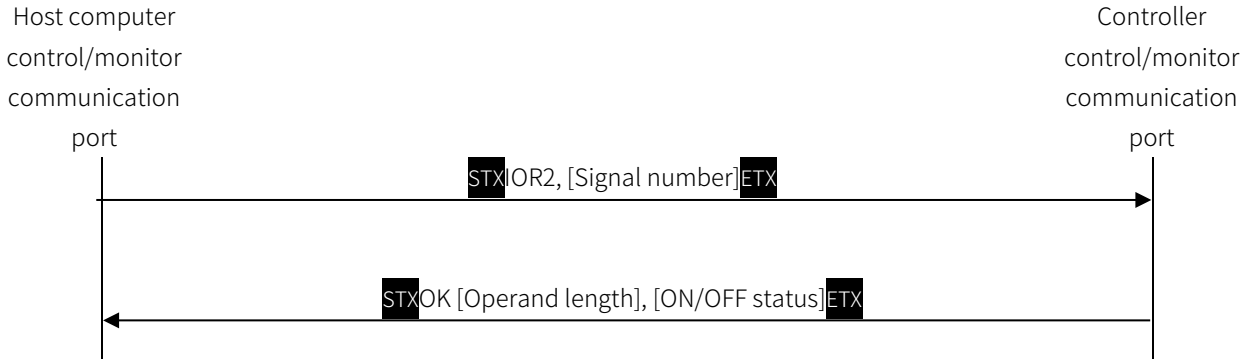
No.	Name	Size (bytes)	Description
1	Signal number	Variable length	Specify one of the following numbers for the signal number. 1 to 64, 101 to 164 201 to 280, 301 to 364 401 to 464, 501 to 564 601 to 664, 701 to 764 801 to 864, 901 to 964 1001 to 1064, 1101 to 1164 1201 to 1264, 1301 to 1364 1401 to 1464, 1501 to 1564 1601 to 1664, 1701 to 1764 1801 to 1864, 1901 to 1964 2001 to 2064, 2101 to 2164 2201 to 2264, 2301 to 2364 2401 to 2464, 2501 to 2564 2601 to 2664, 2701 to 2764 2801 to 2864, 2901 to 2964 3001 to 3064

Response Format

STXOK [Operand length], [ON/OFF status]ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	ON/OFF status	1 Fixed length	Specify one of the following numbers for the ON/OFF instruction. 0: OFF instruction 1: ON instruction

Communication Example



Send/Receive Example

When the Din43 status is obtained, the content is as shown below.

STXIOR2, 43ETX
STXOK␣␣␣␣␣␣␣␣2, 1ETX

IOR3 Read output signal information (Specified Signal Number)

Function

Monitorable command

Transfers I/O output signal information from the controller to the HOST computer.

Request Format

STXIOR3, [Signal number]ETX

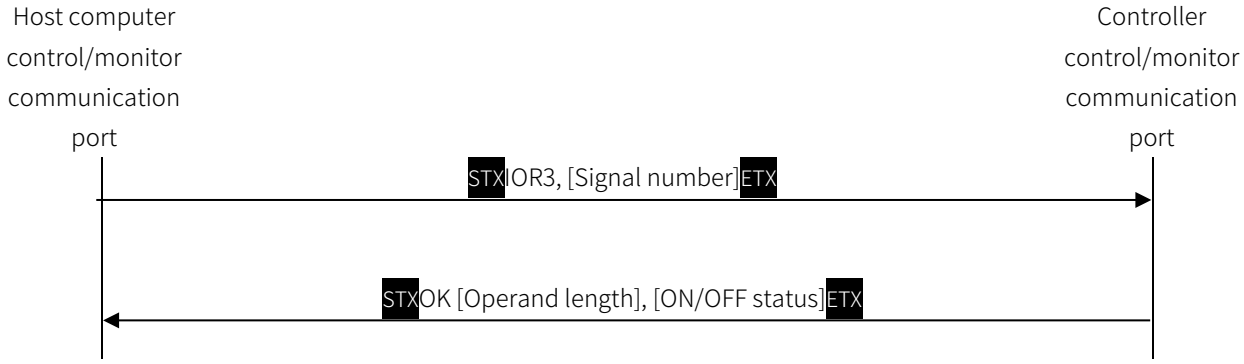
No.	Name	Size (bytes)	Description
1	Signal number	Variable length	Specify one of the following numbers for the signal number. 1 to 64, 101 to 164 201 to 280, 301 to 364 401 to 464, 501 to 564 601 to 664, 701 to 764 801 to 864, 901 to 964 1001 to 1064, 1101 to 1164 1201 to 1264, 1301 to 1364 1401 to 1464, 1501 to 1564 1601 to 1664, 1701 to 1764 1801 to 1864, 1901 to 1964 2001 to 2064, 2101 to 2164 2201 to 2264, 2301 to 2364 2401 to 2464, 2501 to 2564 2601 to 2664, 2701 to 2764 2801 to 2864, 2901 to 2964 3001 to 3064

Response Format

STXOK [Operand length], [ON/OFF status]ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	ON/OFF status	1 Fixed length	Specify one of the following numbers for the ON/OFF instruction. 0: OFF instruction 1: ON instruction

Communication Example



Send/Receive Example

When the Dout43 status is obtained, the content is as shown below.

STXIOR3, 43ETX
STXOK␣␣␣␣␣␣␣␣2, 1ETX

IOW1 Write Signal Information (General-purpose Output)

Function

This forces writing of general-purpose output signals.
If the request cannot be received, an error response is returned.

Request Format

STXIOW1 Output instruction ETX

For the output instruction, store the following data consecutively for the signal count that is controlled.
The maximum number of simultaneous settable signals is 64.
CR[Signal number], [ON/OFF instruction]

When writing only one general-purpose output signal
STXIOW1CR[Signal number], [ON/OFF instruction]ETX

When writing multiple general-purpose output signals
STXIOW1CR[Signal number], [ON/OFF instruction]CR[Signal number], [ON/OFF instruction]
…CR[Signal number], [ON/OFF instruction]ETX

No.	Name	Size (bytes)	Description
1	Signal number	Variable length	Specify one of the following numbers for the signal number. 1 to 64, 101 to 164 201 to 280, 301 to 364 401 to 464, 501 to 564 601 to 664, 701 to 764 801 to 864, 901 to 964 1001 to 1064, 1101 to 1164 1201 to 1264, 1301 to 1364 1401 to 1464, 1501 to 1564 1601 to 1664, 1701 to 1764 1801 to 1864, 1901 to 1964 2001 to 2064, 2101 to 2164 2201 to 2264, 2301 to 2364 2401 to 2464, 2501 to 2564 2601 to 2664, 2701 to 2764 2801 to 2864, 2901 to 2964 3001 to 3064

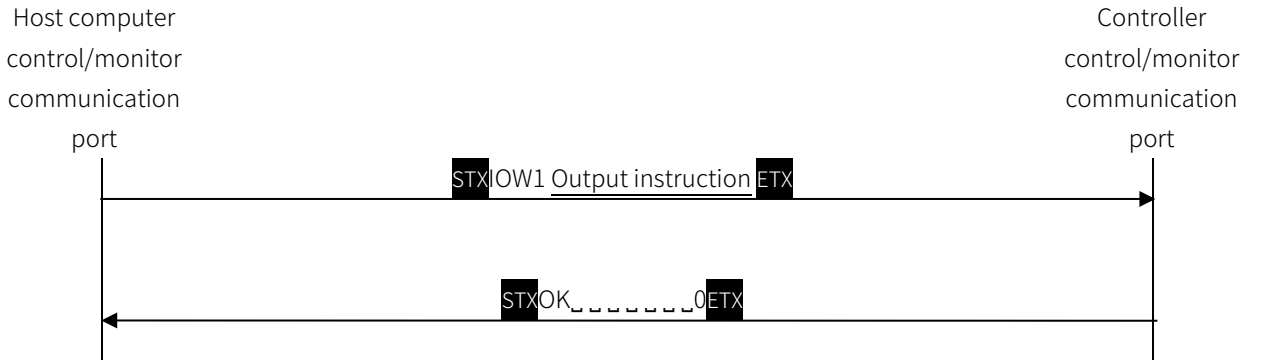
2	ON/OFF instruction	1 Fixed length	Specify one of the following numbers for the ON/OFF instruction. 0: OFF instruction 1: ON instruction

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)

Communication Example



Send/Receive Example

The command when setting Dout43 to ON and Dout42 to OFF and the signal status for other signals is retained is as follows.

STXIOW1CR43, 1CR42, 0ETX
STXOK␣␣␣␣␣␣␣␣0ETX

NOTICE

· Before sending this command, be sure to check the effect of the output signals.
The robot that has performed unexpected motion may collide with peripheral equipment.
Example: If the suction signal is turned off, the clamped object drops.

IOW2 Write Signal Information (General-purpose Output)

Function

This forces writing of general-purpose output signals.

If the request cannot be received, an error response is returned.

Request Format

STXIOW2, [Signal number area], [Target signal], [Output value]ETX

No.	Name	Size (bytes)	Description
1	Signal number area	1 to 2 (Variable length)	Shows the output area of the general-purpose output signals. Specify one of the following numbers for the area number. 0 : Signal number 1 to 64 1 : Signal number 101 to 164 2 : Signal number 201 to 216 3 : Signal number 301 to 364 4 : Signal number 401 to 464 5 : Signal number 501 to 564 6 : Signal number 601 to 664 7 : Signal number 701 to 764 8 : Signal number 801 to 864 9 : Signal number 901 to 964 10 : Signal number 1001 to 1064 11 : Signal number 1101 to 1164 12 : Signal number 1201 to 1264 13 : Signal number 1301 to 1364 14 : Signal number 1401 to 1464 15 : Signal number 1501 to 1564 16 : Signal number 1601 to 1664 17 : Signal number 1701 to 1764 18 : Signal number 1801 to 1864 19 : Signal number 1901 to 1964 20 : Signal number 2001 to 1064 21 : Signal number 2101 to 2164 22 : Signal number 2201 to 2264 23 : Signal number 2301 to 2364

			24 : Signal number 2401 to 2464 25 : Signal number 2501 to 2564

			26 : Signal number 2601 to 2664 27 : Signal number 2701 to 2764 28 : Signal number 2801 to 2864 29 : Signal number 2901 to 2964 30 : Signal number 3001 to 3064
2	Target signal	16 (Fixed length)	Shows the bit position for the setting target of the general-purpose output signal. (Hexadecimal) 0000000000000000 to FFFFFFFFFFFFFFFF [Output value] is changed only for bits that are set to 1, and the signal status is retained regardless of [Output value] for bits that are set to 0. MSB (bit 63) ⋯ Dout64 ... LSB (bit 0) ⋯ Dout1
3	Output value	16 (Fixed length)	Shows the value that is output from the general-purpose output signal. (Hexadecimal) 0000000000000000 to FFFFFFFFFFFFFFFF When set to 1, the target signal is set to ON. When set to 0, the target signal is set to OFF. MSB (bit 63) ⋯ Dout64 ... LSB (bit 0) ⋯ Dout1 * When [Signal number area] is 2, the signal number is a value from 201 to 216. For [Output value] and [Target signal], the lower 16 bits only are valid, and the other bits are invalid.

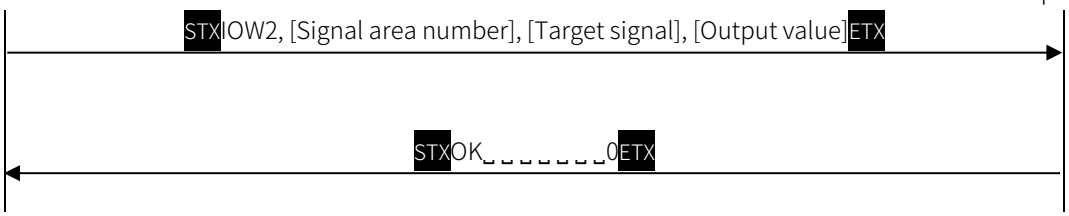
Response Format

STXOK␣␣␣␣␣␣␣0ETX

Communication Example

Host computer control/monitor communication port

Controller control/monitor communication port



Send/Receive Example

The command when setting Dout43 to ON and Dout42 to OFF and the signal status for other signals is retained is as follows.

Signal number	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
Target signal	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
	0				0				0				0			
Output value	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	-				-				-				-			

Signal number	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17
Target signal	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
	0				0				0				0			
Output value	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	-				-				-				-			

Signal number	48	47	46	45	44	43	42	41	40	39	38	37	36	35	34	33
Target signal	OFF	OFF	OFF	OFF	OFF	**ON**	**ON**	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
	0				6				0				0			
Output value	-	-	-	-	-	**ON**	OFF	-	-	-	-	-	-	-	-	-
	-				4				-				-			

Signal number	64	63	62	61	60	59	58	57	56	55	54	53	52	51	50	49
Target signal	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	0				0				0				0			
Output value	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	-				-				-				-			

STXIOW2, 0, 0000060000000000, 0000040000000000ETX
STXOK␣␣␣␣␣␣␣␣0ETX

NOTICE

· Before sending this command, be sure to check the effect of the output signals.
The robot that has performed unexpected motion may collide with peripheral equipment.
Example: If the suction signal is turned off, the clamped object drops.

IPC1 Close IP Port

Function

This closes the Ethernet port for non-protocol communication.

If the specified port was already closed, a normal response is returned.

If the request cannot be received, an error response is returned.

Request Format

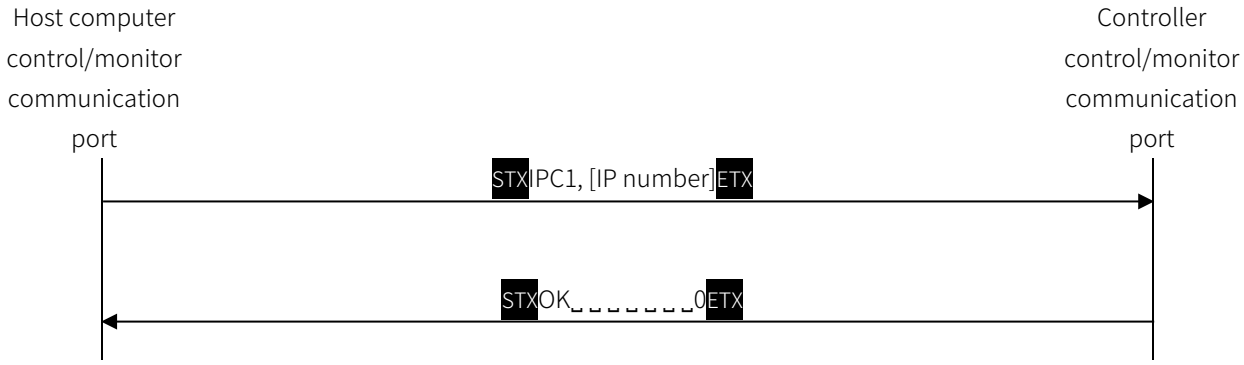
STXIPC1, [IP number]ETX

No.	Name	Size (bytes)	Description
1	IP number	1 Fixed length	Specify the identification number of the IP port of the controller. Set to a number from 1 to 8.

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command for closing IP1 is as follows.

STXIPC1, 1ETX
STXOK␣␣␣␣␣␣␣␣0ETX

NOTICE

· Before sending this command, be sure to check the effect on the peripheral devices.
The robot that has performed unexpected motion may collide with peripheral equipment.

IPD1 Set IP Dump

Function

This sets the target that is dumped by the IP dump function.

If the request cannot be received, an error response is returned.

For details on the usage procedure, see "4.1.3 IP Dump Protocol".

Request Format

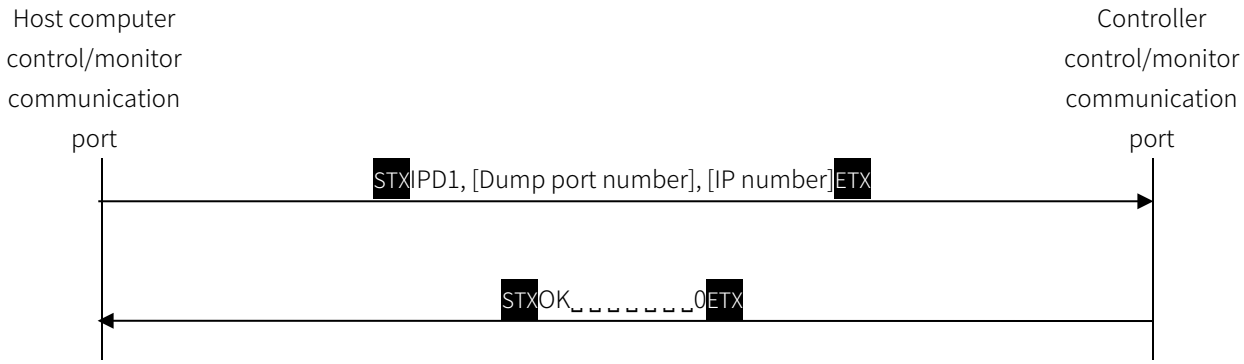
STXIPD1, [Dump port number], [IP number]ETX

No.	Name	Size (bytes)	Description
1	Dump port number	1 Fixed length	Select the dump port to be set. 1: IP dump port 1 2: IP dump port 2
2	IP number	1 Fixed length	Specify the identification number of the IP port of the controller. Set to a number from 1 to 8.

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command when the IP1 communication data is output from IP dump port 1 is as follows.

STXIPD1, 1, 1ETX
STXOK␣␣␣␣␣␣␣␣0ETX

IPO1 Open IP Port

Function

This opens the Ethernet port for non-protocol communication.
If the specified port is already open or if the request cannot be received, an error response is returned.

Request Format

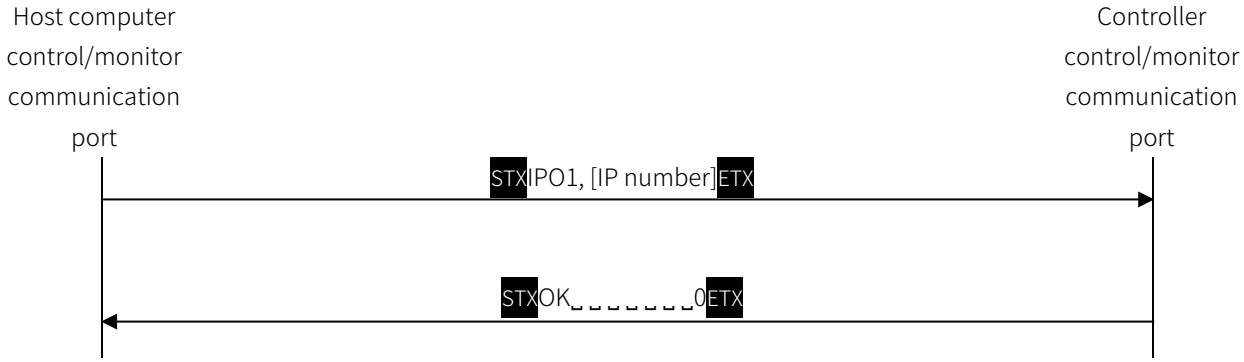
STXIPO1, [IP number]ETX

No.	Name	Size (bytes)	Description
1	IP number	1 Fixed length	Specify the identification number of the IP port of the controller. Set to a number from 1 to 8.

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command for opening IP1 is as follows.

STXIPO1, 1ETX
STXOK␣␣␣␣␣␣␣␣0ETX

NOTICE

· Before sending this command, be sure to check the effect on the peripheral devices.
The robot that has performed unexpected motion may collide with peripheral equipment.

IPR1 IP Dump Results

Function

This command is a response only.

This transfers the dump results for non-protocol communication or dedicated communication to the host computer.

When the connection with the IP dump port is opened, the communication content can be obtained from the non-protocol communication or dedicated communication that was set by the IP dump setting command.

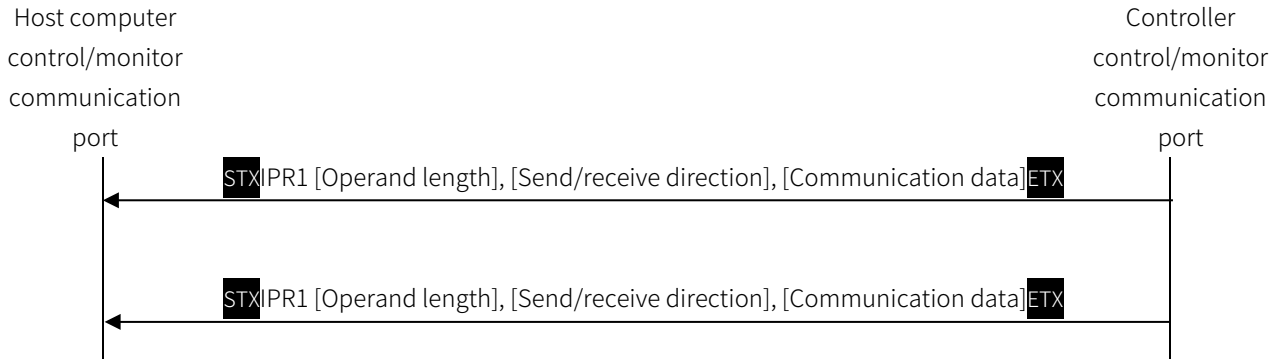
For details on the usage procedure, see "4.1.3 IP Dump Protocol".

Response Format

STXIPR1 [Operand length], [Send/receive direction], [Communication data]ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	Send/receive direction	1 Fixed length	Send/receive direction with respect to controller R: Data received by controller S: Data sent by controller
3	Communication data	Variable length	Data that was sent or received by the controller

Communication Example



Send/Receive Example

The dumped data that was communicated with the controller is as follows.

(When received by controller)

STXIPR1␣␣␣␣␣␣␣6, R, 100ETX

(When sent by controller)

STXIPR1␣␣␣␣␣␣␣5, S, OKETX

JOC1 Select Guidance Coordinates

Function

This selects the controller guidance coordinates.

If the request cannot be received, an error response is returned.

Request Format

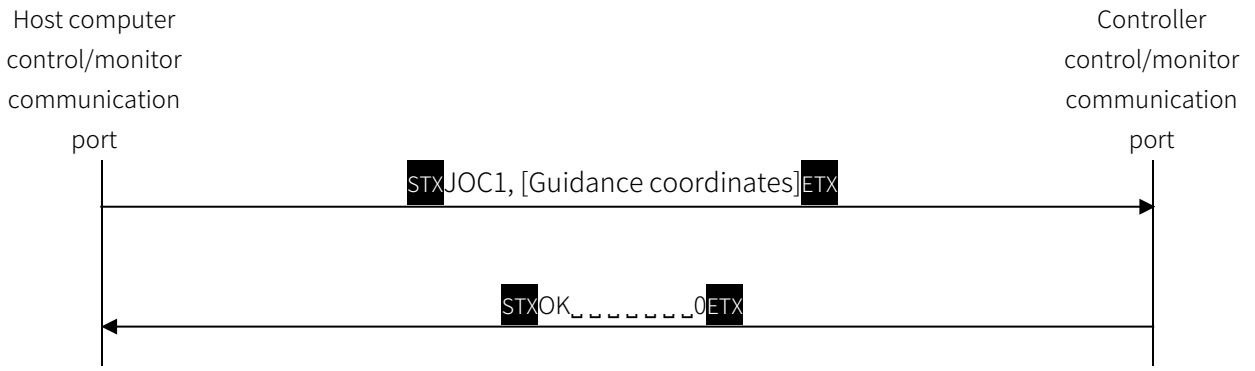
STXJOC1, [Guidance coordinates]ETX

No.	Name	Size (bytes)	Description
1	Guidance coordinates	1 Fixed length	Specify one of the following values for the guidance coordinates. 0: Joint coordinates 1: Tool coordinates 2: Work coordinates 3: World coordinates

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command to select world coordinates as the guidance coordinates is as follows.

STXJOC1, 3ETX
STXOK␣␣␣␣␣␣␣␣0ETX

JOG1 Execute Jogging

Function

This sends a jogging start command to the controller.

To continue execution of jogging, send this command within 200 ms. If the controller does not receive this command within 200 ms, it is determined to be a timeout, and an automatic stop is performed.

If the request cannot be received, an error response is returned.

Request Format

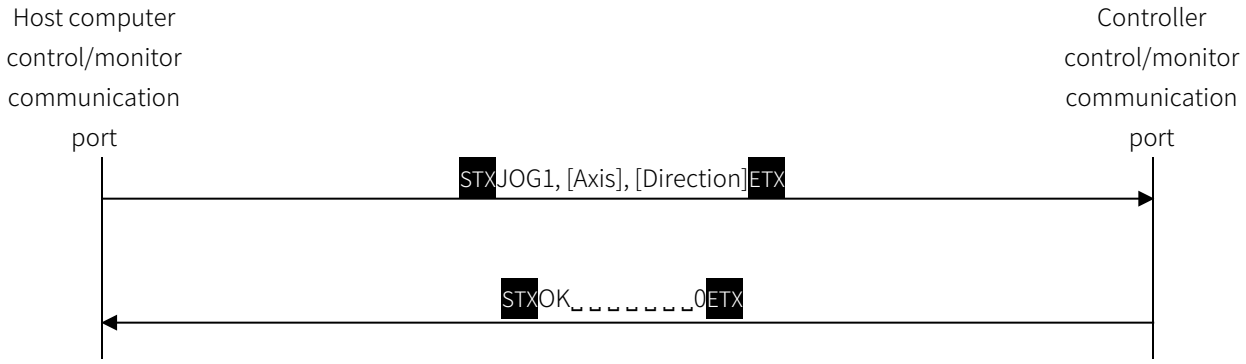
STXJOG1, [Axis], [Direction]ETX

No.	Name	Size (bytes)	Description
1	Axis	1 Fixed length	Specify one of the following numbers for the axis. 1 to 5
2	Direction	1 Fixed length	Specify one of the following characters for the direction. +: Jogging in positive direction - : Jogging in negative direction !: Stop jog operation

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command to start jogging in the positive direction is as follows.



MEMO

** When an operation instruction is received, this command returns an immediate response.*

** Multiple axis jog operations cannot be executed simultaneously.*

NOTICE

- Be sure to always check the guidance coordinates and guidance speed selected before sending this command.
- When tool coordinates are selected for the guidance coordinates, be sure to always check the tool type and tool offset amount that is data of tool coordinates.
  The robot that has performed unexpected motion may collide with peripheral equipment.

⚠ DANGER

- Do not approach the robot during jog operation.
  Otherwise, a collision could occur due to an unexpected movement by the robot.

JOI1 Execute Inching

Function

This performs the inching operation of the controller.

If the request cannot be received, an error response is returned.

Request Format

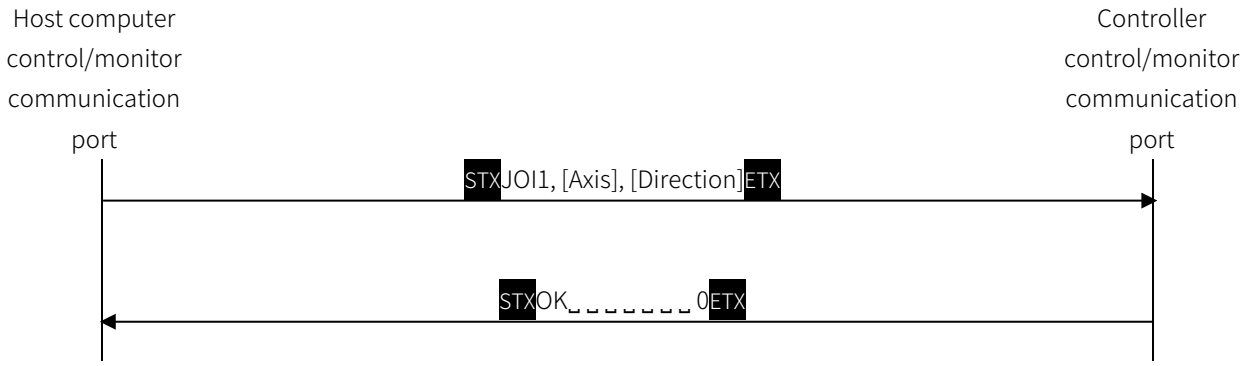
STXJOI1, [Axis], [Direction]ETX

No.	Name	Size (bytes)	Description
1	Axis	1 Fixed length	Specify one of the following numbers for the axis. 1 to 5
2	Direction	1 Fixed length	Specify one of the following characters for the direction. +: Inching in positive direction -: Inching in negative direction

Response Format

STXOK␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command to start inching in the positive direction is as follows.

STXJOI1, 1, +ETX
STXOK_ _ _ _ _ _ _ 0ETX

NOTICE

- Be sure to always check the guidance coordinates and guidance speed selected before sending this command.
- When tool coordinates are selected for the guidance coordinates, be sure to always check the tool type and tool offset amount that is data of tool coordinates.
  Otherwise, a collision could occur due to an unexpected movement by the robot.

⚠ DANGER

- Do not approach the robot during inching operation.
  Otherwise, a collision could occur due to an unexpected movement by the robot.

JOI2 Execute Inching

Function

This performs the inching operation of the controller. If the request cannot be received, an error response is returned.
The units for the movement amount vary depending on the axis and selected guidance coordinates.

Request Format

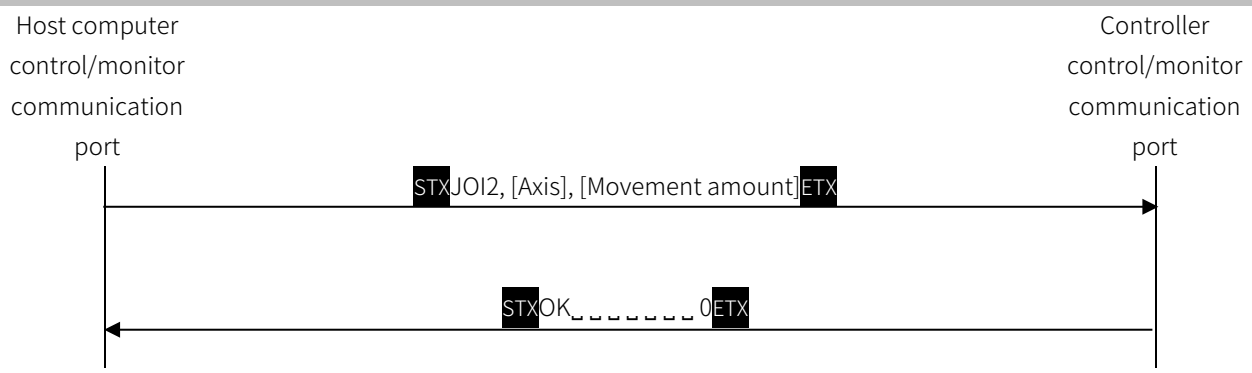
STXJOI2, [Axis], [Movement amount]ETX

No.	Name	Size (bytes)	Description
1	Axis	1 Fixed length	Specify one of the following numbers for the axis. 1 to 5
2	Movement amount	Variable length	Guidance amount and movement direction When a positive value is set, movement is performed in the positive direction. When a negative value is set, movement is performed in the negative direction. The units for the movement amount vary depending on the guidance coordinates selection that is currently set.

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command to start 2.5 inching of the 1st axis is as follows.

STXJOI2, 1, 2.5ETX
STXOK_ _ _ _ _ _ _ _0ETX

NOTICE

- Be sure to always check the guidance coordinates and movement amount selected before sending this command.
- When tool coordinates are selected for the guidance coordinates, be sure to always check that the tool coordinates are set correctly.
  The robot that has performed unexpected motion may collide with peripheral equipment.

⚠ DANGER

- Do not approach the robot during jog operation.
  Otherwise, a collision could occur due to an unexpected movement by the robot.

JOM1 Set Guidance Amount

Function

This sets the controller guidance amount.

If the request cannot be received, an error response is returned.

Request Format

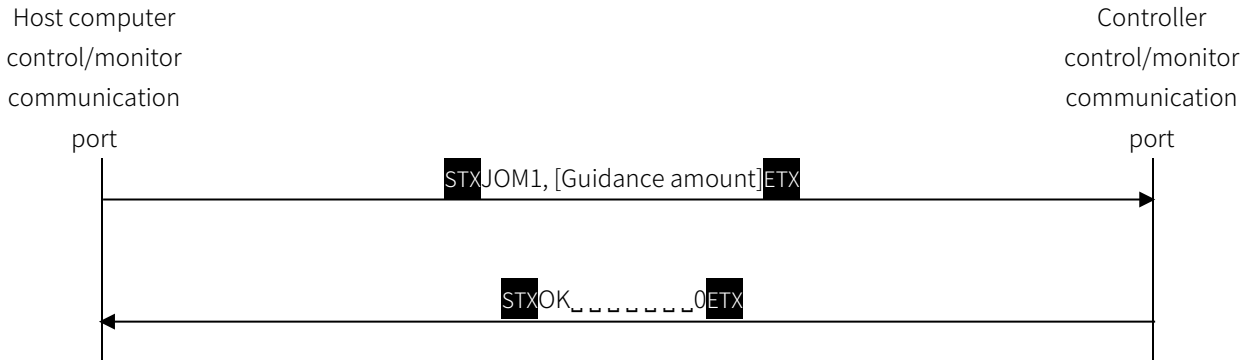
STXJOM1, [Guidance amount]ETX

No.	Name	Size (bytes)	Description
1	Inching Movement amount	1 Fixed length	Specify one of the following values for the guidance amount. 0: Small guidance amount 1: Medium guidance amount 2: Large guidance amount

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command to set the guidance amount to Medium is as follows.

STXJOM1, 1ETX

STXOK␣␣␣␣␣␣␣␣0ETX

JOS1 Set Guidance Speed

Function

This sets the controller guidance speed.

If the request cannot be received, an error response is returned.

Request Format

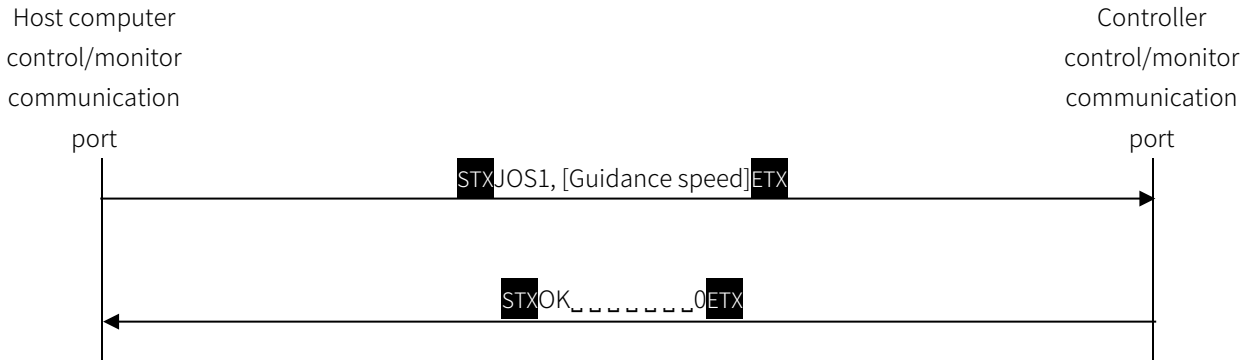
STXJOS1, [Guidance speed]ETX

No.	Name	Size (bytes)	Description
1	Guidance speed	1 Fixed length	Specify one of the following values for the guidance speed. 0: Slow jog speed 1: Medium jog speed 2: Fast jog speed

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command to set the guidance speed to Medium is as follows.

STXJOS1, 1ETX

STXOK␣␣␣␣␣␣␣␣0ETX

JPR1 Obtain Jog Payload

Function

This obtains the payload setting during jog operation.

If the request cannot be received, an error response is returned.

Request Format

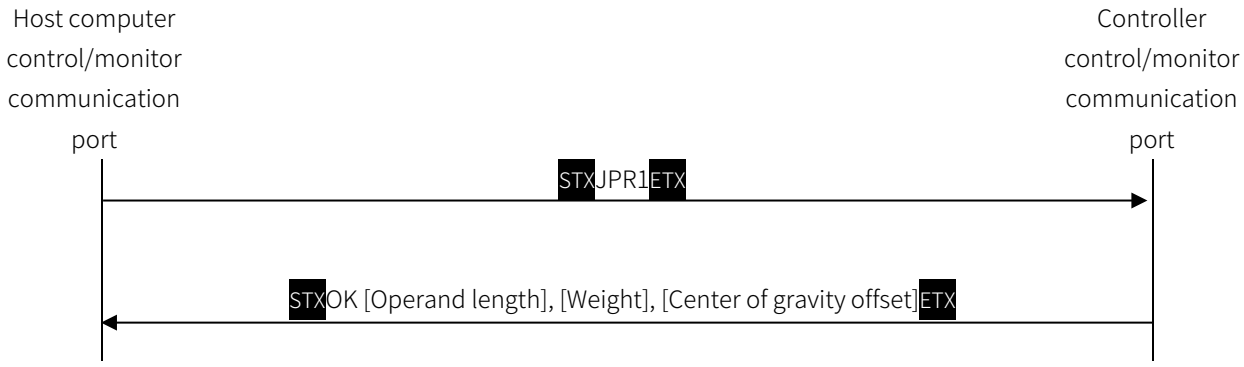
STXJPR1ETX

Response Format

STXOK [Operand length], [Weight], [Center of gravity offset]ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	Weight	Variable length	Weight up to 3 decimal places
3	Center of gravity offset	Variable length	Center of gravity offset up to 3 decimal places

Communication Example



Send/Receive Example

The command to obtain the payload during jog operation is as follows.

STXJPR1ETX
STXOK␣␣␣␣␣␣15, 10.000, 100.000ETX

JPS1 Set Jog Payload

Function

This sets the payload during jog operation.

If the request cannot be received, an error response is returned.

Request Format

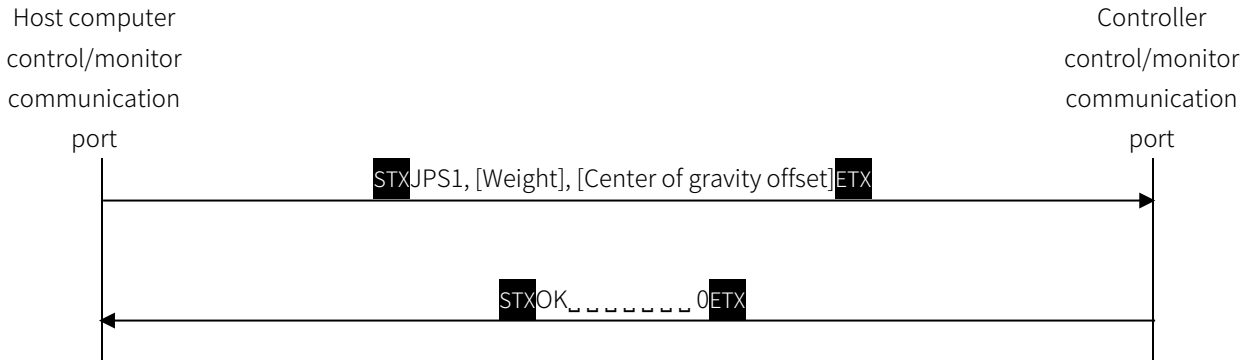
STXJPS1, [Weight], [Center of gravity offset]ETX

No.	Name	Size (bytes)	Description
1	Weight	Variable length	Specify as a real number up to 3 decimal places.
2	Center of gravity offset	Variable length	Specify as a real number up to 3 decimal places.

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command to set the weight to 10 and the center of gravity to 100 for the payload during jog operation is as follows.

STXJPS1, 10, 100ETX
STXOK␣␣␣␣␣␣␣␣0ETX

LCR0 Read Load Cell Information

Function

This obtains load cell data.

If the controller is not equipped with an A/D conversion board or if parameter [U027] is not set, NG response is returned.

*Supported version: X8KCS-02I or later

Request Format

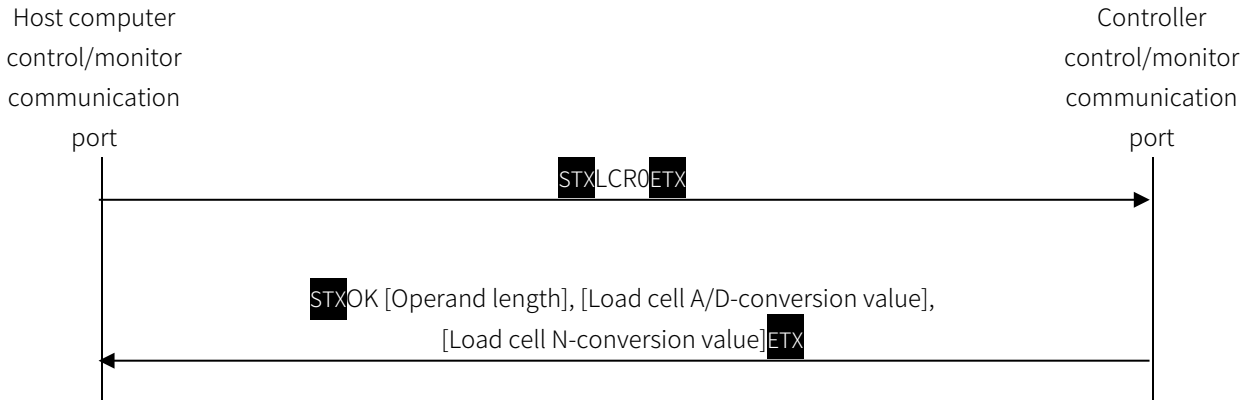
STXLCR0ETX

Response Format

STXOK [Operand length], [Load cell A/D conversion value], [Load cell N conversion value]ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	Load cell A/D conversion value	Variable length	A/D-converted value of load cell output (0 to 65535)
3	Load cell N-conversion value	Variable length	N-converted value of load cell A/D conversion value (Up to 3 decimal places)

Communication Example



Send/Receive Example

The command for obtaining load cell information is as follows.

STXLCR0ETX

STXOK␣␣␣␣␣␣12,32768,0.000ETX

LCZ0 Set Load Cell Origin

Function

This sets the load cell origin in the current load state.

If the controller is not equipped with an A/D conversion board or if parameter [U027] is not set, NG response is returned.

*Supported version: X8KCS-02I or later

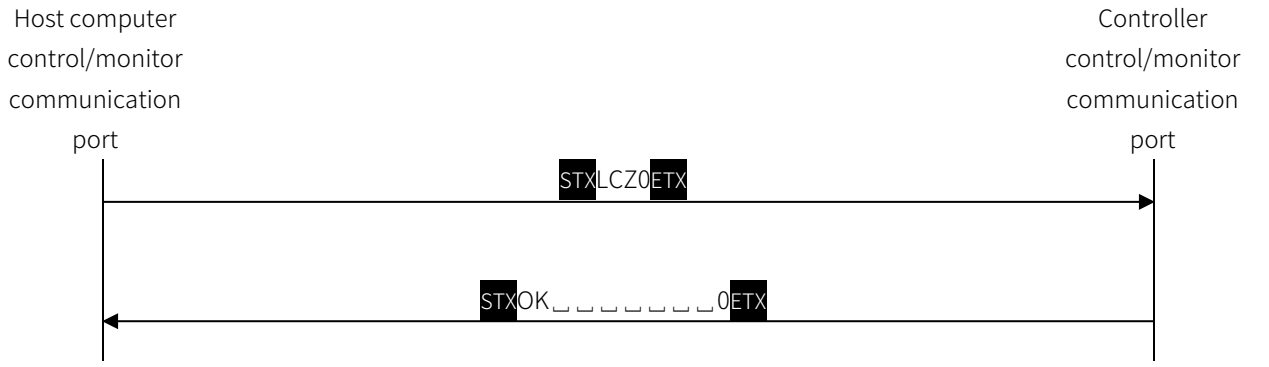
Request Format

STXLCZ0ETX

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command for setting the load cell origin is as follows.

STXLCZ0ETX
STXOK␣␣␣␣␣␣␣␣0ETX

MDE1 Set Execution Mode

Function

This sets the controller execution mode.

If the request cannot be received, an error response is returned.

Request Format

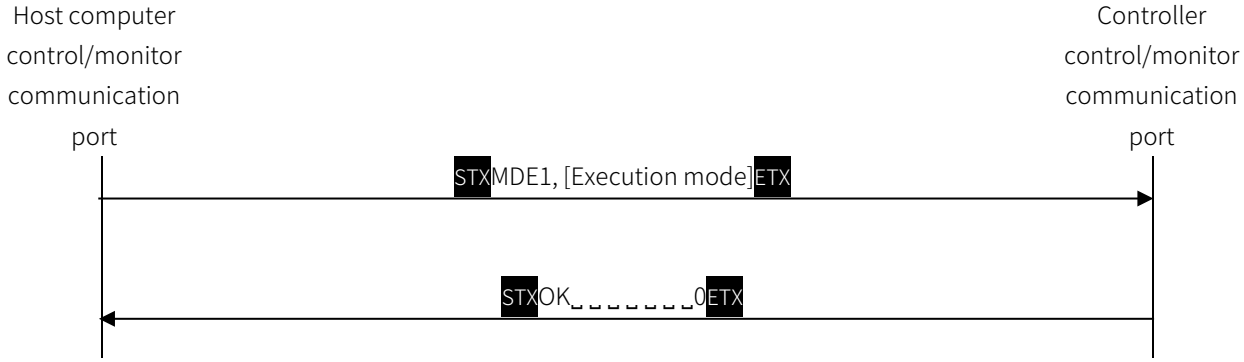
STXMDE1, [Execution mode]ETX

No.	Name	Size (bytes)	Description
1	Execution mode	1	The execution modes that can be set are as follows. 0: Continuous operation mode (CONT) When the program completes one cycle, the program repeats continuously until the next cycle is executed and a stop operation is performed. 1: Cycle operation mode (CYCLE) The program stops after one cycle is completed. 2: Step operation mode (STEP) The program stops after one line is executed. 3: Segment operation mode (SEG) The program stops each time an operation command is executed.

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command to set the controller execution mode to "CONT" is as follows.

STXMDE1, 0ETX
STXOK_ _ _ _ _ _ _ 0ETX

MER1 Read Memory Variable

Function

Monitorable command

This reads the memory variable.

Request Format

STXMER1, [Variable Type], [Element number]ETX

No.	Name	Size (bytes)	Description
1	Variable Type	1 Fixed length	Specify one of the following values for the variable type data. 0: Integer type 3: Coordinate type 1: Real number type 4: Position type 2: Load type 5: Axis type
2	Element number	Variable length	Specify the element number to be read. The range varies for each variable type as follows. Integer type: 1 to 1024 Real number type: 1 to 1024 Load type: 1 to 16 Coordinate type: 1 to 128 Position type: 1 to 1024 Axis type: 1 to 128

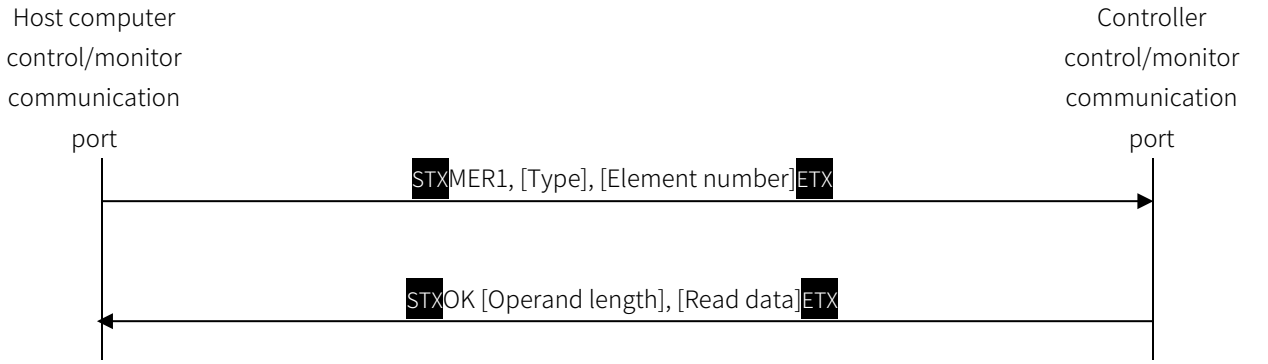
Response Format

STXOK [Operand length], [Read data]ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	Read data	Variable length	The read data varies for each variable type as follows.

Type	Read data		
Integer type	Format	Long type data	
	Quantity	1	
	Type	Long type data	long
Real number type	Format	Float type data	
	Quantity	1	
	Type	Float type	float
Load type	Format	Weight and center of gravity offset	
	Quantity	2	
	Type	Weight	float
		Center of gravity offset	float
Coordinate type	Format	X-coordinate value, Y-coordinate value, Z-coordinate value, C-coordinate value	
	Quantity	4	
	Type	X-coordinate value	float
		Y-coordinate value	float
		Z-coordinate value	float
		C-coordinate value	float
Position type	Format	X-coordinate value, Y-coordinate value, Z-coordinate value, C-coordinate value, T-coordinate value, Orientation	
	Quantity	6	
	Type	X-coordinate value	float
		Y-coordinate value	float
		Z-coordinate value	float
		C-coordinate value	float
		T-coordinate value	float
		Orientation	One of the following numbers 0: FREE 1: LEFTY 2: RIGHTY
Axis type	Format	1st axis, 2nd axis, 3rd axis, 4th axis, 5th axis	
	Quantity	5	
	Type	1st axis	float
		2nd axis	float
		3rd axis	float
		4th axis	float
		5th axis	float

Communication Example



Send/Receive Example

The command when reading the element number 0 of the integer type area for the memory variable is as follows.

STXMER1, 0, 1ETX
STXOK␣␣␣␣␣␣␣␣3, 10ETX

MEW1 Write Memory Variable

Function

This writes to the memory variable.

Request Format

STXMEW1, [Variable Type], [Element number], [Write data]ETX

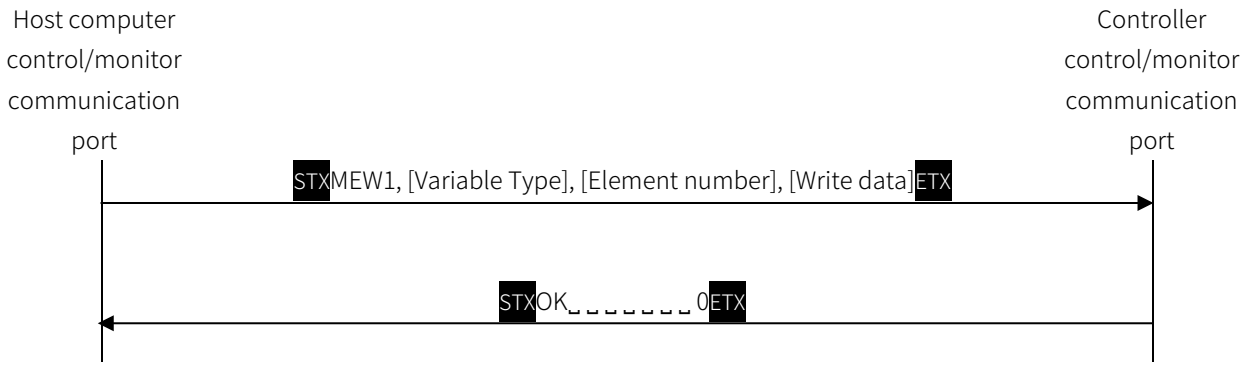
No.	Name	Size (bytes)	Description
1	Variable Type	1 Fixed length	Specify one of the following values for the variable type data. 0: Integer type 3: Coordinate type 1: Real number type 4: Position type 2: Load type 5: Axis type
2	Element number	Variable length	The range varies for each variable type as follows. Integer type: 1 to 1024 Real number type: 1 to 1024 Load type: 1 to 16 Coordinate type: 1 to 128 Position type: 1 to 1024 Axis type: 1 to 128
3	Write data	Variable length	The write data varies for each variable type as follows.

Type	Write data		
Integer type	Format	Long type data	
	Quantity	1	
	Type	Long type data	long
Real number type	Format	Float type data	
	Quantity	1	
	Type	Float type	float
Load type	Format	Weight and center of gravity offset	
	Quantity	2	
	Type	Weight	float
		Center of gravity offset	float
Coordinate type	Format	X-coordinate value, Y-coordinate value, Z-coordinate value, C-coordinate value	
	Quantity	4	
	Type	X-coordinate value	float
		Y-coordinate value	float
		Z-coordinate value	float
		C-coordinate value	float
Position type	Format	X-coordinate value, Y-coordinate value, Z-coordinate value, C-coordinate value, T-coordinate value, Orientation	
	Quantity	6	
	Type	X-coordinate value	float
		Y-coordinate value	float
		Z-coordinate value	float
		C-coordinate value	float
		T-coordinate value	float
		Orientation	One of the following numbers 0: FREE 1: LEFTY 2: RIGHTY
Axis type	Format	1st axis, 2nd axis, 3rd axis, 4th axis, 5th axis	
	Quantity	5	
	Type	1st axis	float
		2nd axis	float
		3rd axis	float
		4th axis	float
		5th axis	float

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command when writing 10 to the element number 0 of the integer type area for the memory variable is as follows.

STXMEW1, 0, 0, 10ETX
STXOK␣␣␣␣␣␣␣␣0ETX

MEMO

** The memory variable is retained even after the AC power is turned off.*

NOTICE

· Before sending this command, be sure to check the effect of the target memory variable on the program. The robot that has performed unexpected motion may collide with peripheral equipment.

MNT1 Obtain Maintenance Information

Function

Monitorable command

This transfers the specified maintenance information to the host computer.

Request Format

STXMNT1, [Maintenance information number]ETX

No.	Name	Size (bytes)	Description
1	Maintenance information number	Variable length	Specify one of the following numbers for the maintenance information number. 1: Motor total rotation count 2: Joint axis movement amount 3: Motor torque cumulative amount 4: Motor torque cumulative amount exceeding rating 5: Peak torque (This is cleared by a communication request.) 6: 3-second load factor 7: 1-minute load factor 8: 15-minute load factor 9: 2-hour load factor 10: Peak torque (This is not cleared by a communication request.)

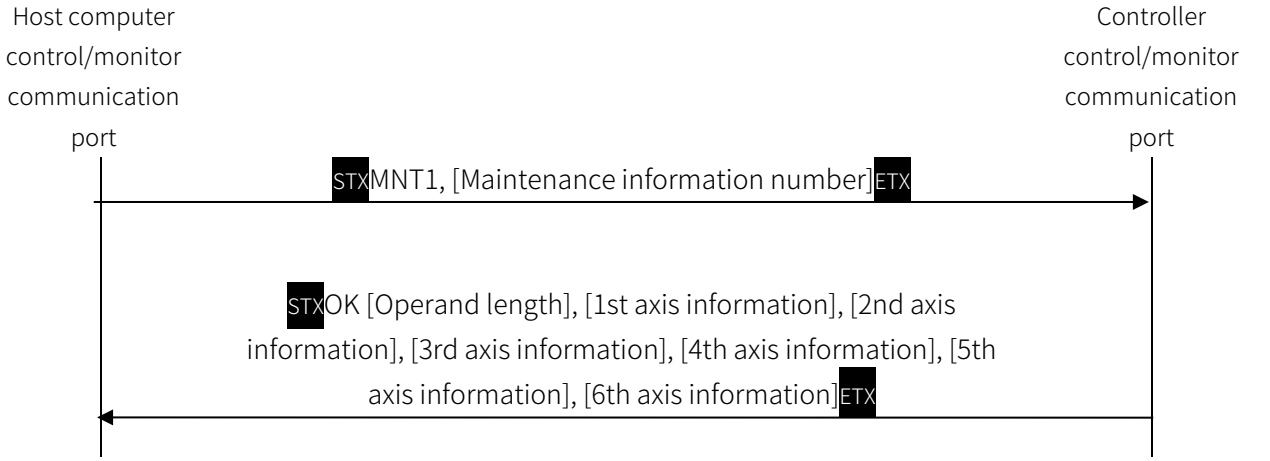
Response Format

STXOK [Operand length], [1st axis information], [2nd axis information], [3rd axis information], [4th axis information], [5th axis information], [6th axis information]ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)

No.	Name	Size (bytes)	Description
2 to 7	Axis information	Variable length	Information for 1st to 6th axes This is received as a real number up to 1 decimal place. * The response content varies depending on the maintenance information number. 1: Motor total rotation count 2: Joint axis movement amount 3: Motor torque cumulative amount 4: Motor torque cumulative amount exceeding rating 5: Peak torque 6: 3-second load factor 7: 1-minute load factor 8: 15-minute load factor 9: 2-hour load factor 10: Peak torque

Communication Example



Send/Receive Example

The command for obtaining the motor total rotation count is as follows.

STXMNT1, 1ETX
STXOK␣␣␣␣␣␣25, 46.4, 0.0, 0.0, 0.0, 0.0, 0.0ETX

MNT2 Obtain Maintenance Information

Function

Monitorable command

This transfers the specified maintenance information to the host computer.

Request Format

STXMNT2, [Maintenance information number]ETX

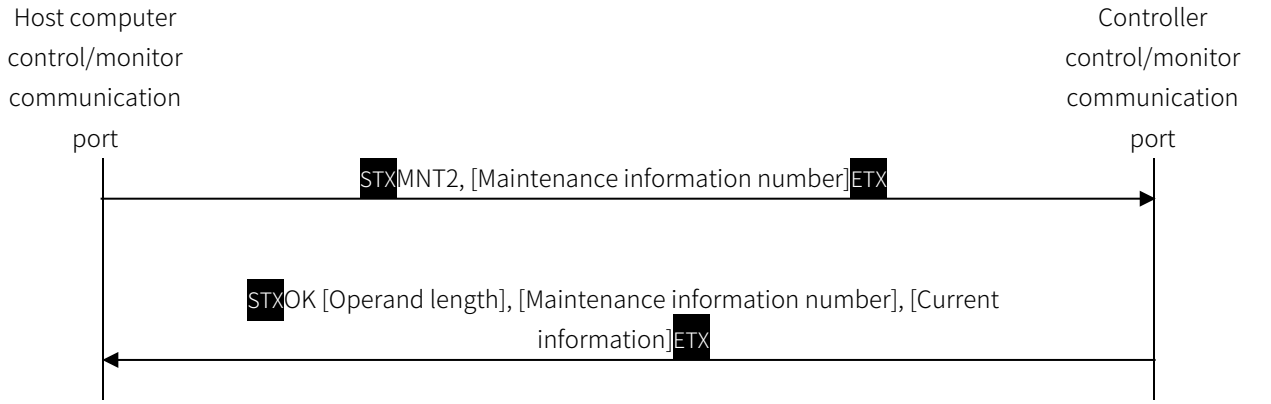
No.	Name	Size (bytes)	Description
1	Maintenance information number	Variable length	Specify one of the following numbers for the maintenance information number. 11: Total time 12: Power-on time 13: AC power-off time 14: Operation time 15: Failure time 16: Servo information (servo off count) 17: Total servo on time

Response Format

STXOK [Operand length], [Current information]ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	Current information	Variable length	Current information. This is received as a real number up to 1 decimal place. *The response content varies depending on the maintenance information number.

Communication Example



Send/Receive Example

The command for the total time is as follows.

STXMNT2, 11ETX
STXOK␣␣␣␣␣␣␣6, 100.0ETX

MNT3 Obtain Maintenance Information

Function

Monitorable command

This transfers the maintenance information to the host computer.

Request Format

STXMNT3ETX

Response Format

STXOK [Operand length] Motor total rotation count Joint axis movement amount Motor torque cumulative amount Motor torque cumulative amount exceeding rating Peak torque 3-second load factor 1-minute load factor 15-minute load factor 2-hour load factor CR[Total time], [Power-on time], [AC power-off time], [Operation time], [Failure time], [Servo information (servo off count)], [Total servo on time]ETX

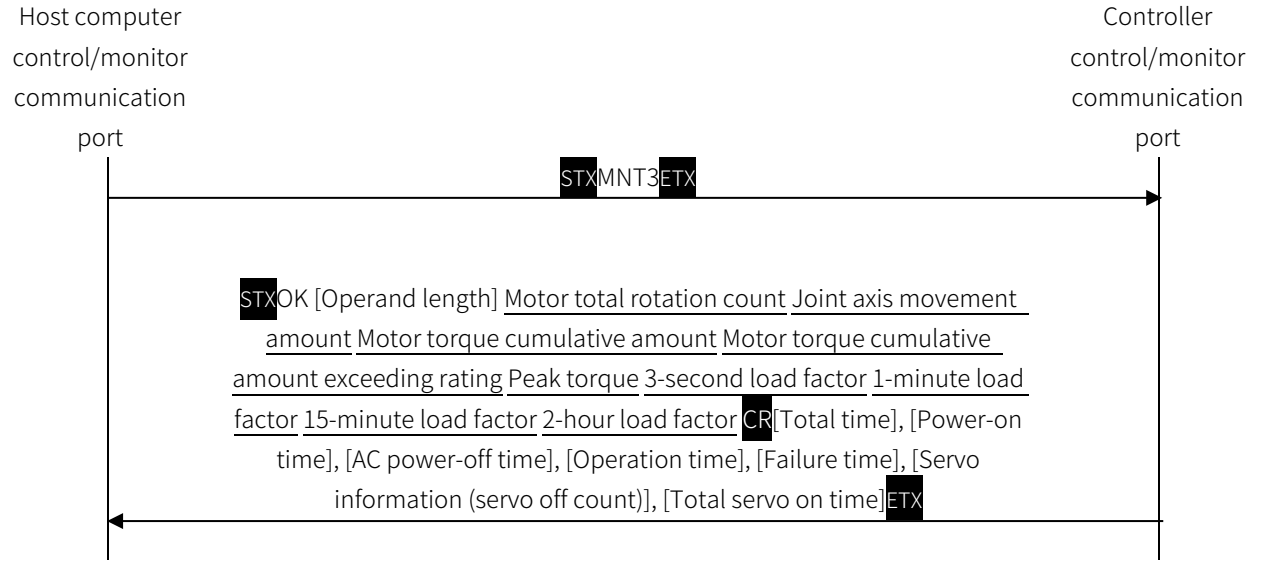
The following information is stored in the Motor total rotation count, Joint axis movement amount, Motor torque cumulative amount, Motor torque cumulative amount exceeding rating, Peak torque, 3-second load factor, 1-minute load factor, 15-minute load factor, and 2-hour load factor.

CR[1st axis information], [2nd axis information], [3rd axis information], [4th axis information], [5th axis information], [6th axis information]

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	Axis information	Variable length	Current information. This is received as a real number up to 1 decimal place. * The response content varies depending on the item.
3	Total time	Variable length	This is received as a real number up to 1 decimal place.
4	Power-on time	Variable length	This is received as a real number up to 1 decimal place.
5	AC power-off time	Variable length	This is received as a real number up to 1 decimal place.
6	Operation time	Variable length	This is received as a real number up to 1 decimal place.

7	Failure time	Variable length	This is received as a real number up to 1 decimal place.
8	Servo information (Servo off count)	Variable length	This is received as a real number up to 1 decimal place.
9	Total servo on time	Variable length	This is received as a real number up to 1 decimal place.

Communication Example



Send/Receive Example

The command for obtaining the maintenance information is as follows.

STXMNT3ETX

STXOK␣␣␣␣␣349,

29586.8, 24236.3, 2983.5, 931.7, 0.0, 0.0, 181677.7, 138606.4, 60292.6, 24876.3, 0.0, 0.0, 62557150.1, 46363774.1, 33706444.3, 6957018.5, 0.0, 0.0, 3894799.3, 2730865.2, 60350.9, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 542907668.0, 47737.9, 542859931.0, 4241.3, 1457.2, 127.0, 5598.7ETX

MVP1 Start Teaching Point Movement (MOVE) (PTP Movement)

Function

This moves to the teaching point using PTP movement.
To continue execution of teaching point movement, send the MVXC command (continue teaching point movement and obtain status command) within 200 ms. If the controller does not receive the MVXC command within 200 ms, it is determined to be a timeout, and an automatic stop is performed.
To immediately abort teaching point movement, send the brake (immediate stop) command.
If the request cannot be received, an error response is returned.

Request Format

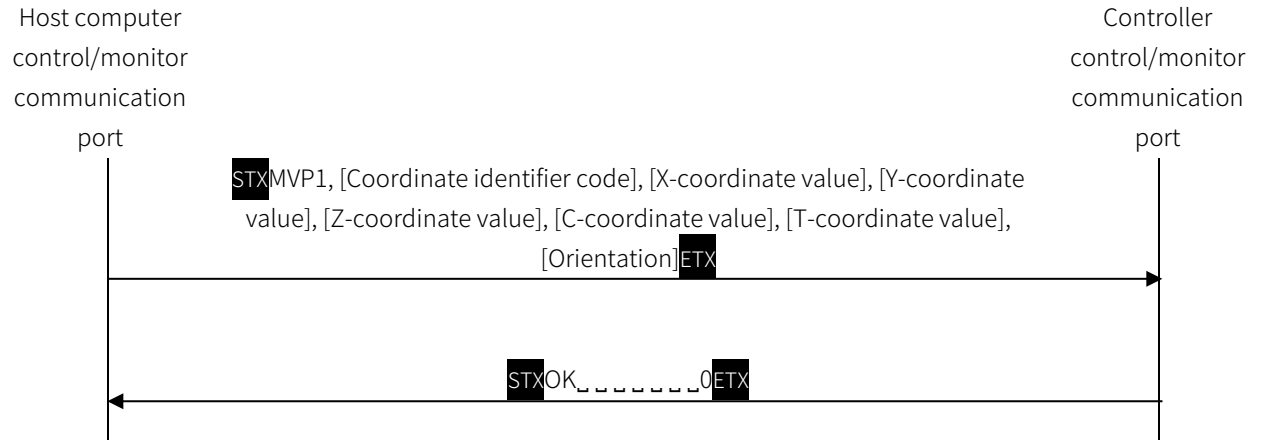
STXMVP1, [Coordinate identifier code], [X-coordinate value], [Y-coordinate value], [Z-coordinate value], [C-coordinate value], [T-coordinate value], [Orientation]ETX

No.	Name	Size (bytes)	Description
1	Coordinate identifier code	1 Fixed length	Specify one of the following numbers for the coordinate identifier code. 0: Normal movement operation 1: Move Z-axis at current position
2 to 6	Coordinate value	Variable length	Specify the coordinate for the teaching point movement as a real number up to 3 decimal places. The axes use the following units. X-axis: mm Y-axis: mm Z-axis: mm C-axis: deg T-axis: Check the specifications of the device.
7	Orientation	1 Fixed length	Specify one of the following characters for the teaching orientation. 0: FREE 1: LEFTY 2: RIGHTY

Response Format

STXOK_ _ _ _ _ _ _0ETX

Communication Example



Send/Receive Example

Teaching point conditions

Coordinate identifier code	: Normal movement operation
X-axis	: 300.0
Y-axis	: 370.0
Z-axis	: 700.0
C-axis	: 160.0
Orientation	: FREE

The command when teaching point movement is performed using the above teaching point conditions is as follows.

STXMVP1, 0, 300.0, 370.0, 700.0, 160.0, 0ETX
STXOK␣␣␣␣␣␣␣␣0ETX
STXMVXCETX
STXOK␣␣␣␣␣␣␣␣2, 1ETX
STXMVXCETX
STXOK␣␣␣␣␣␣␣␣2, 0ETX

MEMO

** When an operation instruction is received, this command returns an immediate response.*

NOTICE

- Before sending this command, be sure to check that the target position is set correctly.
- Be sure to always check the selected guidance speed before sending this command.
- Before sending this command, be sure to check that the selected coordinates are set correctly.
- Before sending this command, be sure to check that the selected tool coordinates and work coordinates are set correctly.
  The robot that has performed unexpected motion may collide with peripheral equipment.

MVP2 Start Teaching Point Movement (PTP Movement)

Function

This moves to the teaching joint using PTP movement.
To continue execution of teaching point movement, send the MVXC command (continue teaching point movement and obtain status command) within 200 ms. If the controller does not receive the MVXC command within 200 ms, it is determined to be a timeout, and an automatic stop is performed.
To immediately abort teaching point movement, send the brake (immediate stop) command.
If the request cannot be received, an error response is returned.

Request Format

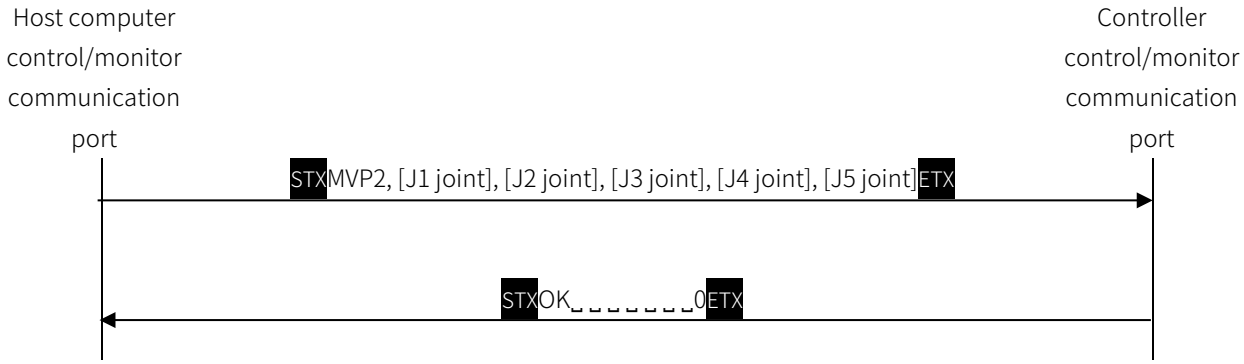
STXMVP2, [J1 joint], [J2 joint], [J3 joint], [J4 joint], [J5 joint]ETX

No.	Name	Size (bytes)	Description
1 to 5	Joint value	Variable length	Specify the teaching point joint for the movement target as a real number up to 3 decimal places. The joints use the following units. J1 joint: deg J2 joint: deg J3 joint: mm J4 joint: deg J5 joint: deg

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

Teaching point conditions

J1 joint	: 100.0
J2 joint	: 000.0
J3 joint	: 000.0
J4 joint	: 000.0
J5 joint	: 000.0

The command when teaching point movement is performed using the above teaching point conditions is as follows.

STXMVP2, 100.0, 000.0, 000.0, 000.0, 000.0ETX
STXOK_ _ _ _ _ _ _ _0ETX
STXMVXCETX
STXOK_ _ _ _ _ _ _ _2, 1ETX
STXMVXCETX
STXOK_ _ _ _ _ _ _ _2, 0ETX

MEMO

* *When an operation instruction is received, this command returns an immediate response.*

NOTICE

- Before sending this command, be sure to check that the target position is set correctly.
- Be sure to always check the selected guidance speed before sending this command.
- Before sending this command, be sure to check that the selected coordinates are set correctly.
- Before sending this command, be sure to check that the selected tool coordinates and work coordinates are set correctly.
  The robot that has performed unexpected motion may collide with peripheral equipment.

MVS1 Start teaching point movement (MOVES) (CP movement)

Function

This moves to the teaching point using CP movement.
To continue execution of teaching point movement, send the MVXC command (continue teaching point movement and obtain status command) within 200 ms. If the controller does not receive the MVXC command within 200 ms, it is determined to be a timeout, and an automatic stop is performed.
To immediately abort teaching point movement, send the brake (immediate stop) command.
If the request cannot be received, an error response is returned.

Request Format

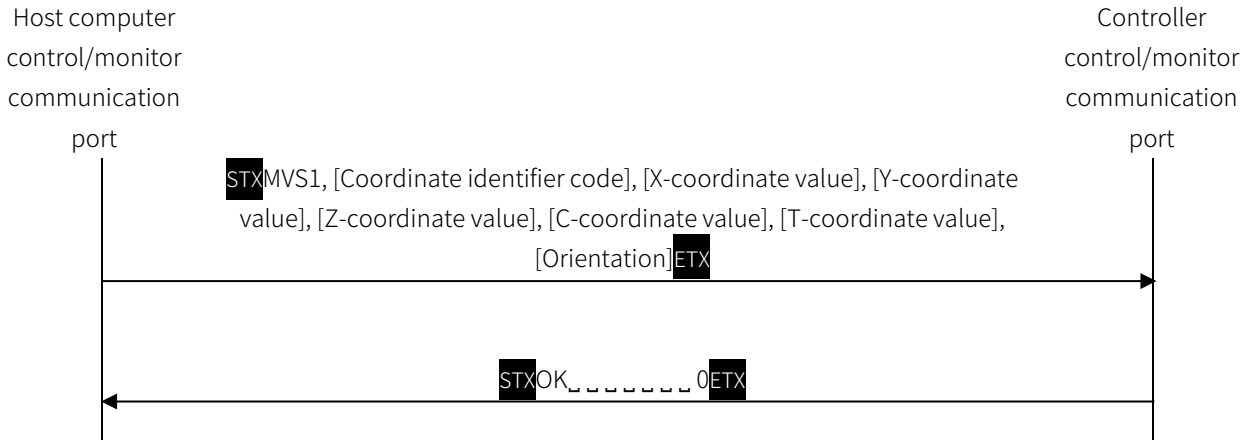
STXMVS1, [Coordinate identifier code], [X-coordinate value], [Y-coordinate value], [Z-coordinate value], [C-coordinate value], [T-coordinate value], [Orientation]ETX

No.	Name	Size (bytes)	Description
1	Coordinate identifier code	1 Fixed length	Specify one of the following numbers for the coordinate identifier code. 0: Normal movement operation 1: Move Z-axis at current position
2 to 6	Coordinate value	Variable length	Specify the coordinate for the teaching point movement as a real number up to 3 decimal places. The axes use the following units. X-axis: mm Y-axis: mm Z-axis: mm C-axis: deg T-axis: Check the specifications of the device.
7	Orientation	1 Fixed length	Specify one of the following characters for the teaching orientation. 0: FREE 1: LEFTY 2: RIGHTY

Response Format

STXOK_ _ _ _ _ _ _ _0ETX

Communication Example



Send/Receive Example

Teaching point conditions

Coordinate identifier code	: Normal movement operation
X-axis	: 300.0
Y-axis	: 370.0
Z-axis	: 700.0
C-axis	: 160.0
Orientation	: FREE

The command when teaching point movement is performed using the above teaching point conditions is as follows.

STXMVS1, 0, 300.0, 370.0, 700.0, 160.0, 0ETX
STXOK_ _ _ _ _ _ _ _0ETX
STXMVXCETX
STXOK_ _ _ _ _ _ _ _2, 1ETX
STXMVXCETX
STXOK_ _ _ _ _ _ _ _2, 0ETX

MEMO

** When an operation instruction is received, this command returns an immediate response.*

NOTICE

- Before sending this command, be sure to check that the target position is set correctly.
- Be sure to always check the selected guidance speed before sending this command.
- Before sending this command, be sure to check that the selected coordinates are set correctly.
- Before sending this command, be sure to check that the selected tool coordinates and work coordinates are set correctly.

  The robot that has performed unexpected motion may collide with peripheral equipment.

MVXC Continue Teaching Point Movement and Obtain Status

Function

This continues execution of the teaching point movement operation.
After starting the each teaching point movement, send this command within 200 ms from completion of movement to the specified teaching point. If the controller does not receive this command within 200 ms, it is determined to be a timeout, and an automatic stop is performed.
To abort teaching point movement, send the abort command.
If the request cannot be received, an error response is returned.

Request Format

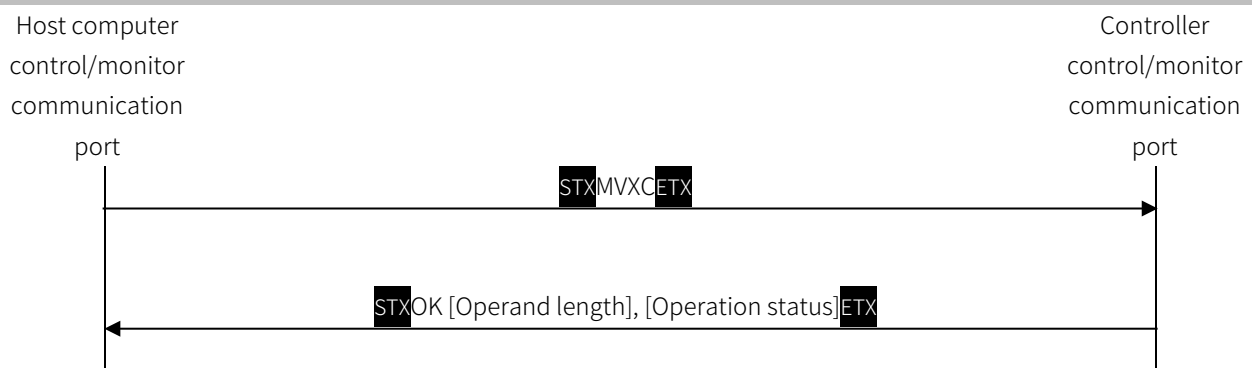
STXMVXCETX

Response Format

STXOK [Operand length], [Operation status]ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	Operation status	Variable length	Status of teaching point movement operation that was last executed 1 or more: Percentage of remaining operations (1 to 100) 0: Operation completed -1: Terminated by Stop -2: Terminated by Break

Communication Example



Send/Receive Example

Teaching point conditions

Coordinate identifier code	: Normal movement operation
X-axis	: 300.0
Y-axis	: 370.0
Z-axis	: 700.0
C-axis	: 160.0 0
Orientation	: FREE

The command when teaching point movement is performed using the above teaching point conditions is as follows.

STXMVP1, 0, 300.0, 370.0, 700.0, 160.0, 0ETX
STXOK_ _ _ _ _ _ _0ETX
STXMVXCETX
STXOK_ _ _ _ _ _ _2, 1ETX
STXMVXCETX
STXOK_ _ _ _ _ _ _2, 0ETX

MEMO

** When an operation instruction is received, this command returns an immediate response.*

NOTICE

· Send this command in a state where the emergency stop switch can be accessed at any time.
The robot that has performed unexpected motion may collide with peripheral equipment.

OVR1 Set Override

Function

This sets the override of the controller.

If the request cannot be received, an error response is returned.

Request Format

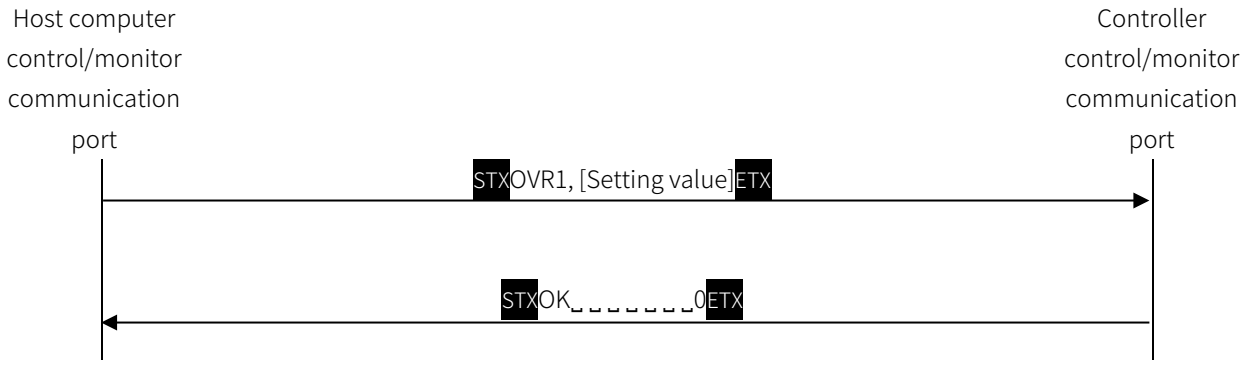
STXOVR1, [Setting value]ETX

No.	Name	Size (bytes)	Description
1	Setting value	Variable length	This sets the operation speed. Set to a number from 0 to 100.

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command to set the override to 20 is as follows.

STXOVR1, 20ETX

STXOK␣␣␣␣␣␣␣␣0ETX

PLC1 TCmini (Internal PLC) Monitor

Function

Monitorable command

This transfers the register information of the TCmini (internal PLC) installed in the controller to the host computer.

Request Format

STXPLC1, [Register name]ETX

No.	Name	Size (bytes)	Description
1	Register name	1 Fixed length	Set one of the following for the register name. X: Input relay Y: Output relay G: Interface relay (input) H: Interface relay (output)

Response Format

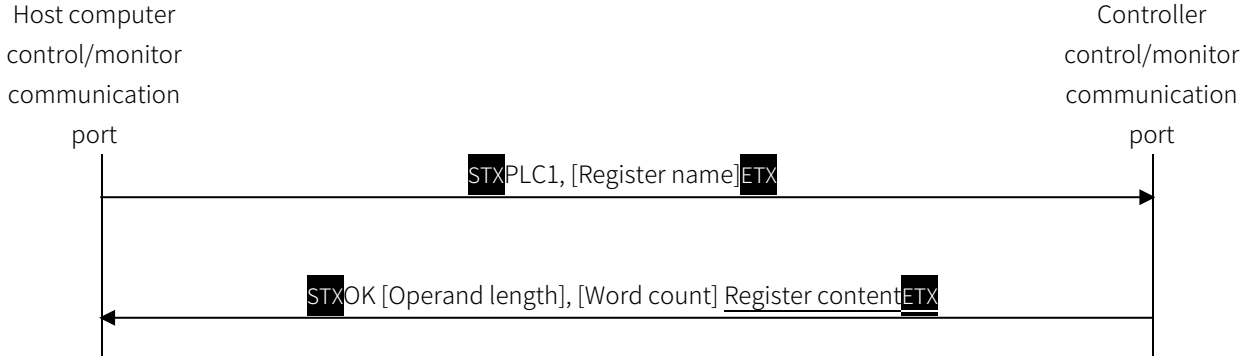
STXOK [Operand length], [Word count] Register content ETX

In the register content, the following information is stored consecutively for the quantity in [Word count].

, [Register value]

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	Word count	Variable length	Word count for register for each register type
3	Register value	4 Fixed length	Hexadecimal character string 0000 to FFFF

Communication Example



Send/Receive Example

The command when obtaining the R00W register information for the TCmini (internal PLC) register is as follows.

STXPLC1, XETX

STXOK␣␣␣␣␣␣83, 16, 0000, 001D, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000ETX

PLC2 Obtain TCmini (Internal PLC) Status

Function

Monitorable command

This transfers the status information of the TCmini (internal PLC) installed in the controller to the host computer.

Request Format

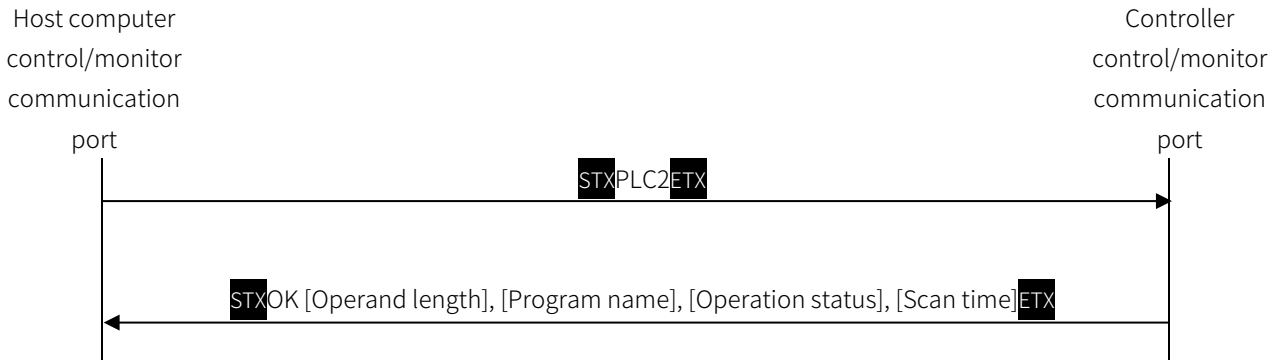
STXPLC2ETX

Response Format

STXOK [Operand length], [Program name], [Operation status], [Scan time]ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	Program name	16 Fixed length	This stores the name of the program that is executed by TCmini (internal PLC).
3	Operation status	1 Fixed length	This stores the operation status of the TCmini (internal PLC). 0: Stopped 1: Running
4	Scan time	Variable length	This stores the scan time of the TCmini (internal PLC).

Communication Example



Send/Receive Example

The command for obtaining the currently-registered base, work, and tool coordinate information is as follows.

STXPLC2ETX

STXOK␣␣␣␣␣␣21, TS5KSEQ00␣␣␣␣␣␣␣␣, 1, 2ETX

PMA1 Register Periodic Monitor

Function

This registers the command that is monitored periodically.

If the request cannot be received, an error response is returned.

For details on the usage procedure, see "4.1.2 Periodic Monitor Protocol".

Request Format

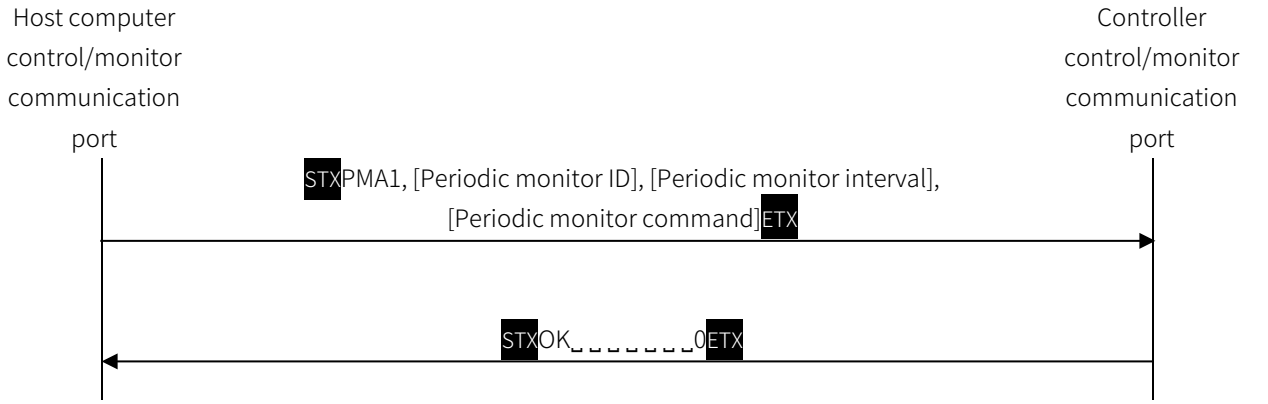
STXPMA1, [Periodic monitor ID], [Periodic monitor interval], [Periodic monitor command]ETX

No.	Name	Size (bytes)	Description
1	Periodic monitor ID	Variable length	ID to be registered by the periodic monitor (0 to 127) To execute multiple commands, set the periodic monitor ID to a unique number. If the periodic monitor cycle is duplicated with other commands, the data that was stored from the response results of the command with the smaller periodic monitor ID is sent from the periodic monitor port as the periodic monitor results.
2	Periodic monitor interval	Variable length	Monitor interval of command registered in the periodic monitor The units are milliseconds. Set in 10 millisecond steps in the range from 100 to 10, 000 milliseconds. The digit in the 1 millisecond place is discarded.
3	Periodic monitor command	Variable length	Command message that is periodically monitored See the respective command for the command format. Do not include STXETX in the periodic monitor command.

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command for obtaining the general-purpose input status at 100 ms intervals and the robot tip speed at 200 ms intervals is as follows.

STXPMA1, 0, 100, IOR1ETX
STXOK␣␣␣␣␣␣␣␣0ETX
STXPMA1, 1, 200, VLW2, 0ETX
STXOK␣␣␣␣␣␣␣␣0ETX

PMD1 Delete Periodic Monitor

Function

This deletes the command that is monitored periodically.

If the request cannot be received, an error response is returned.

For details on the usage procedure, see "4.1.2 Periodic Monitor Protocol".

Request Format

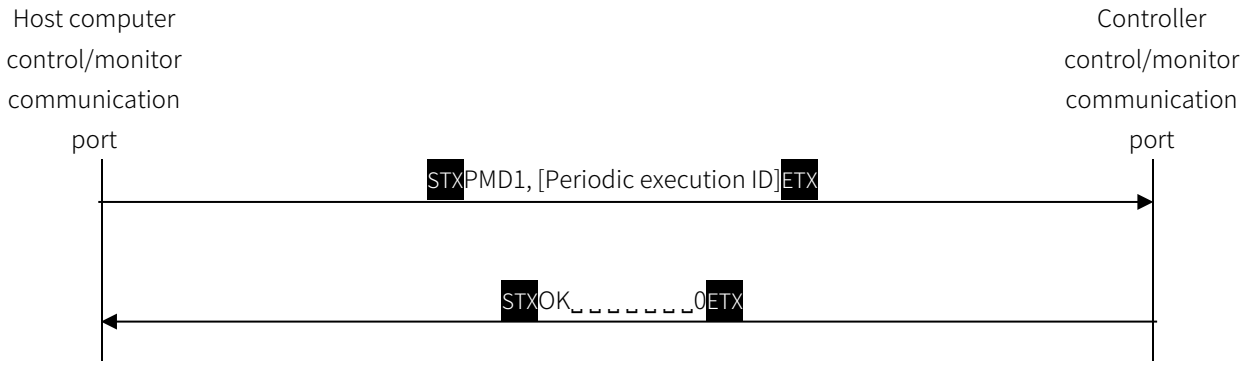
STXPMD1, [Periodic execution ID]ETX

No.	Name	Size (bytes)	Description
1	Periodic monitor ID	Variable length	ID to be registered by the periodic monitor (-1 to 127) This deletes the periodic monitor command of the specified ID. If -1 is specified, this deletes all periodic monitor commands.

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command for deleting the command registered to periodic monitor ID3 is as follows.

STXPMD1, 3ETX
STXOK␣␣␣␣␣␣␣␣0ETX

PME1 End Periodic Monitor Operation

Function

This ends the periodic monitor operation.

If the request cannot be received, an error response is returned.

For details on the usage procedure, see "4.1.2 Periodic Monitor Protocol".

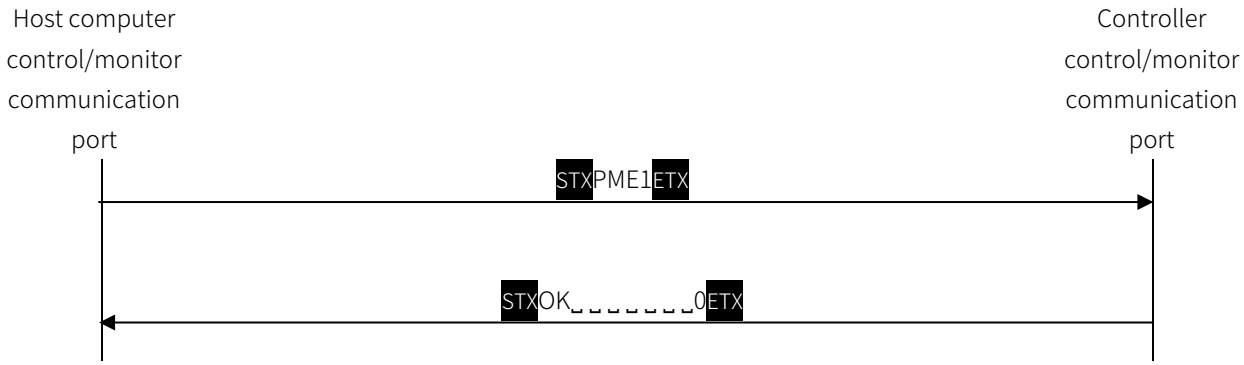
Request Format

STXPME1ETX

Response Format

STXOK_ _ _ _ _ _ _ _0ETX

Communication Example



Send/Receive Example

The command for ending the periodic monitor operation is as follows.

STXPME1ETX
STXOK_ _ _ _ _ _ _ _0ETX

PMF1 Instant Execute Periodic Monitor

Function

This instantly executes the command registered for the periodic monitor regardless of the periodic monitor interval.

If the periodic monitor operation has not been started, it is executed only once.

The controller sends the execution results to the periodic monitor port.

If the request cannot be received, an error response is returned.

Request Format

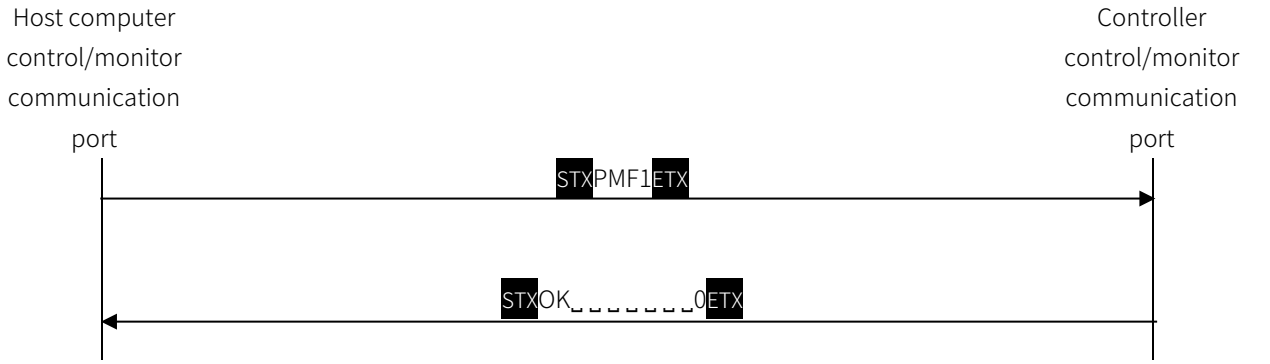
STXPMF1ETX

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)

Communication Example



Send/Receive Example

The command for instantly executing the periodic monitor is as follows.

STXPMF1ETX
STXOK␣␣␣␣␣␣␣␣0ETX

PMR1 Periodic Monitor Results

Function

This command is a response only.

This transfers the periodic monitor results to the host computer.

When a Start Periodic Monitor Operation command is issued, the response results for the command registered by the periodic monitor registration command can be obtained periodically from the periodic monitor port.

For details on the usage procedure, see "4.1.2 Periodic Monitor Protocol".

Response Format

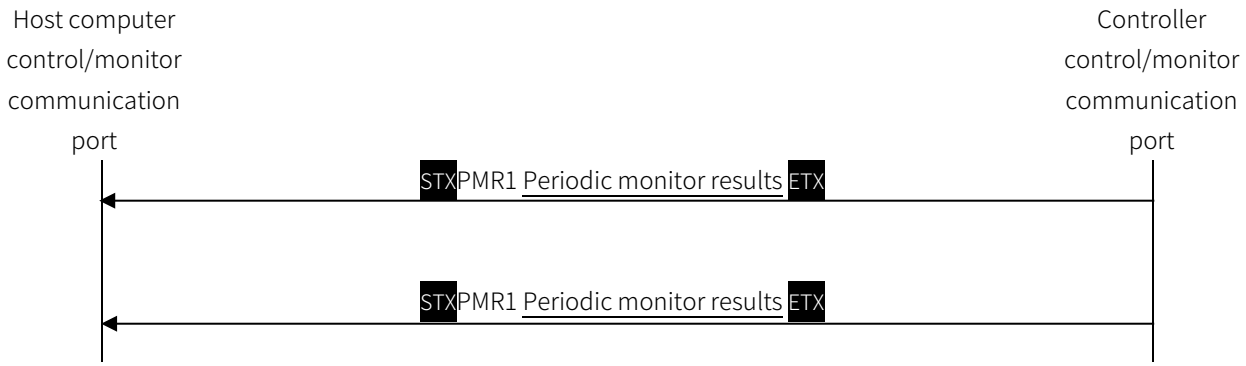
STXPMR1 Periodic monitor results ETX

In the periodic monitor results, a plurality of the following item will be stored.

CR[Periodic execution ID], [Command response results]

No.	Name	Size (bytes)	Description
1	Periodic monitor ID	Variable length	Periodic monitor ID that was set when registering and linked to the command response results
2	Command response results	Variable length	The execution results for the command registered to the periodic monitor ID are stored. For details, see the respective response format. STX ETX is not included in the command response results.

Communication Example



Send/Receive Example

The command for obtaining the general-purpose input status at 100 ms intervals and the robot tip speed at 200 ms intervals is as follows.

STXPMR1CR0, OK␣␣␣␣␣␣17, FEDCBA9876543210CR1, OK␣␣␣␣␣␣␣6, 3.520ETX

(After 100 ms has elapsed)

STXPMR1CR0, OK␣␣␣␣␣␣17, FEDCBA9876543210ETX

(After 100 ms has elapsed)

STXPMR1CR0, OK␣␣␣␣␣␣17, FEDCBA9876543210CR1, OK␣␣␣␣␣␣␣8, 120.560ETX

PMS1 Start Periodic Monitor Operation

Function

This starts the periodic monitor operation.

If the request cannot be received, an error response is returned.

For details on the usage procedure, see "4.1.2 Periodic Monitor Protocol".

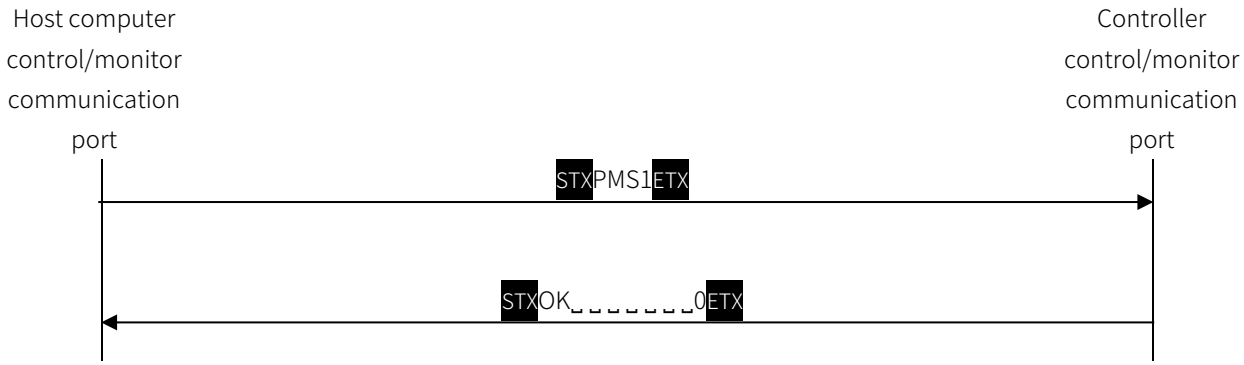
Request Format

STXPMS1ETX

Response Format

STXOK_ _ _ _ _ _ _ _0ETX

Communication Example



Send/Receive Example

The command for starting the periodic monitor operation is as follows.

STXPMS1ETX
STXOK_ _ _ _ _ _ _ _0ETX

PSJ1 Obtain Joint Position Information

Function

Monitorable command

This transfers the current value information of the joint to the host computer.

Request Format

STXPSJ1, [Coordinate system]ETX

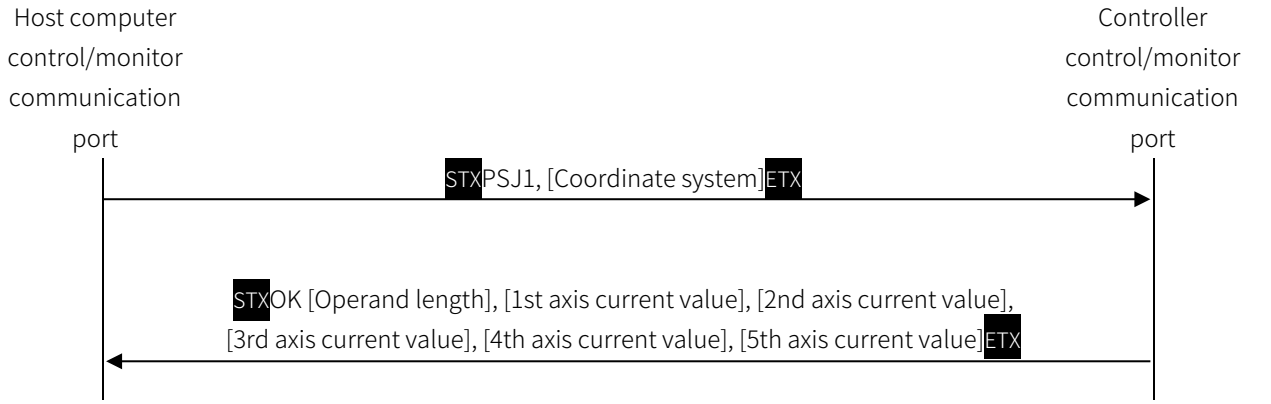
No.	Name	Size (bytes)	Description
1	Coordinate system	1 Fixed length	Specify one of the following numbers for the coordinate system. 0: Joint coordinates 10: Joint feedback coordinates

Response Format

STXOK [Operand length], [1st axis current value], [2nd axis current value], [3rd axis current value], [4th axis current value], [5th axis current value]ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2 to 5	Current value of each axis	9 Fixed length	Current values for 1st axis to 4th axis Units: degrees This is received as a real number up to 3 decimal places.
6	Current value for 5th axis	9 Fixed length	Current values for the 5th axis Units: degrees or position (deg or mm) This is received as a real number up to 3 decimal places.

Communication Example



Send/Receive Example

The command for obtaining the current value information for the joint coordinates is as follows.

STXPSJ1, 0ETX
STXOK␣␣␣␣␣␣86, ␣␣300.000, ␣␣370.000, ␣␣699.999, ␣␣170.000, -␣␣55.000, ␣␣160.000, ␣␣␣␣0.000, ␣␣␣␣0.000, 12111ETX

PSW1 Obtain World Position Information

Function

Monitorable command

This transfers the current value information in the specified coordinate system to the host computer.

Request Format

STXPSW1, [Coordinate system]ETX

No.	Name	Size (bytes)	Description
1	Coordinate system	1 Fixed length	Specify one of the following numbers for the coordinate system. 2: Work coordinates 3: World coordinates 12: Work feedback coordinates 13: World feedback coordinates

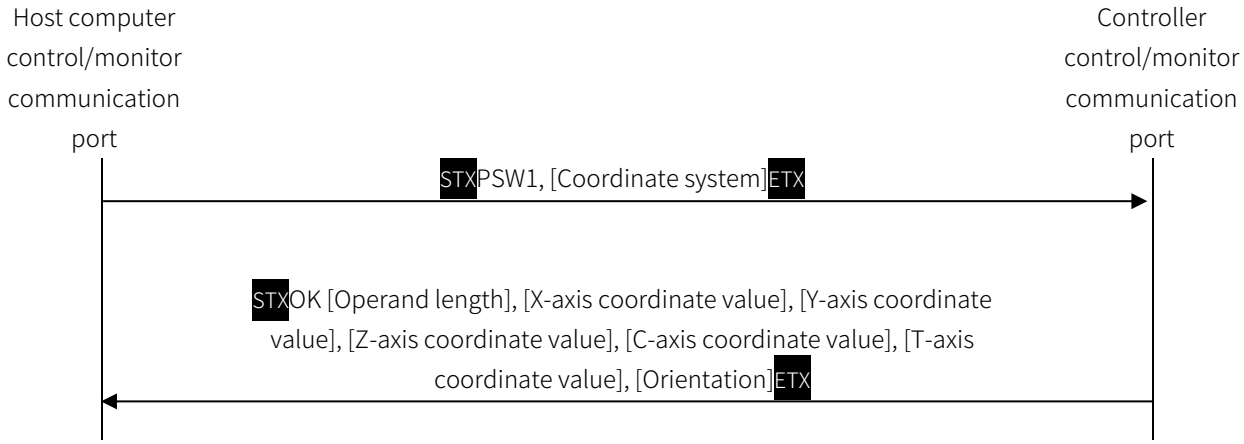
Response Format

STXOK [Operand length], [X-axis coordinate value], [Y-axis coordinate value], [Z-axis coordinate value], [C-axis coordinate value], [T-axis coordinate value], [Orientation]ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2 to 4	X-axis coordinate value Y-axis coordinate value Z-axis coordinate value	Variable length	Current X, Y, and Z-coordinate values. Units: Position (mm) This is received as a real number up to 3 decimal places.
5	C-axis coordinate value	Variable length	Current C-coordinate value. Units: Position (mm) This is received as a real number up to 3 decimal places.
6	T-axis coordinate value	Variable length	Current T-coordinate value. Units: degrees or position (deg or mm) This is received as a real number up to 3 decimal places.

7	Orientation	1 Fixed length	One of the following numbers 0: FREE 1: LEFTY 2: RIGHTY

Communication Example



Send/Receive Example

The command for obtaining the current value information is as follows.

STXPSW1, 1ETX

STXOK␣␣␣␣␣␣86, ␣␣300.000, ␣␣370.000, ␣␣699.999, ␣␣170.000, -␣␣55.000, ␣␣160.000, ␣␣␣␣0.000, ␣␣␣␣0.000, 1ETX

RCH1 Determine If Specified Position Reached

Function

This determines whether the teaching point can be reached using PTP movement.

Request Format

STXRCH1, [X-coordinate value], [Y-coordinate value], [Z-coordinate value], [C-coordinate value], [T-coordinate value], [Orientation]ETX

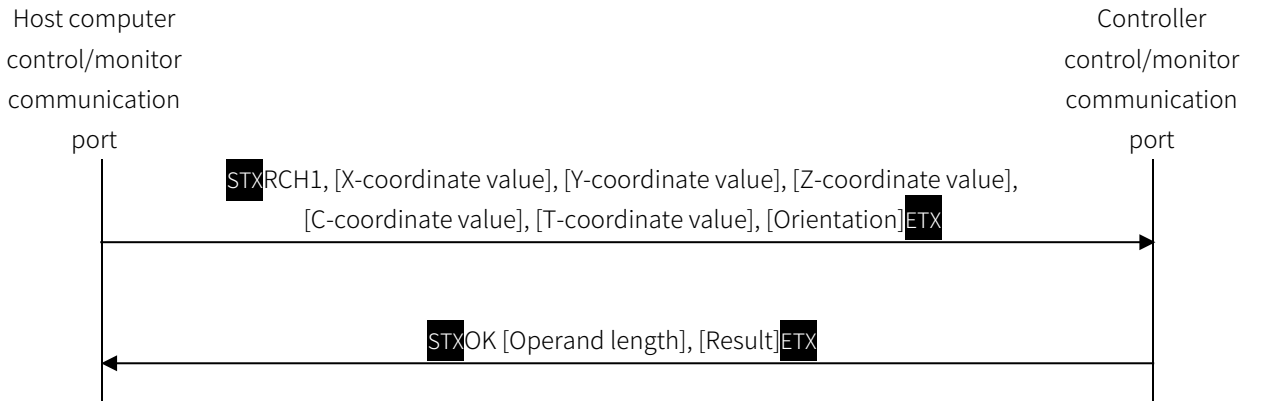
No.	Name	Size (bytes)	Description
1 to 5	Coordinate value	Variable length	Specify the coordinate for the teaching point movement as a real number up to 3 decimal places. The axes use the following units. X-axis: mm Y-axis: mm Z-axis: mm C-axis: deg T-axis: Check the specifications of the device.
7	Orientation	1 Fixed length	Specify one of the following characters for the teaching orientation. 0: FREE 1: LEFTY 2: RIGHTY

Response Format

STXOK [Operand length], [Result]ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	Result	1 Fixed length	1: Can reach teaching point 0: Cannot reach teaching point

Communication Example



Send/Receive Example

Teaching point conditions

X-axis	: 300.0
Y-axis	: 370.0
Z-axis	: 700.0
C-axis	: 160.0
Orientation	: FREE

The following shows the case when the teaching point can be reached using the above teaching point conditions.

STXRCH1, 300.0, 370.0, 700.0, 160.0, 0ETX
STXOK␣␣␣␣␣␣␣␣2, 1ETX

RCH2 Determine If Specified Joint Angle Reached

Function

This determines whether the specified joint angle can be reached using PTP movement.

Request Format

STXRCH2, [J1 position], [J2 position], [J3 position], [J4 position], [J5 position]ETX

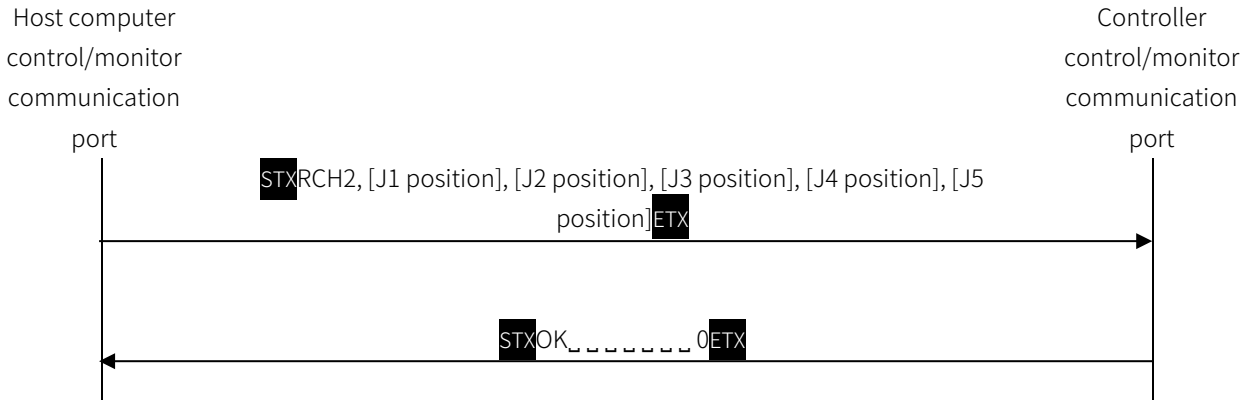
No.	Name	Size (bytes)	Description
1 to 5	Joint value	Variable length	Specify the joint position for the movement target as a real number up to 3 decimal places. The axes use the following units. J1 position: deg J2 position: deg J3 position: mm J4 position: deg J5 position: deg or mm

Response Format

STXOK [Operand length], [Result]ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed Length	Byte size after [Operand length] until ETX (not including ETX)
2	Result	1 Fixed Length	1: Can reach teaching point 0: Cannot reach teaching point

Communication Example



Send/Receive Example

Teaching point conditions

J1 position : 100.0
J2 position : -50.0
J3 position : 0.0
J4 position : 0.0
J5 position : 0.0

The following shows the case when the teaching point can be reached using the above teaching point conditions.

STXRCH2, 100.0, -50.0, 0.0, 0.0, 0.0ETX
STXOK␣␣␣␣␣␣␣␣2, 1ETX

REN1 Rename File

Function

This renames a file in the controller.

If the request cannot be received, an error response is returned.

Request Format

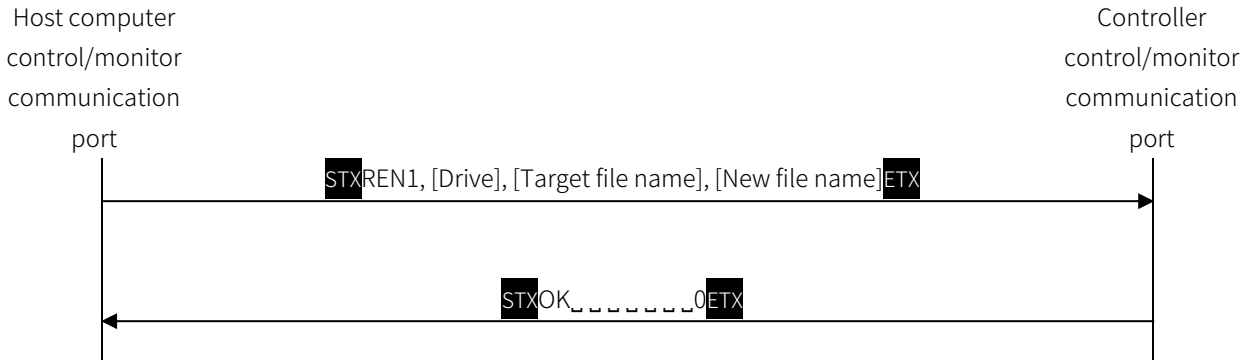
STXREN1, [Drive], [Target file name], [New file name]ETX

No.	Name	Size (bytes)	Description
1	Drive	1 Fixed length	Specifies the drive containing the file to be renamed. 1: ROM drive 2: RAM drive 3: SD card drive
2	Target file name	Variable length	Name of file to be renamed The file name must be in one of the following formats. · name.extension · name The name must be from 1 to 8 characters. The extension must be from 0 to 3 characters. When omitting the extension, the period (.) must also be omitted.
3	New file name	Variable length	New file name The file name must be in one of the following formats. · name.extension · name The name must be from 1 to 8 characters. The extension must be from 0 to 3 characters. When omitting the extension, the period (.) must also be omitted.

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command when renaming the file "PRG1" in the ROM drive of the controller to the file "PRG2" is as follows.

STXREN1, 1, PRG1, PRG2ETX
STXOK␣␣␣␣␣␣␣␣0ETX

NOTICE

- Do not rename parameter files.
  Otherwise, the robot that has performed unexpected motion may collide with peripheral equipment.
  Renaming the parameter file will reset the parameters.

RST1 Reset

Function

This resets the various statuses of the controller.

If the request cannot be received, an error response is returned.

Request Format

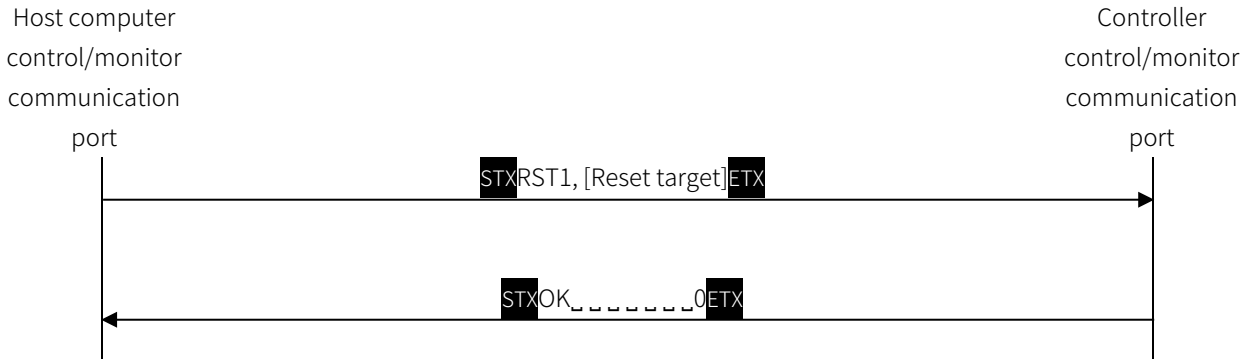
STXRST1, [Reset target]ETX

No.	Name	Size (bytes)	Description
1	Reset target	1 Fixed length	Specify one of the following characters for the reset target. 0: PROGRAM Sets the step number to 1 and resets all of the program information. 1: STEP The step number only is set to 1. The program information (variables, etc.) remains unchanged. 2: CYCLE This sets the step number to the specified label (RCYCLE). The program information remains unchanged. 3: SIGNAL This sets the digital output to non-active. 4: SEL This resets a target executable file.

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command to set all program information to the initial status is as follows.

STXRST1, 0ETX
STXOK_ _ _ _ _ _ _ _0ETX

MEMO

· *The command is not received unless operation is stopped.*

NOTICE

· Before sending a signal reset using this command, be sure to check the effect of the output signals.
The robot that has performed unexpected motion may collide with peripheral equipment.
Example: If the suction signal is turned off, the clamped object drops.

RUN1 Start Automatic Operation

Function

This starts automatic operation of a selected program.
When this command is sent again after the program is stopped by an automatic operation stop command, the robot restarts operation from the step following the step where it was stopped.
If the request cannot be received, an error response is returned.

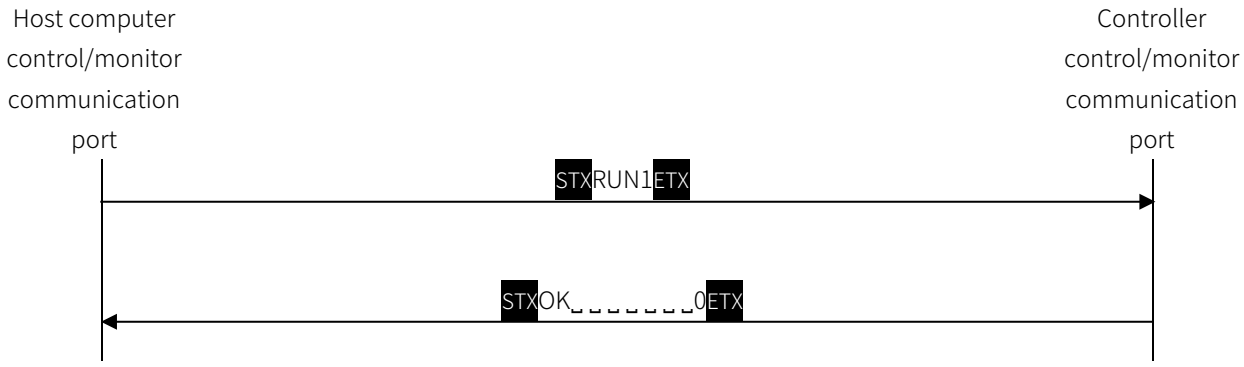
Request Format

STXRUN1ETX

Response Format

STXOK␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command to start automatic operation is as follows.

STXRUN1ETX
STXOK␣␣␣␣␣␣␣0ETX

NOTICE

- Before sending this command, be sure to check the program execution line number.
- Before sending this command, be sure to check the selected guidance coordinates and guidance speed.
- When tool coordinates are selected for the guidance coordinates, be sure to check the tool type and tool offset amount that is data of tool coordinates.
  The robot that has performed unexpected motion may collide with peripheral equipment.

⚠DANGER

- Never enter the hazard zone during automatic operation.
  Otherwise, a collision could occur due to an unexpected movement by the robot.

SAL1 Save Alarm History to SD Card

Function

This saves the alarm history to an SD card loaded in the controller.

The file name is "ALARM.TXT", and it is saved by overwriting.

Request Format

STXSAL1ETX

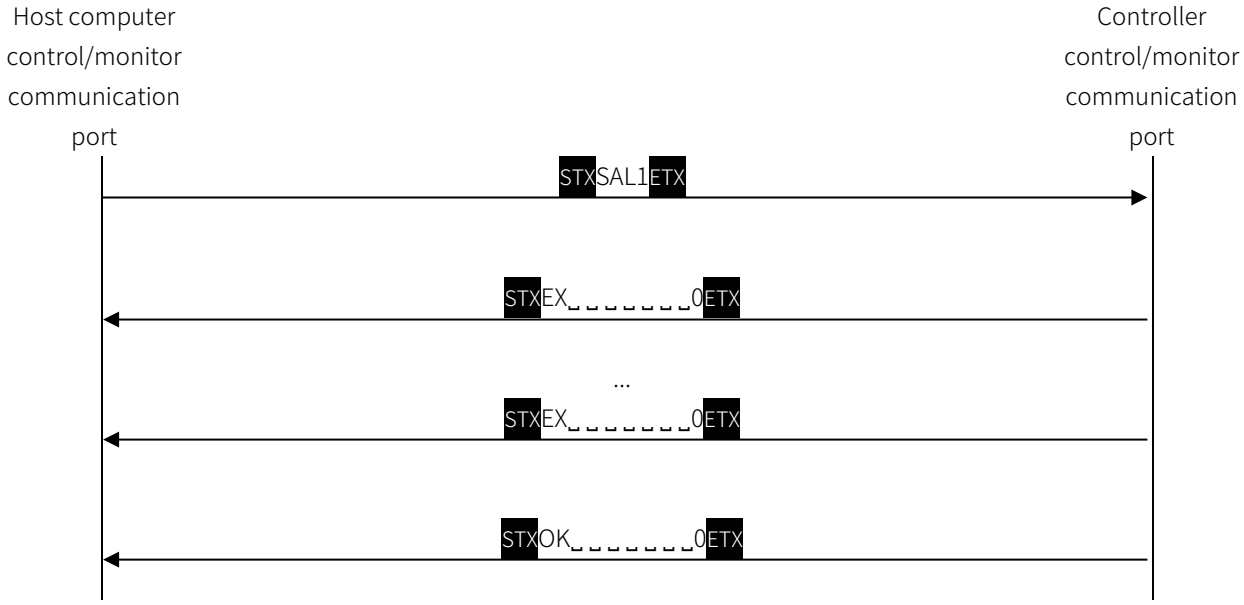
Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

STXEX␣␣␣␣␣␣␣␣0ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)

Communication Example



Send/Receive Example

The command when saving the alarm history to an SD card is as follows.

STXSAL1ETX

STXEX␣␣␣␣␣␣␣␣0ETX

STXOK␣␣␣␣␣␣␣␣0ETX

SEL1 Select Program

Function

This is used to select the program that is executed by automatic operation.
If the request cannot be received, an error response is returned.

Request Format

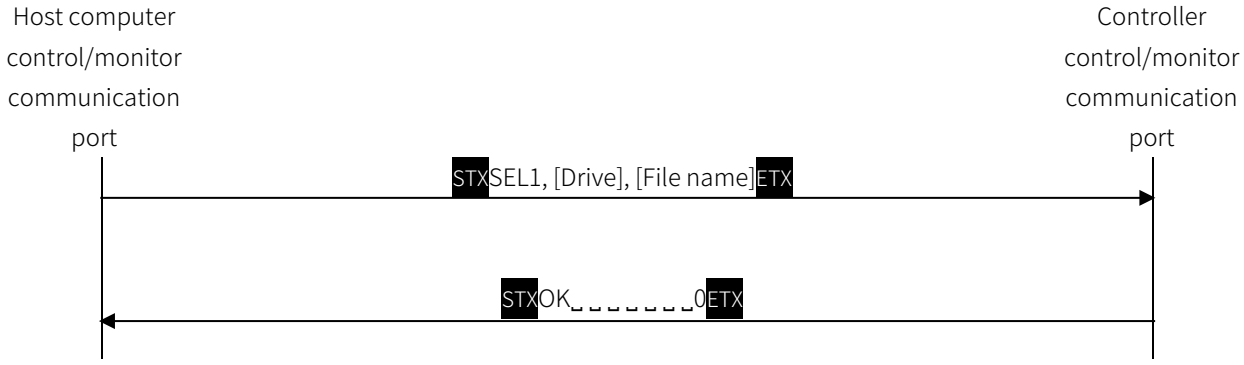
STXSEL1, [Drive], [File name]ETX

No.	Name	Size (bytes)	Description
1	Drive	1 Fixed length	This specifies the drive containing the file to be executed. 1: ROM drive 2: RAM drive 3: SD card drive
2	File name	Variable length	Name of file to be executed The file name must be in one of the following formats. · name.extension · name The name must be from 1 to 8 characters. The extension must be from 0 to 3 characters. When omitting the extension, the period (.) must also be omitted.

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command when selecting the program "PRG1" executed by automatic operation is as follows.

STXSEL1, 1, PRG1ETX

STXOK␣␣␣␣␣␣␣0ETX

SEL2 Select Program

Function

This is used to select the program that is executed by automatic operation.
If the request cannot be received, an error response is returned.
If the Select Program process takes longer than 2 seconds, the controller sends a "Process execution in progress" response command at 2-second intervals to indicate that the process is being executed.

Request Format

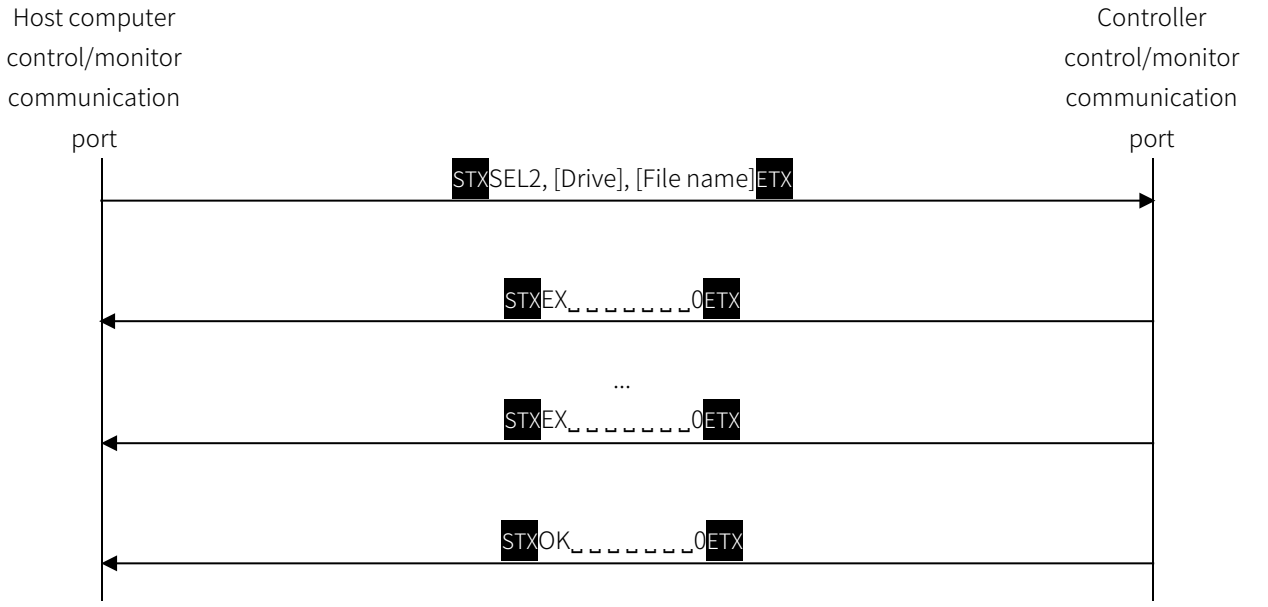
STXSEL2, [Drive], [File name]ETX

No.	Name	Size (bytes)	Description
1	Drive	1 Fixed length	This specifies the drive containing the file to be executed. 1: ROM drive 2: RAM drive 3: SD card drive
2	File name	Variable length	Name of file to be executed The file name must be in one of the following formats. · name.extension · name The name must be from 1 to 8 characters. The extension must be from 0 to 3 characters. When omitting the extension, the period (.) must also be omitted.

Response Format

STXOK␣␣␣␣␣␣␣0ETX
STXEX␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command when selecting the program "PRG1" executed by automatic operation is as follows.

STXSEL2, 1, PRG1ETX
STXEX_ _ _ _ _ _ _ _0ETX
STXOK_ _ _ _ _ _ _ _0ETX

SOF1 Servo Off

Function

This turns off the power for the servo in the controller.

If the request cannot be received, an error response is returned.

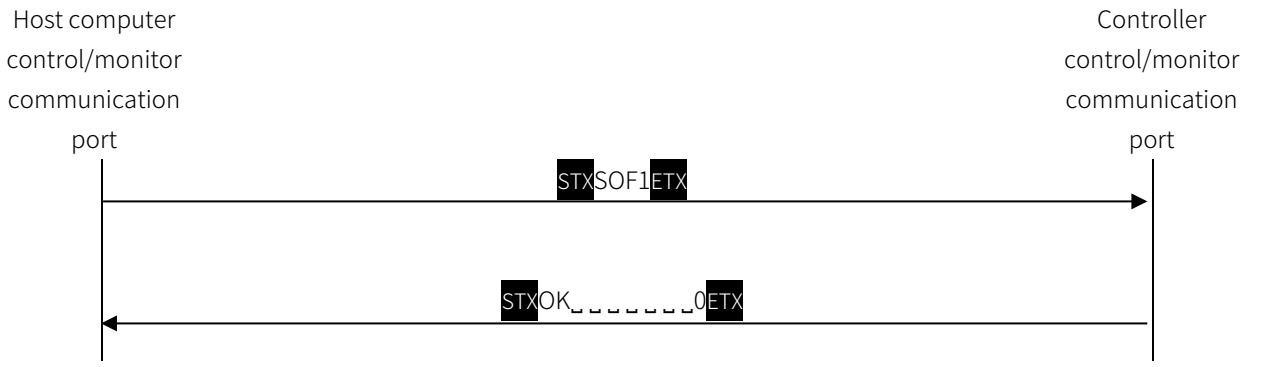
Request Format

STXSOF1ETX

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command to execute Servo Off is as follows.

STXSOF1ETX
STXOK␣␣␣␣␣␣␣␣0ETX

SON1 Servo On

Function

This turns on the power for the servo in the controller.

If the request cannot be received, an error response is returned.

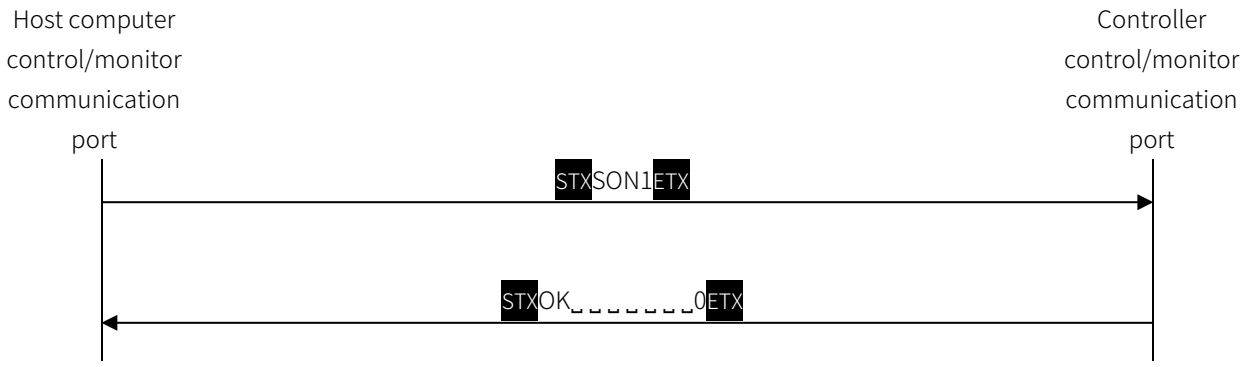
Request Format

STXSON1ETX

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command to execute Servo On is as follows.

STXSON1ETX
STXOK␣␣␣␣␣␣␣␣0ETX

MEMO

· *If the Servo On and Servo Off operations are executed repeatedly, the controller may return "NG" to prevent burning.*

Wait and try to send the Servo On command again later.

⚠ DANGER

· When turning on the servo power, be sure to exit from the maximum space of the robot. Otherwise, an accidental collision with the robot in unexpected motion could occur.

STB1 Obtain Board Information

Function

Monitorable command

This obtains the board information.

If the request cannot be received, an error response is returned.

Request Format

STXSTB1ETX

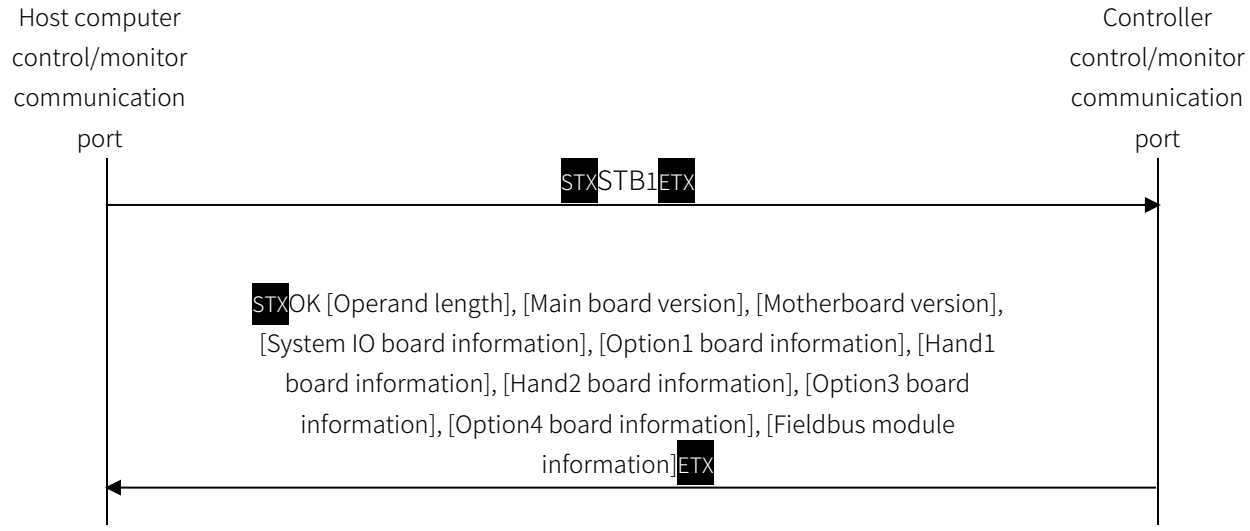
Response Format

STXOK [Operand length], [Main board version], [Motherboard version], [System IO board information], [Option1 board information], [Hand1 board information], [Hand2 board information], [Option3 board information], [Option4 board information], [Fieldbus module information]ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	Main board version	1 Fixed length	Main board version information 0 to 7
3	Motherboard version	1 Fixed length	Motherboard version information 0 to 7
4	System IO board information	Variable length	System IO board information ENABLE-P : TYPE-P mode ENABLE-N : TYPE-N mode DISABLE : Not connected ERROR : Error detected
5	Option1 IO board information	Variable length	System IO board information ENABLE-P : TYPE-P mode ENABLE-N : TYPE-N mode DISABLE : Not connected ERROR : Error detected
6	Hand1 board information	Variable length	Hand1 board information ENABLE : Connected DISABLE : Not connected
7	Hand2 board information	Variable length	Hand2 board information ENABLE : Connected DISABLE : Not connected

No.	Name	Size (bytes)	Description
8	Option3 board information	Variable length	Option3 board information ENABLE: Connected DISABLE: Not connected
9	Option4 board information	Variable length	Option4 board information RS232: RS232C communication RS485: RS485 communication RS422: RS422 communication EXxxx: Expansion board information The connected expansion board information is stored in x. DISABLE: Not connected
10	Fieldbus module information	Variable length	Fieldbus module information M30: M30 module M40: M40 module DISABLE: Not connected

Communication Example



Send/Receive Example

The command for obtaining the board information is as follows.

STXSTB1ETX

STXOK␣␣␣␣␣␣56, 1, 1, ENABLE-P, DISABLE, ENABLE, DISABLE, DISABLE, DISABLE, M40ETX

STB2 Obtain Board Information

Function

Monitorable command

This obtains the board information.

If the request cannot be received, an error response is returned.

This command is an extension of the "FIELDBUS module information" function of STB1 information.

Request Format

STXSTB2ETX

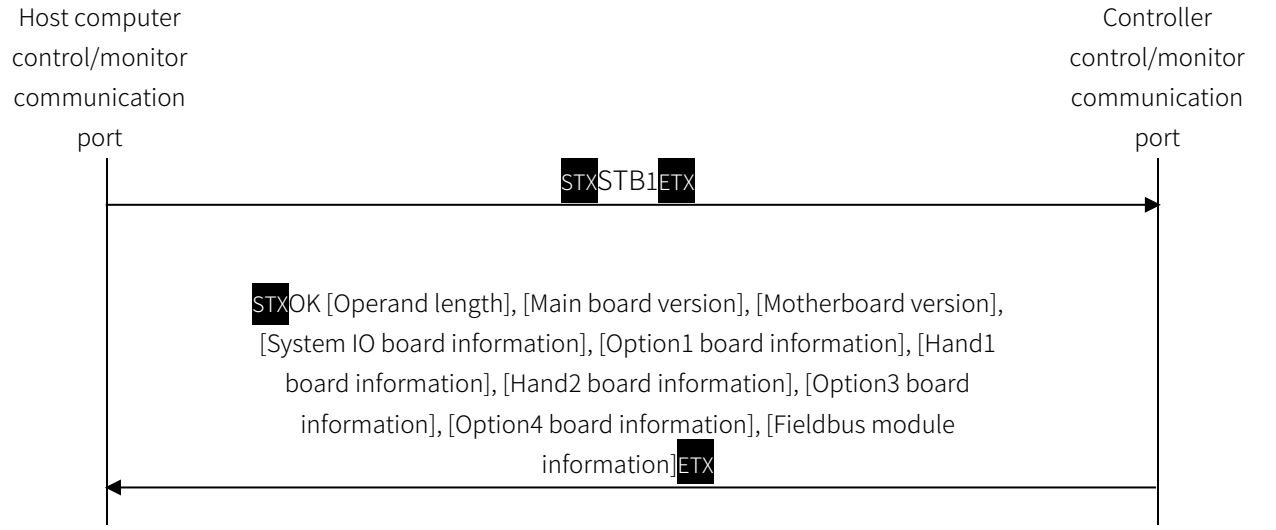
Response Format

STXOK [Operand length], [Main board version], [Motherboard version], [System IO board information], [Option1 board information], [Hand1 board information], [Hand2 board information], [Option3 board information], [Option4 board information], [Fieldbus module information]ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	Main board version	1 Fixed length	Main board version information 0 to 7
3	Motherboard version	1 Fixed length	Motherboard version information 0 to 7
4	System IO board information	Variable length	System IO board information ENABLE-P : TYPE-P mode ENABLE-N : TYPE-N mode DISABLE : Not connected ERROR : Error detected
5	Option1 IO board information	Variable length	System IO board information ENABLE-P : TYPE-P mode ENABLE-N : TYPE-N mode DISABLE : Not connected ERROR : Error detected
6	Hand1 board information	Variable length	Hand1 board information ENABLE : Connected DISABLE : Not connected
7	Hand2 board information	Variable length	Hand2 board information ENABLE : Connected DISABLE : Not connected

No.	Name	Size (bytes)	Description
8	Option3 board information	Variable length	Option3 board information ENABLE: Connected DISABLE: Not connected
9	Option4 board information	Variable length	Option4 board information RS232: RS232C communication RS485: RS485 communication RS422: RS422 communication EXxxx: Expansion board information The connected expansion board information is stored in x. DISABLE: Not connected
10	Fieldbus module information	Variable length	Fieldbus module information is stored. If the fieldbus modules are disabled, "DISABLE" is stored. If it cannot be recognized, "NOT FOUND" is stored. When a fieldbus module is enabled, it is stored in the format [module type name]␣[fieldbus type]␣[module version]. [Module type name] M30: M30 module M40: M40 module [Fieldbus type] IP: Ethernet/IP PN: PROFINET CC: CC-Link PB: PROFIBUS CA: EtherCAT DN: DeviceNet [Module version] The version of the fieldbus module is stored.

Communication Example



Send/Receive Example

The command for obtaining the board information is as follows.

STXSTB2ETX

STXOK␣␣␣␣␣␣56, 1, 1, ENABLE-P, DISABLE, ENABLE, DISABLE, DISABLE, DISABLE, M40␣PN␣2.4.1ETX

STC1 Obtain Controller Status Information

Function

Monitorable command

This transfers the internal status of the controller to the host computer.

Request Format

STXSTC1ETX

Response Format

STXOK [Operand length], Status information ETX

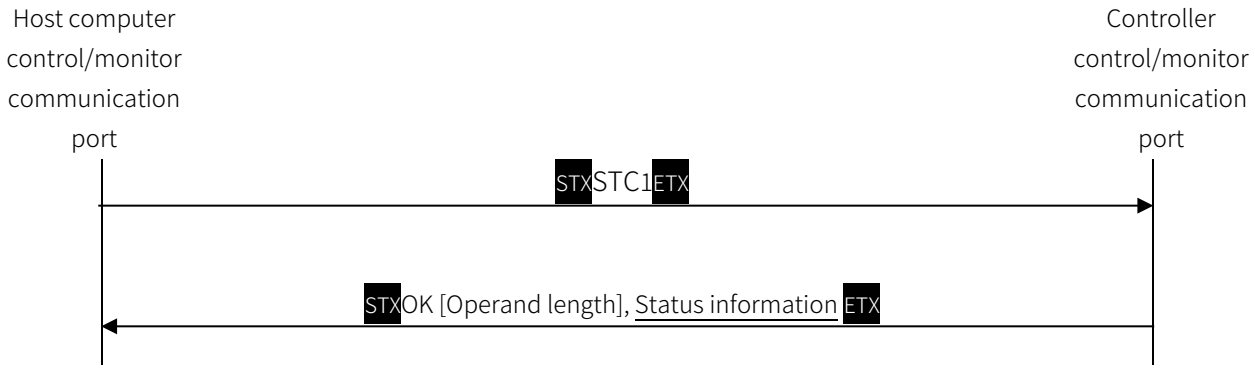
In the status information, the following information is stored.
[Controller mode], [Execution mode], [File name], [Speed override], [Execution status], [Servo power status], [Emergency stop switch status], [Program execution line], [Program analysis line], [Program execution task], [Program analysis task], [Feed hold status], [Guidance coordinate status], [Guidance speed], [Inching amount], [Work coordinate name], [Tool coordinate name], [Base coordinate name], [1(X)-axis guidance mode status] [2(Y)-axis guidance mode status] [3(Z)-axis guidance mode status] [4(C)-axis guidance mode status] [5(T)-axis guidance mode status]

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	Master mode	1 Fixed length	Current controller master mode 0: MANUAL (Test operation mode) 1: AUTO (External automatic mode)
3	Execution mode	1 Fixed length	Current execution mode 0: Continuous operation mode (CONT) 1: Cycle operation mode (CYCLE) 2: Step operation mode (STEP) 3: Segment operation mode (SEG)
4	File name	Variable length	Currently-selected file name The drive number and the file name are separated by a slash (/) to indicate the file name, as shown below. 1: ROM drive 2: RAM drive 3: SD card drive [Drive number]/[File name]
5	Speed override	Variable length	The speed override is expressed as a percentage.

No.	Name	Size (bytes)	Description
6	Execution status	Variable length	Current program execution status 0: STOP (RESET) Reset mode stopped (Same status as program reset) 1: RUN During automatic operation 2: STOP (RETRY) Retry mode stopped (Restart after an aborted operation) 3: STOP (CONT) Continuous mode stopped (Resume from current step)
7	Servo power status	1 Fixed length	Current servo status 0: Servo Off 1: Servo On
8	Emergency stop switch status	1 Fixed length	Current emergency stop status 0: Emergency stop off 1: Emergency stop on
9	Program execution line	Variable length	Program line number currently being executed
10	Program analysis line	Variable length	Program line number currently being analyzed
11	Program execution task	Variable length	Program task number currently being executed
12	Program analysis task	Variable length	Program task number currently being executed
13	Feed hold status	1 Fixed length	Current feed hold status 0: Feed hold off 1: Feed hold on
14	Guidance coordinate status	1 Fixed length	Currently-selected guidance coordinate status 0: Joint coordinates 1: Tool coordinates 2: Work coordinates 3: World coordinates
17	Guidance speed	1 Fixed length	Currently-set guidance speed 0: Slow jog speed 1: Medium jog speed 2: Fast jog speed
18	Tool coordinate name	Variable length	Tool coordinate name

No.	Name	Size (bytes)	Description
19	Base coordinate name	Variable length	Base coordinate name
20 to 24	Guidance mode status	1 Fixed length	Guidance mode status for 1st to 5th axes 0: Locked 1: Free

Communication Example



Send/Receive Example

The command for obtaining the internal information of the controller is as follows.

STXSTC1ETX

STXOK␣␣␣␣␣␣66, 0, 0, 1/PRG1, 20, 1, 1, 0, 5, 6, 1, 2, 0, 0, 0, 0, .world, .system, .system, 00000ETX

STE1 Obtain Ethernet Status Information

Function

Monitorable command

This transfers the internal Ethernet status of the controller to the host computer.

Request Format

STXSTE1ETX

Response Format

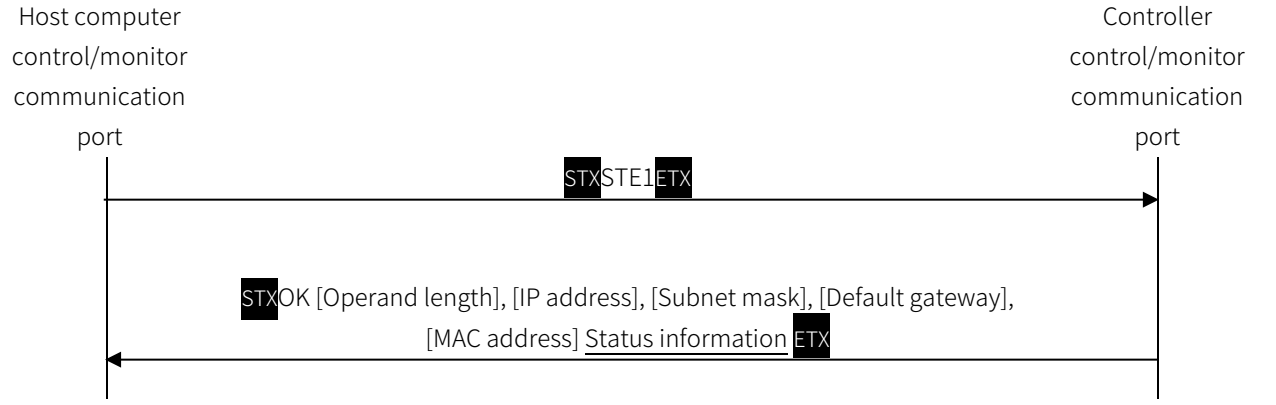
STXOK [Operand length], [IP address], [Subnet mask], [Default gateway], [MAC address] Status information ETX

In the status information, the following information is stored consecutively for the number of IPs.
CR[IP name], [Communication mode], [IP status], [Controller port number], [Connection target IP address], [Connection target port number]

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	IP Address	Variable length	Controller IP address
3	Subnet mask	Variable length	Controller subnet mask
4	Default gateway	Variable length	Controller default gateway
5	MAC address	17 Fixed length	Controller MAC address
6	IP name	Variable length	IP name
7	Communication mode	1 Fixed length	Communication mode set to the IP 0: Not used 1: TCP server 2: TCP client 3: UDP

No.	Name	Size (bytes)	Description
8	IP status	Variable length	The Ethernet status is one of the following. For a detailed description of the meaning of each status, refer to a book or other reference that explains TCP/IP (available commercially). Close Closed Listen Syn-Sent Syn-Received Established Fin-Wait1 Fin-Wait2 Close␣wait Closing Last␣ACK Time␣Wait During UDP communication, the status is "－."
9	Controller port number	Variable length	Port number used by controller
10	Connection target IP address	Variable length	IP address of connection target device If not connected, "-" is stored.
11	Connection target port number	Variable length	Port number of connection target device If not connected, "-" is stored.

Communication Example



Send/Receive Example

STXSTE1ETX

STXOK␣␣␣␣␣334, 192.168.1.124, 02-00-00-00-00-01CRIP1, 1, Listen, 10000, 192.168.0.150, 1000CRIP2, 1, Listen, 10001, 192.168.0.10, 50000CRIP3, 1, Listen, 10002, 192.168.0.152, 1002CRIP4, 1, Listen, 10003, 192.168.0.153, 1003CRIP5, 1, Listen, 10004, 192.168.0.154, 10004CRIP6, 1, Listen, 10005, 192.168.0.155, 10005CRIP7, 1, Listen, 10006, 192.168.0.156, 10006CRIP8, 1, Listen, 10007, 192.168.0.157, 10007ETX

STP1 Stop Automatic Operation

Function

This stops automatic operation.

If the request cannot be received, an error response is returned.

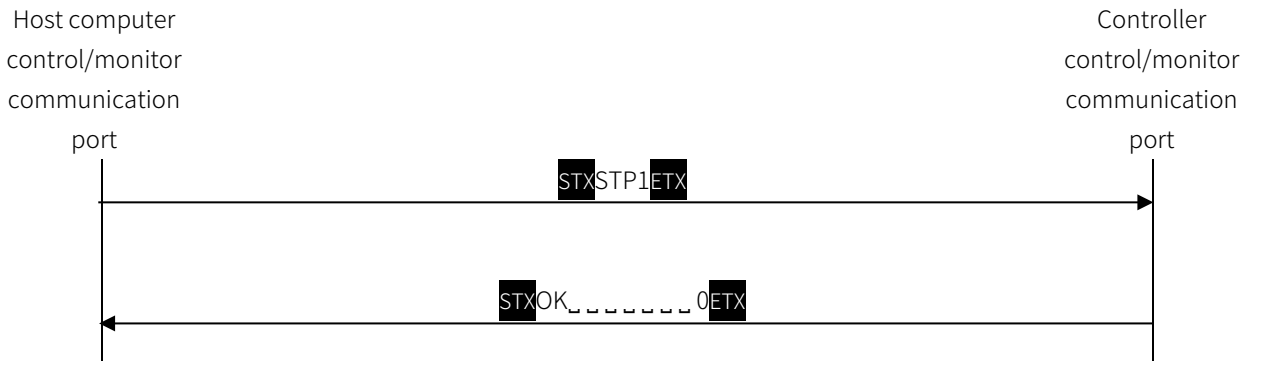
Request Format

STXSTP1ETX

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command to stop automatic operation is as follows.

STXSTP1ETX
STXOK␣␣␣␣␣␣␣␣0ETX

TRQ1 Obtain Motor Torque information

Function

Monitorable command

This transfers the current motor torque information to the host computer.

Request Format

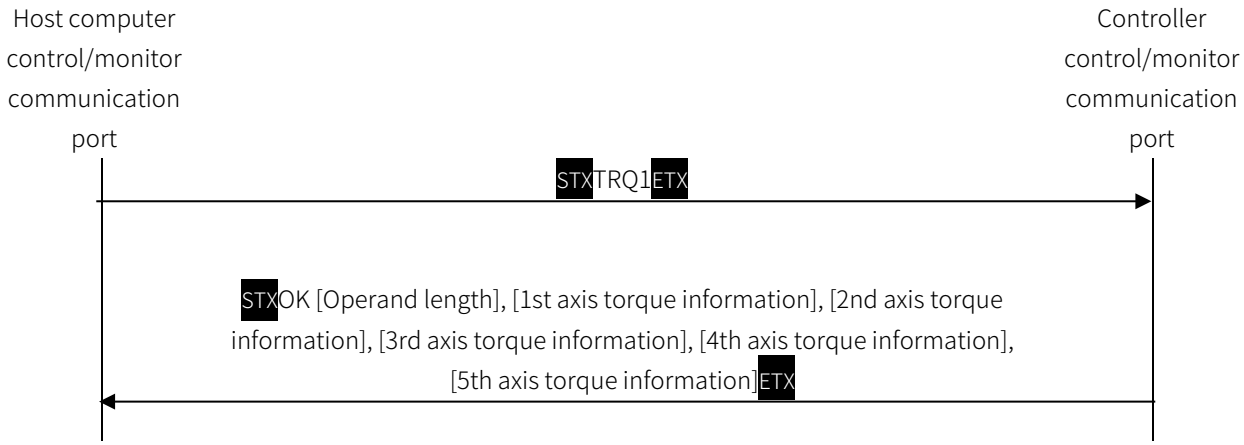
STXTRQ1ETX

Response Format

STXOK [Operand length], [1st axis torque information], [2nd axis torque information], [3rd axis torque information], [4th axis torque information], [5th axis torque information]ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2 to 6	Torque information	Variable length	Current motor torque value for 1st axis to 5th axis. Units: Percentage (%) of rated motor torque This is received as a real number up to 3 decimal places.

Communication Example



Send/Receive Example

The command for obtaining the current value information is as follows.

STXTRQ1ETX

STXOK␣␣␣␣␣␣50,␣␣␣␣0.000,␣␣␣␣0.000,␣␣␣␣0.000,␣␣␣␣0.000,␣␣␣␣0.000ETX

VER1 Read Version

Function

Monitorable command

This transfers the system version information to the host computer.

Request Format

STXVER1ETX

Response Format

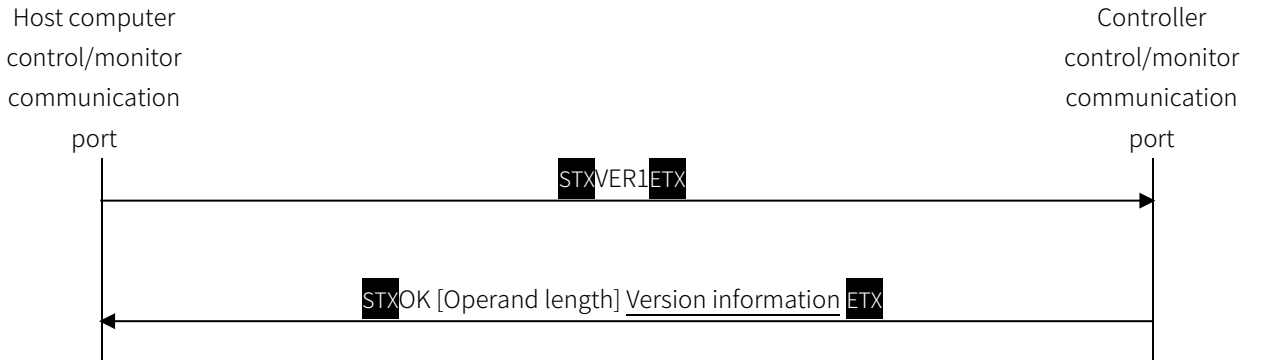
STXOK [Operand length] Version information ETX

In the version information, a plurality of the following item will be stored.

CR[System name], [Date]␣[Time], [Checksum]

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	System name	10 Fixed length	This shows the system name.
3	Date	11 Fixed length	This shows the date that the system was created. “Mon-DD-YYYY” Mon: Month January: Jan, February: Feb, March: Mar, April: Apr, May: May, June: Jun, July: Jul, August: Aug, September: Sep, October: Oct, November: Nov, December: Dec DD: Date YYYY: Year (Western calendar year)
4	Time	8 Fixed length	This shows the time that the system was created. “HH:MM:SS” HH: Hours (24-hour format) MM: Minutes SS: Seconds
5	Checksum	4 Fixed length	This shows the checksum value of the system file. (Hexadecimal)

Communication Example



Send/Receive Example

The command for obtaining the version information is as follows.

STXVER1ETX

STXOK␣␣␣␣␣178CRX8KCB-Y01, Jul-08-2019 14:28:00, CB6ECRS068A, Jul-02-2019 15:42:00, FBE7CRX8KCS-Y02, Jul-16-2019 13:45:00, 60A0CRX8KCRB-01A, Jul-09-2019 16:26:00, CB6ECRX8KCRX-01A, Jul-09-2019 16:39:00, 7FEDETX

VLJ1 Obtain Joint Speed Information

Function

Monitorable command

This transfers the robot joint speed information to the host computer.

Request Format

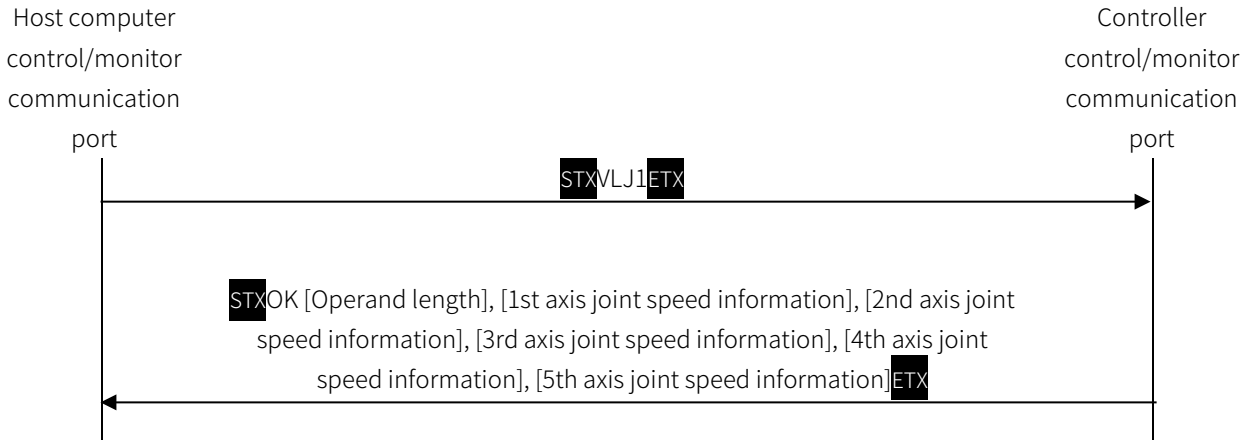
STXVLJ1ETX

Response Format

STXOK [Operand length], [1st axis joint speed information], [2nd axis joint speed information], [3rd axis joint speed information], [4th axis joint speed information], [5th axis joint speed information]ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2 to 6	Joint speed	Variable length	Information for 1st to 5th axes. Units: Angular velocity (deg/s or mm/s) This is received as a real number up to 3 decimal places.

Communication Example



Send/Receive Example

The command for obtaining the robot joint speed information is as follows.

STXVLJ1ETX

STXOK␣␣␣␣␣␣49, 2.089, 1.804, 0.045, -4.215, 2.498, 2.264, 0.000, 0.000ETX

VLW1 Obtain Hand Speed

Function

Monitorable command

This transfers the hand speed information to the host computer.

Request Format

STXVLW1, [Speed type]ETX

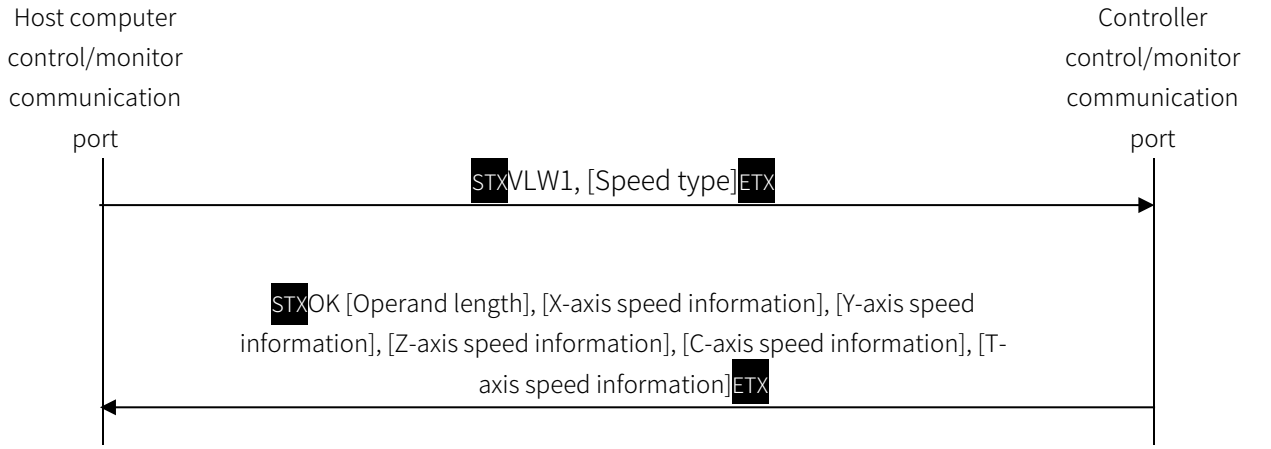
No.	Name	Size (bytes)	Description
1	Speed type	1 Fixed length	Specify one of the following numbers for the speed type. 2: Work coordinate system (tool tip) 3: World coordinate system (tool tip) 4: Mechanical interface (TCP)

Response Format

STXOK [Operand length], [X-axis speed information], [Y-axis speed information], [Z-axis speed information], [C-axis speed information], [T-axis speed information]ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2 to 6	Speed information	Variable length	Hand speed information Units: Angular velocity (deg/s or mm/s) This is received as a real number up to 3 decimal places. Because the T-axis is an additional axis, it has the same value as the joint speed.

Communication Example



Send/Receive Example

The command for obtaining the hand speed (mechanical interface) information is as follows.

STXVLW1, 0ETX
STXOK␣␣␣␣␣␣49, 2.089, 1.804, 0.045, -4.215, 2.498, 2.264, 0.000, 0.000ETX

VLW2 Obtain Hand Combined Speed

Function

Monitorable command

This transfers the hand combined speed information to the host computer.

Request Format

STXVLW2, [Speed type]ETX

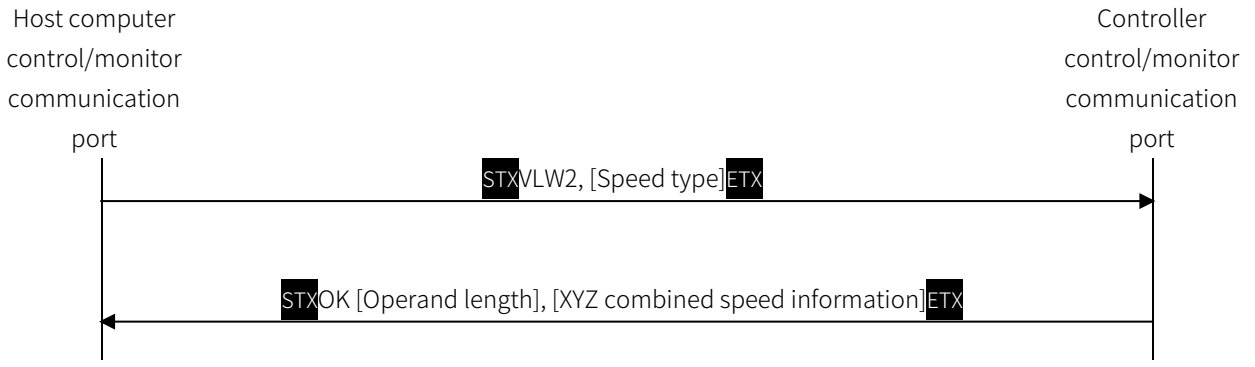
No.	Name	Size (bytes)	Description
1	Speed type	1 Fixed length	Specify one of the following numbers for the speed type. 0: Mechanical interface (TCP) 1: Tool

Response Format

STXOK [Operand length], [XYZ combined speed information]ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	XYZ combined speed information	Variable length	Hand combined speed information. Units: speed (mm/s) This is received as a real number up to 3 decimal places.

Communication Example



Send/Receive Example

The command for obtaining the robot tip speed information is as follows.

STXVLW2, 0ETX

STXOK␣␣␣␣␣␣␣6, 0.000ETX

WHL1 Obtain Global Variable List

Function

This obtains a list of the global variables defined in the currently-selected program file.

Request Format

STXWHL1ETX

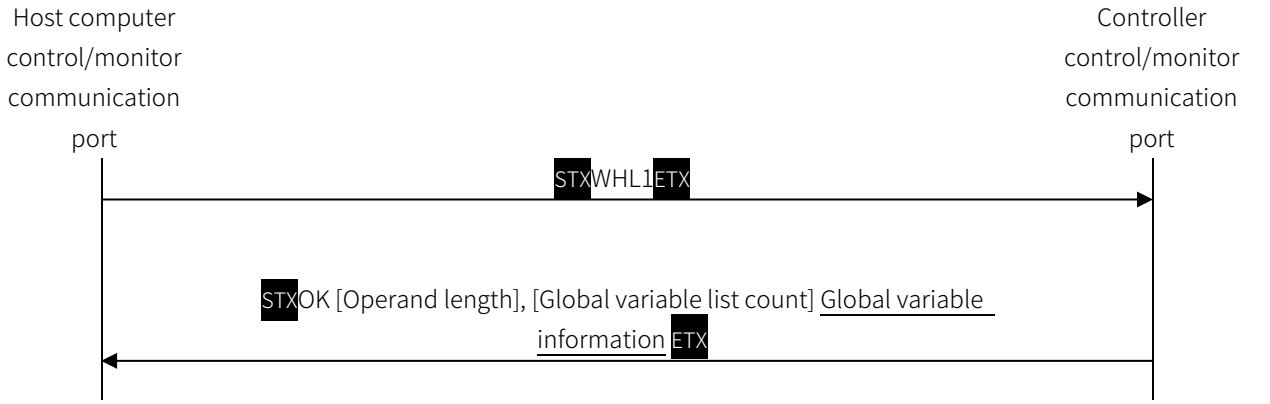
Response Format

STXOK [Operand length], [Global variable list count] Global variable information ETX

In the global variable information, the following information is stored consecutively for the quantity in [Global variable list count].
CR[Variable name], [Variable type]

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	Global variable list count	Variable length	The number of global variable lists defined in the program file is stored.
3	Variable name	Variable length	The global variable name is stored. Array variables use array elements ("variable name (*, *, ...)").
4	Variable type	Variable length	Specify one of the following values for the variable type data. 0: Integer type 1: Real number type 2: Load type 3: Coordinate type 4: Position type 5: Axis type 6: Character string type -1: Unknown 10: Array integer type 11: Array real number type 12: Array load type 13: Array coordinate type 14: Array position type 15: Array axis type 16: Array character string type

Communication Example



Send/Receive Example

The command for global variables when using a real number type called "A", character string type called "S", and array real number type called "B" is as follows.

STXWHL1ETX

STXOK␣␣␣␣␣␣19, 3CRA, 0CRS, 6CRB(10), 10ETX

WHR1 Read Global Variable

Function

Monitorable command

This reads the values of the global variables defined in the program during automatic operation.

Request Format

STXWHR1, [Variable name], [Variable type]ETX

No.	Name	Size (bytes)	Description
1	Variable name	Variable length	Specify the name of the global variable to be read. For array variables, specify the array elements ("variable name (*, *, ...)").
2	Variable type	Variable length	Specify one of the following values for the variable type data. 0: Integer type 1: Real number type 2: Load type 3: Coordinate type 4: Position type 5: Axis type 6: Character string type -1: Unknown 10: Array integer type 11: Array real number type 12: Array load type 13: Array coordinate type 14: Array position type 15: Array axis type 16: Array character string type

Response Format

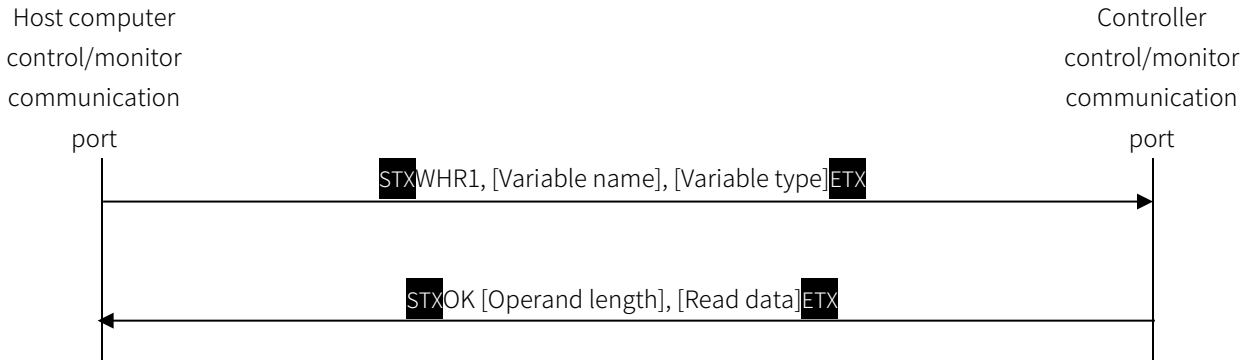
STXOK [Operand length], [Read data]ETX

No.	Name	Size (bytes)	Description
1	Operand length	8 Fixed length	Byte size after [Operand length] until ETX (not including ETX)
2	Read data	Variable length	The read variable data varies for each variable type as follows. The float data is expressed as value up to 3 decimal places.

Type	Read data		
Integer type	Format	[Long type data]	
	Quantity	1	
	Type	Long type data	long
Real number type	Format	[Float type data]	
	Quantity	1	
	Type	Float type data	float
Load type	Format	[Weight], [Center of gravity offset]	
	Quantity	2	
	Type	Weight	float
		Center of gravity offset	float
Coordinate type	Format	[X-coordinate value], [Y-coordinate value], [Z-coordinate value], [C-coordinate value]	
	Quantity	4	
	Type	X-coordinate value	float
		Y-coordinate value	float
		Z-coordinate value	float
		C-coordinate value	float
Position type	Format	[X-coordinate value], [Y-coordinate value], [Z-coordinate value], [C-coordinate value], [T-coordinate value], [Orientation]	
	Quantity	6	
	Type	X-coordinate value	float
		Y-coordinate value	float
		Z-coordinate value	float
		C-coordinate value	float
		T-coordinate value	float
		Orientation	One of the following numbers 0: FREE 1: LEFTY 2: RIGHTY
Axis type	Format	1st axis, 2nd axis, 3rd axis, 4th axis, 5th axis	
	Quantity	5	
	Type	1st axis	float
		2nd axis	float
		3rd axis	float
		4th axis	float
		5th axis	float
Character string type	Format	[Character string type data]	
	Quantity	1	
	Type	Character string type data	char

Type	Read data		
Array integer type	Format	[Long type data]	
	Quantity	1	
	Type	Long type data	long
Array real number type	Format	[Float type data]	
	Quantity	1	
	Type	Float type data	float
Array load type	Format	[Weight], [Center of gravity offset]	
	Quantity	2	
	Type	Weight	float
		Center of gravity offset	float
Array coordinate type	Format	[X-coordinate value], [Y-coordinate value], [Z-coordinate value], [C-coordinate value]	
	Quantity	4	
	Type	X-coordinate value	float
		Y-coordinate value	float
		Z-coordinate value	float
		C-coordinate value	float
Array position type	Format	[X-coordinate value], [Y-coordinate value], [Z-coordinate value], [C-coordinate value], [T-coordinate value], [Orientation]	
	Quantity	6	
	Type	X-coordinate value	float
		Y-coordinate value	float
		Z-coordinate value	float
		C-coordinate value	float
		T-coordinate value	float
		Orientation	One of the following numbers 0: FREE 1: LEFTY 2: RIGHTY
Array axis type	Format	1st axis, 2nd axis, 3rd axis, 4th axis, 5th axis	
	Quantity	5	
	Type	1st axis	float
		2nd axis	float
		3rd axis	float
		4th axis	float
		5th axis	float
Array character string type	Format	[Character string type data]	
	Quantity	1	
	Type	Character string type data	char

Communication Example



Send/Receive Example

The command for reading the value of the integer type global variable "X" is as follows.

STXWHR1ETX

STXOK␣␣␣␣␣␣␣4, 100ETX

The command for reading the value of the integer type 2D array global variable "X(1, 2)" is as follows.

STXWHR1, X(1, 2), 0ETX

STXOK␣␣␣␣␣␣␣4, 100ETX

WHW1 Write Global Variable

Function

This writes data to the global variables defined in the program during automatic operation.

Request Format

STXWHW1, [Variable name], [Variable type], [Write data]ETX

No.	Name	Size (bytes)	Description
1	Variable name	Variable length	Specify the name of the global variable to be written. For array variables, specify the array elements ("variable name (*, *, ...)").
2	Variable type	1 Fixed length	Specify one of the following values for the variable type data. 0: Integer type 1: Real number type 2: Load type 3: Coordinate type 4: Position type 5: Axis type 6: Character string type 10: Array integer type 11: Array real number type 12: Array load type 13: Array coordinate type 14: Array position type 15: Array axis type 16: Array character string type
3	Write data	Variable length	The write variable data varies for each variable type as follows.

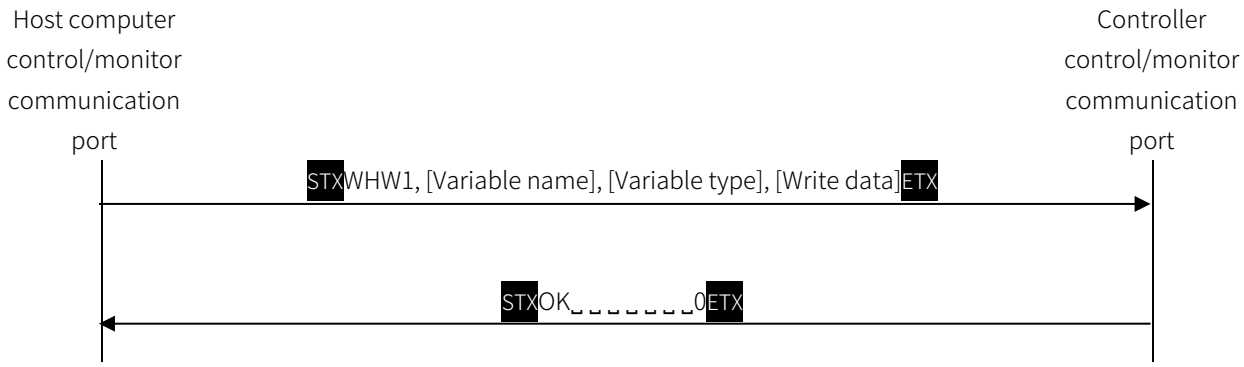
Type	Write data		
Integer type	Format	[Long type data]	
	Quantity	1	
	Type	Long type data	long
Real number type	Format	[Float type data]	
	Quantity	1	
	Type	Float type data	float
Load type	Format	[Weight], [Center of gravity offset]	
	Quantity	2	
	Type	Weight	float
		Center of gravity offset	float
Coordinate type	Format	[X-coordinate value], [Y-coordinate value], [Z-coordinate value], [C-coordinate value]	
	Quantity	4	
	Type	X-coordinate value	float
		Y-coordinate value	float
		Z-coordinate value	float
		C-coordinate value	float
Position type	Format	[X-coordinate value], [Y-coordinate value], [Z-coordinate value], [C-coordinate value], [T-coordinate value], [Orientation]	
	Quantity	6	
	Type	X-coordinate value	float
		Y-coordinate value	float
		Z-coordinate value	float
		C-coordinate value	float
		T-coordinate value	float
		Orientation	One of the following numbers 0: FREE 1: LEFTY 2: RIGHTY
Axis type	Format	1st axis, 2nd axis, 3rd axis, 4th axis, 5th axis	
	Quantity	5	
	Type	1st axis	float
		2nd axis	float
		3rd axis	float
		4th axis	float
		5th axis	float
Character string type	Format	[Character string type data]	
	Quantity	1	
	Type	Character string type data	char

Type	Read data		
Array integer type	Format	[Long type data]	
	Quantity	1	
	Type	Long type data	long
Array real number type	Format	[Float type data]	
	Quantity	1	
	Type	Float type data	float
Array load type	Format	[Weight], [Center of gravity offset]	
	Quantity	2	
	Type	Weight	float
		Center of gravity offset	float
Array coordinate type	Format	[X-coordinate value], [Y-coordinate value], [Z-coordinate value], [C-coordinate value]	
	Quantity	4	
	Type	X-coordinate value	float
		Y-coordinate value	float
		Z-coordinate value	float
		C-coordinate value	float
Array position type	Format	[X-coordinate value], [Y-coordinate value], [Z-coordinate value], [C-coordinate value], [T-coordinate value], [Orientation]	
	Quantity	6	
	Type	X-coordinate value	float
		Y-coordinate value	float
		Z-coordinate value	float
		C-coordinate value	float
		T-coordinate value	float
		Orientation	One of the following numbers 0: FREE 1: LEFTY 2: RIGHTY
Array axis type	Format	1st axis, 2nd axis, 3rd axis, 4th axis, 5th axis	
	Quantity	5	
	Type	1st axis	float
		2nd axis	float
		3rd axis	float
		4th axis	float
		5th axis	float
Array character string type	Format	[Character string type data]	
	Quantity	1	
	Type	Character string type data	char

Response Format

STXOK_ _ _ _ _ _ _ _0ETX

Communication Example



Send/Receive Example

The command for writing the integer value "100" to the integer type global variable "X" is as follows.

STXWHR1, X, 0, 100ETX
STXOK_ _ _ _ _ _ _ _0ETX

NOTICE

· Before sending this command, be sure to check the effect of the target global variable on the program. The robot that has performed unexpected motion may collide with peripheral equipment.

WHW2 Write Global Variable

Function

This writes data to the global variables defined in the program during automatic operation.

Request Format

STXWHW2 Write instruction ETX

For the write instruction, store the following item consecutively for the number of variables to be written.
The maximum number of simultaneous settable variables is 64.
CR[Variable name], [Variable type], [Write data]

When writing only one variable
STXWHW2CR[Variable name], [Variable type], [Write data]ETX

When writing multiple variables
STXWHW2CR[Variable name], [Variable type], [Write data] CR[Variable name], [Variable type], [Write data] …CR[Variable name], [Variable type], [Write data]ETX

No.	Name	Size (bytes)	Description
1	Variable name	Variable length	Specify the name of global variable to be written. For array variables, specify the array elements (“Variable name (*, *, …)”).
2	Variable type	1 Fixed length	Specify one of the following values for the variable type data 0: Integer type 10: Array integer type 1: Real number type 11: Array real number type 2: Load type 12: Array load type 3: Coordinate type 13: Array coordinate type 4: Position type 14: Array position type 5: Axis type 15: Array axis type 6: Character string type 16: Array character string type
3	Write data	Variable length	The variable write data varies for each variable type as follows.

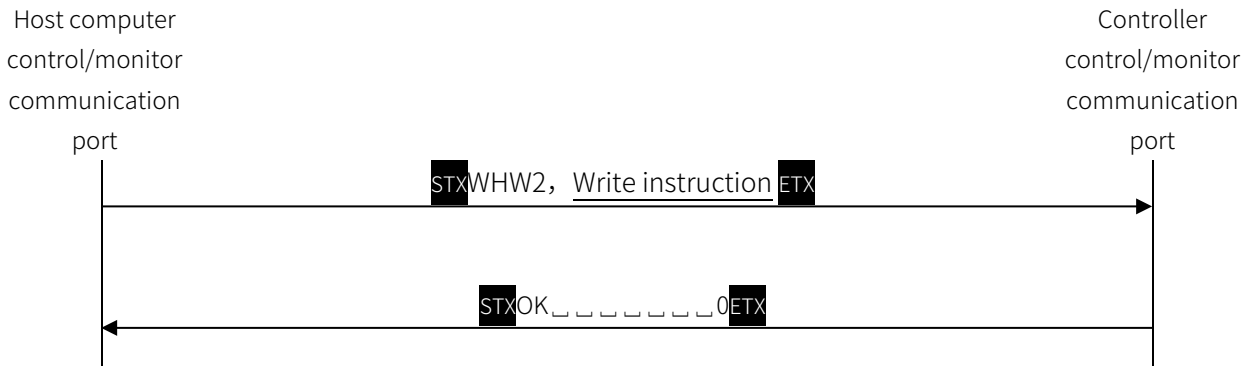
Type	Write data		
Integer type	Format	[Long type data]	
	Quantity	1	
	Type	Long type data	long
Real number type	Format	[Float type data]	
	Quantity	1	
	Type	Float type data	float
Load type	Format	[Weight], [Center of gravity offset]	
	Quantity	2	
	Type	Weight	float
		Center of gravity offset	float
Coordinate type	Format	[X-coordinate value], [Y-coordinate value], [Z-coordinate value], [C-coordinate value]	
	Quantity	4	
	Type	X-coordinate value	float
		Y-coordinate value	float
		Z-coordinate value	float
		C-coordinate value	float
Position type	Format	[X-coordinate value], [Y-coordinate value], [Z-coordinate value], [C-coordinate value], [T-coordinate value], [Orientation]	
	Quantity	6	
	Type	X-coordinate value	float
		Y-coordinate value	float
		Z-coordinate value	float
		C-coordinate value	float
		T-coordinate value	float
		Orientation	One of the following numbers 0: FREE 1: LEFTY 2: RIGHTY
Axis type	Format	1st axis, 2nd axis, 3rd axis, 4th axis, 5th axis	
	Quantity	5	
	Type	1st axis	float
		2nd axis	float
		3rd axis	float
		4th axis	float
		5th axis	float
Character string type	Format	[Character string type data]	
	Quantity	1	
	Type	Character type data	char

Type	Read data		
Array integer type	Format	[Long type data]	
	Quantity	1	
	Type	Long type data	long
Array real number type	Format	[Float type data]	
	Quantity	1	
	Type	Float type data	float
Array load type	Format	[Weight], [Center of gravity offset]	
	Quantity	2	
	Type	Weight	float
		Center of gravity offset	float
Array coordinate type	Format	[X-coordinate value], [Y-coordinate value], [Z-coordinate value], [C-coordinate value]	
	Quantity	4	
	Type	X-coordinate value	float
		Y-coordinate value	float
		Z-coordinate value	float
		C-coordinate value	float
Array position type	Format	[X-coordinate value], [Y-coordinate value], [Z-coordinate value], [C-coordinate value], [T-coordinate value], [Orientation]	
	Quantity	6	
	Type	X-coordinate value	float
		Y-coordinate value	float
		Z-coordinate value	float
		C-coordinate value	float
		T-coordinate value	float
		Orientation	One of the following numbers 0: FREE 1: LEFTY 2: RIGHTY
Array axis type	Format	1st axis，2nd axis，3rd axis，4th axis，5th axis	
	Quantity	5	
	Type	1st axis	float
		2nd axis	float
		3rd axis,	float
		4th axis	float
		5th axis	float
Array character g type	Format	[Character string type data]	
	Quantity	1	
	Type	Character string type	char

Response Format

STXOK␣␣␣␣␣␣␣␣0ETX

Communication Example



Send/Receive Example

The command for writing the value “3” to the integer type global variable and the position “(10,20,30,0,0,FREE)” to the position type global variable “P” is as follows.

STXWHW2CRX,0,100CRP,4,10,20,30,0,0,0ETX

STXOK␣␣␣␣␣␣␣␣0ETX

NOTICE

· Before sending this command, be sure to check the effect of the target global variable on the program. The robot that has performed unexpected motion may collide with peripheral equipment.

MEMO

* *The global variables are written in the order of set variable names. If writing multiple global variables fails while they are being written, writing the failed global variable and later global variables is interrupted and only global variables that have been successfully written are written.*

5. Robot Operation Sequence Using External Communication

Operating procedures for the robot in the External Control Mode are basically the same as those in the Internal Control Mode.

5.1 Basic Operation

The flow sequence when a program is selected and is executed repeatedly is shown in “Fig. 5.3 Basic operation sequence”.

5.2 Program Download

The flow sequence when each cycle program is downloaded and executed is shown in “Fig. 5.4 Operation sequence including program download”.

5.3 Reinitializing and Starting a Stopped Program

The flow sequence when the program is stopped, and the program is executed again from the initial status is shown in “Fig. 5.5 Restart after program interruption”.

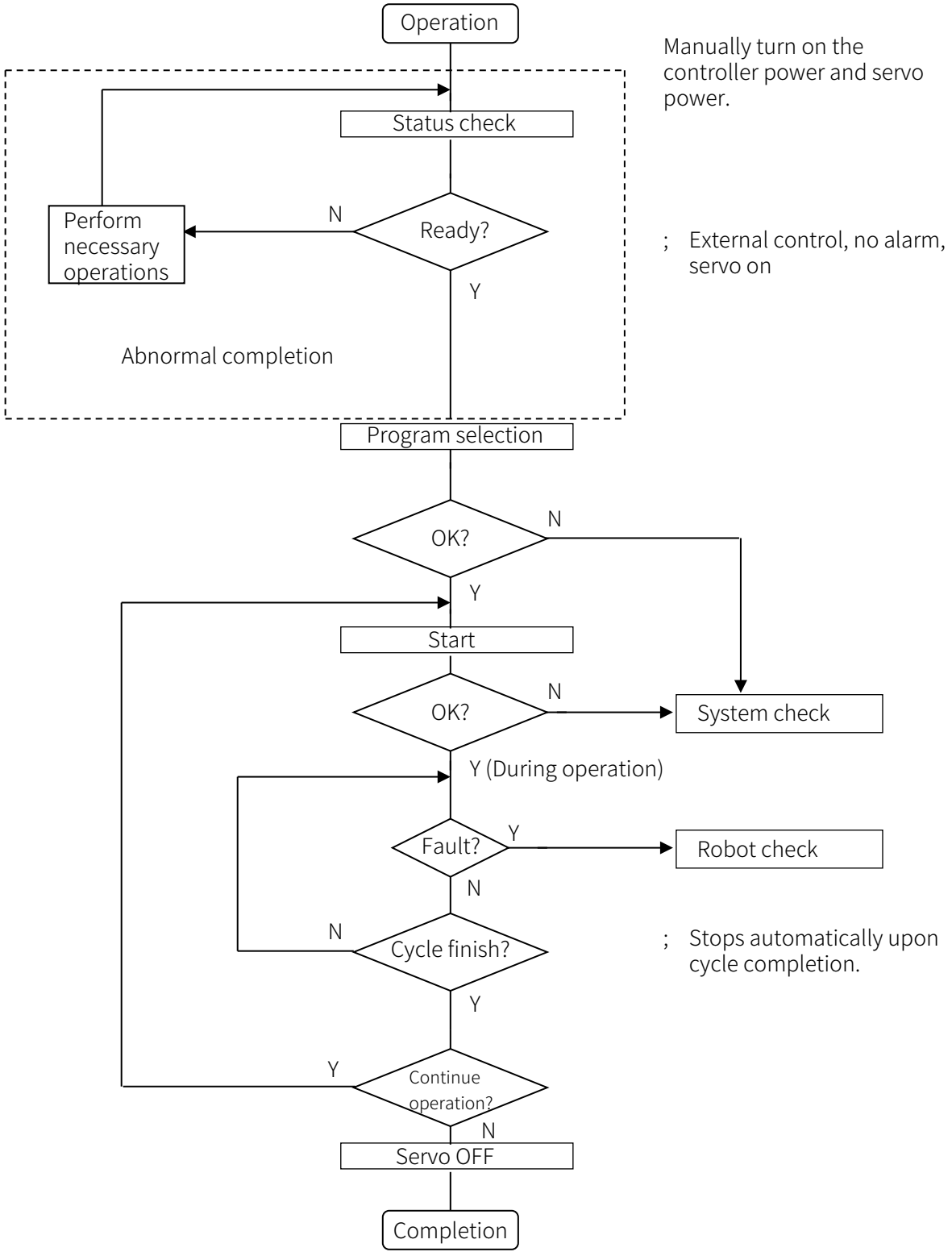


Fig. 5.3 Basic operation sequence

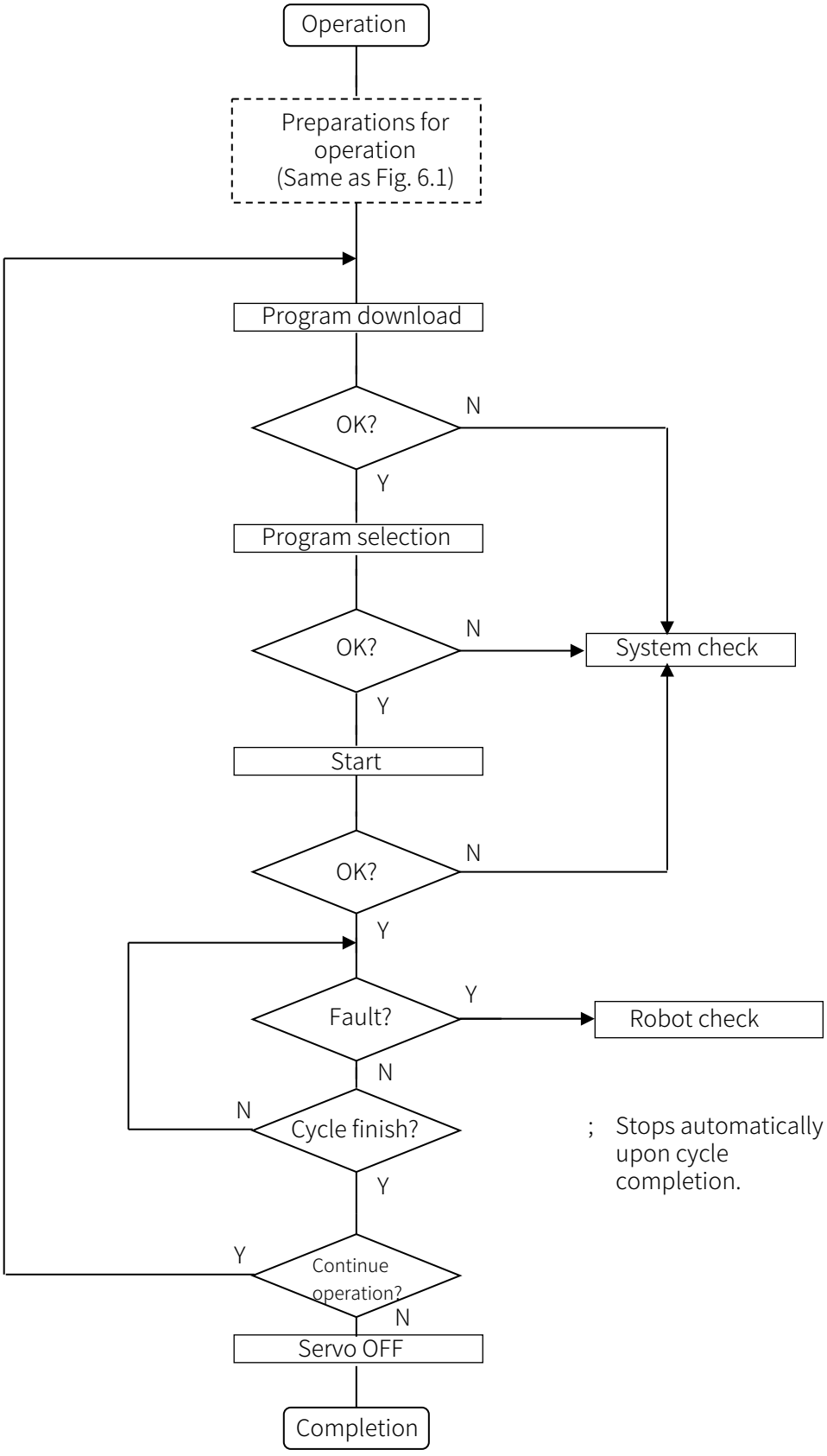


Fig. 5.4 Operation sequence including program download

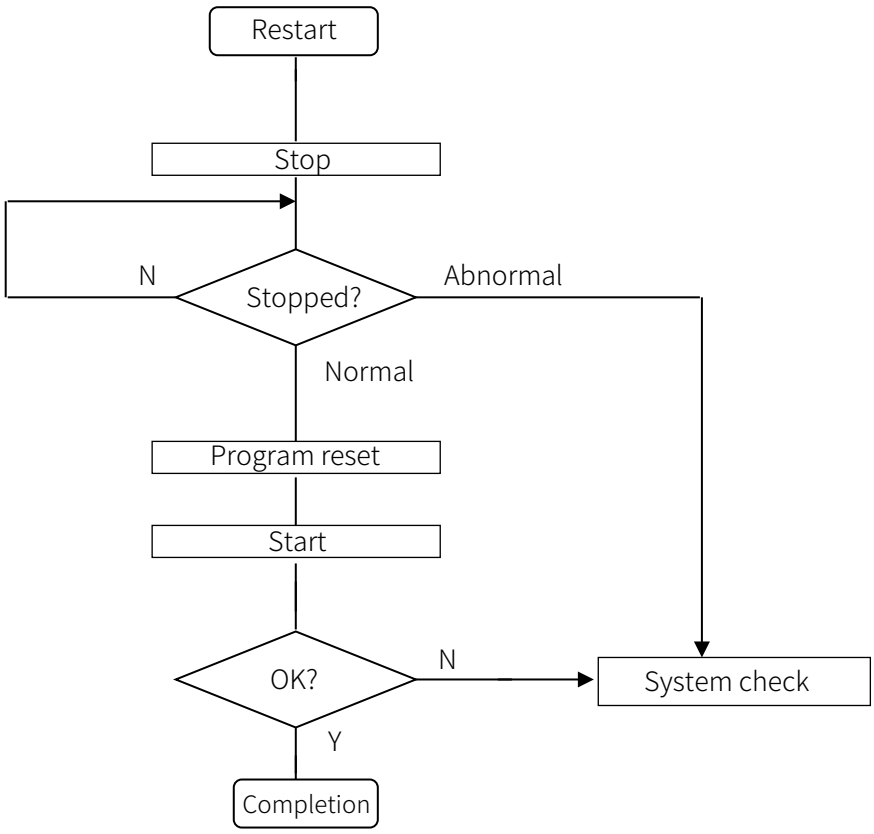


Fig. 5.5 Restart after program interruption

6. Appendix

6.1 ASCII Code Table

Low-order 4 bits (Hexadecimal number)	High-order 4 bits (Hexadecimal number)							
	0	1	2	3	4	5	6	7
0	NUL Null	DLE Data link escape	␣ Space	0	@ At sign	P	` Grave accent	p
1	SOH Start of heading	DC1 Device control 1	! Exclamation mark	1	A	Q	a	q
2	STX Start of text	DC2 Device control 2	" Quotation mark	2	B	R	b	r
3	ETX End of text	DC3 Device control 3	# Number sign	3	C	S	c	s
4	EOT End of transmission	DC4 Device control 4	$ Dollar sign	4	D	T	d	t
5	ENQ Enquiry	NAK Negative acknowledge	% Percent sign	5	E	U	e	u
6	ACK Acknowledge	SYN Synchronous idle	& Ampersand	6	F	V	f	v
7	BEL Bell	ETB End of transmission block	' Single quote	7	G	W	g	w
8	BS Backspace	CAN Cancel	(Left parenthesis	8	H	X	h	x
9	HT Horizontal tab	EM End of medium	) Right parenthesis	9	I	Y	i	y
A	LF Line feed	EOF End of file	* Asterisk	: Colon	J	Z	j	z
B	VT Vertical tab	ESC Escape	+ Plus sign	; Semicolon	K	[Left square bracket	k	{ Left curly bracket
C	FF Form feed	FS File separator	, Comma	< Less-than sign; Inequality	L	\ Yen sign	l	| Vertical line
D	CR Carriage return	GS Group separator	- Hyphen-minus	= Equals sign	M	] Right square bracket	m	} Right curly bracket
E	SO Shift out	RS Record separator	. Period	> Greater-than sign; Inequality	N	^ Caret	n	~ Tilde
F	SI Shift in	US Unit separator	/ Slash	? Question mark	O	_ Underscore	o	DEL Delete

* Codes 00 to 1F correspond to control characters.
These characters will either be interpreted as a space or as a code having a specific meaning.

6.2 TS3000 Communication Command Correspondence Table

No.	Command		Function
	Old	New	
1	AC	ALN1	Transfers a maximum of 10 currently-occurring alarms to the host computer with messages.
2	AH	ALH1	Transfers the alarm history to the host computer with messages.
3	BR	SOF1	Issues an instruction for servo power off.
4	CA	DIR1	Transfers the directory information of the controller to the host computer.
5	DL	FLW1	Reads the content of a file in the controller and transfers it to the host computer.
6	DO	DOC1	Executes a direct command from the host computer.
7	EC	BRK1	Immediately stops automatic operation of the controller.
		MDE1	Executes a controller internal command.
		OVR1	Sets the override of the controller.
8	EM	-	Issues a change instruction for the EXTERNAL mode of the controller.
9	ER	DEL1	Deletes files from the controller.
10	EU	ALH2	Transfers the alarm history information to the host computer.
11	FD	FOF1	Turns off the feed hold of the controller.
		FON1	Turns on the feed hold of the controller.
12	HI	IOR1	Transfers the general-purpose input signal information of the controller to the host computer.
13	HO	IOW1	Forces writing of general-purpose output signals.
		IOW2	Forces writing of general-purpose output signals.
14	IO	IOR1	Transfers the I/O information of the specified model to the host computer.
15	IW	IOW1	Forces writing of general-purpose output signals.
		IOW2	Forces writing of general-purpose output signals.
16	JG	JOG1	Issues a jogging start command to the controller.
		JOI1	Performs the inching operation of the controller.
17	MD	-	Sets the controller guidance mode.
18	MN	MNT1	Transfers the specified maintenance information to the host computer.
		MNT2	Transfers the specified maintenance information to the host computer.
19	MP	MVP1	Moves to the teaching point using PTP movement.
		MVP2	Moves to the teaching point using CP movement.

No.	Command		Function
	Old	New	
20	MR	WHR1	Transfers the read data of the defined global variables to the host computer.
21	MW	WHW1	Writes data to the global variables defined in the controller.
22	PR	PSJ1	Transfers the current position information in the specified coordinate system to the host computer.
		PSW1	Transfers the current value information in the specified coordinate system to the host computer.
23	PS	PSJ1	Transfers the current value information of the joint to the host computer.
		PSW1	Transfers the current value information in the specified coordinate system to the host computer.
		STC1	Transfers the internal status of the controller to the host computer.
		TRQ1	Transfers the current motor torque information to the host computer.
24	RM	MNR1	Issues a reset command for the specified maintenance information to the controller.
25	RN	RUN1	Issues a program operation start command to the controller.
26	RS	ARH1	Clears the alarm history of the controller.
		ARN1	Clears the current alarm in the controller.
		RST1	Issues a reset command for the various statuses to the controller.
27	RT	JOS1	Sets the inching movement amount of the controller.
		JOM1	Sets the controller guidance rate.
28	SC	JOC1	Sets the controller guidance coordinates.
29	SF	STC1	Transfers the internal status of the controller to the host computer.
30	SL	SEL1	Used to select the program that is executed by automatic operation.
31	SM	-	Transfers the controller operation status to the host computer.
32	SO	SON1	Issues a servo on command to the controller.
33	SP	STP1	Issues an operation stop command to the controller.
34	SU	STC1	Transfers the internal status of the controller to the host computer.
35	UL	FLR1	Transfers the file content and writes it to the controller.
36	VL	VLJ1	Transfers the robot joint speed information to the host computer.
		VLW1	Transfers the hand speed information to the host computer.
		VLW2	Transfers the hand combined speed information to the host computer.
37	VR	VER1	Transfers the system version information to the host computer.
38	WD	-	Sets the watchdog timer of the controller.
39	ZS	CDS1	Sets the tool coordinates and work coordinates of the controller.